

Cambridge International AS & A Level

PHYSICS**9702/42**

Paper 4 A Level Structured Questions

February/March 2024**MARK SCHEME**Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **14** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

PUBLISHED**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	(gravitational) potential is zero at infinity	B1
	(gravitational force between two masses is attractive so)	
	either work is done on a mass to move it away from another mass or work is done on a mass to move it to infinity	B1
1(b)(i)	$M = (-) \text{ gradient} / G$	C1
	e.g. $M = (1.76 \times 10^8) / (3.0 \times 10^{-8} \times 6.67 \times 10^{-11}) = 8.8 \times 10^{25} \text{ kg}$	A1
1(b)(ii)	either $GMm / r^2 = mr\omega^2$ and $\omega = 2\pi / T$ or $GMm / r^2 = mv^2 / r$ and $v = 2\pi r / T$ or $GMm / r^2 = 4\pi^2 mr / T^2$	C1
	$R^3 = 6.67 \times 10^{-11} \times 8.8 \times 10^{25} \times (0.72 \times 24 \times 60 \times 60)^2 / 4\pi^2$	C1
	$R = 8.3 \times 10^7 \text{ m}$	A1
1(b)(iii)	$\Delta E = (GMm / r) - \frac{1}{2}mv^2$ kinetic energy = $(\frac{1}{2} \times 1200 \times 8400^2)$	C1
	potential energy = $(-)[(6.67 \times 10^{-11} \times 8.8 \times 10^{25} \times 1200) / (8.3 \times 10^7)]$	C1
	$\Delta E = [(6.67 \times 10^{-11} \times 8.8 \times 10^{25} \times 1200) / (8.3 \times 10^7)]$ $\quad - (\frac{1}{2} \times 1200 \times 8400^2)$ $= 4.3 \times 10^{10} \text{ J}$	A1

PUBLISHED

Question	Answer	Marks
2(a)	total kinetic energy associated with random motion of molecules	M1
	plus total potential energy (of molecules) but potential energy is zero	A1
2(b)(i)	$W = p\Delta V$	C1
	$= 1.01 \times 10^5 \times 5.20 \times 10^{-5}$ $= (+)5.25 \text{ J}$	A1
2(b)(ii)	$V \propto T$ or $V/T = \text{constant}$	C1
	$1.24 / (273 + 20) = (1.24 + 0.520) / T$	A1
	$T = 416 \text{ K}$	
2(b)(iii)	$c = Q / m\Delta T$	C1
	$= 960 / (0.016 \times (416 - 293))$ $= 490 \text{ J kg}^{-1} \text{ K}^{-1}$	A1
2(c)	no change in volume so no work is done (by the gas)	B1
	(same temperature change so) same change in internal energy	B1
	less thermal energy needs to be supplied so c is less	B1

Question	Answer	Marks
3(a)	$E = \frac{1}{2} m \omega^2 x_0^2$	C1
	$2.2 \times 10^{-4} = \frac{1}{2} \times 24 \times 10^{-3} \times (14 \times 10^{-3} / 4)^2 \times \omega^2$	C1
	$\omega = 39 \text{ rad s}^{-1}$	A1

PUBLISHED

Question	Answer	Marks
3(b)(i)	use of acceleration = 9.81 m s^{-2}	C1
	$x_0 = 9.81 / 39^2$ $= 6.4 \times 10^{-3} \text{ m}$	A1
3(b)(ii)	at top of oscillation	B1
	<i>any one point from:</i> where the downward acceleration first exceeds free-fall acceleration where the greatest downwards acceleration occurs where the resultant force is the maximum downwards where the contact force is a minimum	B1

Question	Answer	Marks
4(a)	combined capacitance of parallel capacitors = $30 \text{ } (\mu\text{F})$	C1
	total capacitance = $(1 / 45 + 1 / 30)^{-1}$ $= 18 \text{ } \mu\text{F}$	A1
4(b)	$E = \frac{1}{2} CV^2$	C1
	$\Delta E = \frac{1}{2} \times 45 \times 10^{-6} (9.6^2 - 8.0^2)$ $= 6.3 \times 10^{-4} \text{ J}$	A1
4(c)(i)	gaps in circuit closed and correct symbol for capacitor shown in parallel with load resistor	B1
4(c)(ii)	two correct pairs of values of t and V read off from within same discharge cycle, e.g. (5.0, 4.0) and (13.0, 3.2)	C1
	correct substitution of values of V , V_0 and Δt into $V = V_0 \exp(-\Delta t / \tau)$ e.g. $3.2 = 4.0 \exp(-8.0 / \tau)$	C1
	$\tau = 36 \text{ ms}$	A1

PUBLISHED

Question	Answer	Marks
4(d)(i)	8.0 W	A1
4(d)(ii)	4.0 W	A1

Question	Answer	Marks
5(a)	<i>any 2 points from:</i> • (angular) displacement • velocity • momentum • (centripetal) acceleration • (resultant) force	B2
5(b)(i)	$Bqv = mv^2 / r$	M1
	$v = 2\pi r / T$	M1
	completion of algebra leading to $B = 2\pi m / qT$	A1
5(b)(ii)	$B = (2\pi \times 4 \times 1.66 \times 10^{-27}) / (2 \times 1.60 \times 10^{-19} \times 2.5 \times 10^{-6})$	C1
	= 0.052 T	A1
5(b)(iii)	either the same because T is independent of r or the same because B , q and m are unchanged or the same because both radius and speed have doubled	B1
5(b)(iv)	$qE = Bqv$	C1
	$E = Bv = 0.052 \times 1.1 \times 10^6$ $= 5.7 \times 10^4 \text{ N C}^{-1}$	A1

PUBLISHED

Question	Answer	Marks
6(a)(i)	non-zero horizontal straight line from X to Y	B1
6(a)(ii)	constant flux density (inside coil)	B1
	either (magnetic) flux linkage proportional to flux density or $\Phi = BAN$ and B , A and N are all constant	B1
6(a)(iii)	$\Phi = BAN$	C1
	$= 0.080 \times 0.71 \times 10^{-4} \times 64$ $= 3.6 \times 10^{-4} \text{ Wb}$	A1
6(a)(iv)	<i>sketch showing:</i> E is zero from time 0 to time t and non-zero after time t	B1
	E has constant non-zero magnitude between time t and time $4t$	B1
	E has non-zero value of one sign between time t and time $2t$, and non-zero value of the opposite sign between time $2t$ and time $4t$	B1
6(b)	current in spring creates a magnetic field around the spring	B1
	either (magnetic) fields around adjacent turns interact to cause a force to be exerted (between the turns) or current in one turn interacts with (magnetic) field due to adjacent turns to cause force to be exerted (between the turns)	B1
	(magnetic force) is attractive so distance (between turns) decreases	B1

PUBLISHED

Question	Answer	Marks
7(a)	$p = E / c$	C1
	$= (3.11 \times 10^{-19}) / (3.00 \times 10^8)$ $= 1.04 \times 10^{-27} \text{ N s}$	A1
7(b)(i)	$E = hf$ and $c = f\lambda$ so energy of one photon $= hc / \lambda$ $350 \times 10^{-3} = N \times (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (640 \times 10^{-9})$	C1
	$N = 1.1 \times 10^{18}$	A1
7(b)(ii)	$F = (\text{change in}) \text{ momentum} / \text{time}$	M1
	Clear use of $p = E / c$ and $t = E / P$ to complete the algebra and arrive at the final equation: e.g. $F = [E / c] / [E / P] = P / c$	A1
7(c)(i)	maximum wavelength (of electromagnetic radiation) that causes electrons to be emitted (from surface of metal)	B1
7(c)(ii)	work function $= 2.26 \times 1.60 \times 10^{-19} \text{ (J)}$ $E = hc / \lambda$ so $(2.26 \times 1.60 \times 10^{-19}) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda_0$	C1
	$\lambda_0 = 5.50 \times 10^{-7} \text{ m}$	A1

Question	Answer	Marks
8(a)	(minimum) energy required to separate the nucleons (of a nucleus)	M1
	to infinity	A1
8(b)(i)	4	A1

PUBLISHED

Question	Answer	Marks
8(b)(ii)	energy = $(142 \times 8.37) + (90 \times 8.72) - (235 \times 7.59)$	C1
	= 190 MeV	A1
8(b)(iii)	either it has too many neutrons (for the number of protons) or its neutron to proton ratio is too high	B1
8(b)(iv)	(when $t = 6.0$ s), $N / N_0 = 1 / 32$	C1
	either $(1 / 32) = \exp(-\ln 2 \times 6.0 / t_{1/2})$	C1
	$t_{1/2} = 1.2$ s	A1
	or $32 / 2^n = 1$ so $n = 5$ (half-lives)	(C1)
	$t_{1/2} = 6.0 / 5$ = 1.2 s	(A1)

Question	Answer	Marks
9(a)(i)	$eV = hc / \lambda$	C1
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (84 \times 10^3 \times 1.60 \times 10^{-19})$	A1
	= 1.5×10^{-11} m	
9(a)(ii)	either (some) kinetic energy (of electrons) is converted to thermal energy at target or some X-rays are absorbed by the target so its temperature increases	B1
	(tungsten) has higher melting point so does not melt quickly / easily	B1

PUBLISHED

Question	Answer	Marks
9(b)	$I = I_0 \exp(-\mu t)$	C1
	$0.13 = [\exp(-3.0x)] \times [\exp(-0.22x)]$ $= \exp(-3.22x)$	C1
	$x = 0.63 \text{ cm}$	A1
9(c)(i)	product of density and speed	M1
	speed of ultrasound in medium	A1
9(c)(ii)	$I_R / I_0 = (7.8 - 1.7)^2 / (7.8 + 1.7)^2$ $= 0.41$ fraction transmitted = $1.00 - 0.41 = 0.59$	C1
	percentage transmitted = 59%	A1
9(c)(iii)	more than one <u>boundary</u> so more reflections	B1
	some ultrasound is attenuated in matter	B1

Question	Answer	Marks
10(a)(i)	$L = 4\pi\sigma r^2 T^4$	C1
	$3.85 \times 10^{26} = 4\pi \times 5.67 \times 10^{-8} \times r^2 \times 5780^4$	
	$r = 6.96 \times 10^8 \text{ m}$	A1

PUBLISHED

Question	Answer	Marks
10(a)(ii)	$F = L / 4\pi d^2$ $= (3.85 \times 10^{26}) / (4\pi \times (1.50 \times 10^{11})^2)$	C1
	$= 1.36 \times 10^3 \text{ W m}^{-2}$	A1
10(a)(iii)	line of same shape showing peak intensity at greater wavelength	B1
	line of same shape showing lower peak intensity	B1
10(b)(i)	5 lines in same pattern shifted to longer wavelengths	B1
10(b)(ii)	$\Delta\lambda / \lambda = v / c$ $\Delta\lambda = (21400 / 300000) \times 656$ $= 46.8 \text{ nm}$	C1
	wavelength = $656 + 46.8$ = 703 nm	A1
10(b)(iii)	(peak) wavelength too high so temperature too low	B1

Cambridge International AS & A Level

PHYSICS**9702/52**

Paper 5 Planning, Analysis and Evaluation

February/March 2024**MARK SCHEME**Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	L is the independent variable and f is the dependent variable, or vary L and measure f .	1
	Keep ρ <u>constant</u>	1
	Methods of data collection	
	Labelled diagram of workable experiment including: - rod supported by string / elastic bands from a clamp - clamp attached to stand, with stand on bench - two labels from stand, clamp, hammer, microphone, rod, string.	1
	Diagram showing labelled microphone connected to labelled oscilloscope.	1
	Method to measure L , e.g. use a metre rule	1
	Method to measure mass (m) (of metal rod), e.g. use a (top-pan) balance	1
	Method of Analysis	
	Plots a graph of $\log f$ against $\log L$ or equivalent e.g. $\log f$ against $\log \frac{1}{L}$	1
	$n = -$ gradient (for $\log f$ against $\log \frac{1}{L}$: $n =$ gradient)	1
	$E = 4\rho \times 10^{2 \times y\text{-intercept}}$ (for $\lg f$ vs $\lg \frac{1}{L}$: $E = 4\rho \times 10^{2 \times y\text{-intercept}}$) (for $\ln f$ against $\ln L$; $E = 4\rho \times e^{2 \times y\text{-intercept}}$)	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations Any six from: D1 Precaution linked to <u>falling rod</u>, e.g. sand tray / cushion (in case <u>rod</u> falls) OR gently hit rod prevent <u>rod</u> falling D2 Method to determine area of rod (A) e.g. measure <u>diameter</u> (d) of rod using a micrometer / calipers D3 Repeat measurements of diameter along the length of rod / around the rod and average diameter D4 Method to determine ρ from experimental method, e.g. $\rho = \frac{m}{AL}$ and $A = \frac{\pi d^2}{4}$ or $\rho = \frac{4m}{\pi d^2 L}$ or $r = \frac{d}{2}$ and $\rho = \frac{m}{\pi r^2 L}$ D5 Perform experiment in a quiet room D6 Reasoned method to prevent rod hitting microphone, e.g. have a gap between rod and microphone / gently hit rod or method to obtain measurable signal from the microphone, e.g. use a cone to increase the sound detected by the microphone D7 Method to determine frequency from oscilloscope, e.g. $T = \text{time-base} \times (\text{horizontal}) \text{ length (of one wave)}$ and $f = 1/T$ D8 Method to reduce uncertainties e.g. use large values of L to reduce (percentage) uncertainty in L or adjust time-base to display as few waves as possible or Z waves on oscilloscope and divide time by Z or wait for the wave(form) / frequency to stabilise (and reach resonance) D9 Repeat measurements of f for each value of L and average f	6

PUBLISHED

Question	Answer	Marks
1	D10 Relationship is valid <u>if</u> a straight line is produced (passing through $\log\left(\frac{1}{2}\sqrt{\frac{E}{\rho}}\right)$ OR $\frac{1}{2}\log\left(\sqrt{\frac{E}{4\rho}}\right)$). Do not accept passing through the origin.	

Question	Answer	Marks							
2(a)	Gradient = $\frac{3}{E}$ y-intercept = $\frac{4Z}{E}$	1							
2(b)	<table><tr><td>$\frac{1}{I} / \text{A}^{-1}$</td></tr><tr><td>4440 or 4444</td></tr><tr><td>5410 or 5405</td></tr><tr><td>6250 or 6250</td></tr><tr><td>7140 or 7143</td></tr><tr><td>8000 or 8000</td></tr><tr><td>8700 or 8696</td></tr></table>	$\frac{1}{I} / \text{A}^{-1}$	4440 or 4444	5410 or 5405	6250 or 6250	7140 or 7143	8000 or 8000	8700 or 8696	1
$\frac{1}{I} / \text{A}^{-1}$									
4440 or 4444									
5410 or 5405									
6250 or 6250									
7140 or 7143									
8000 or 8000									
8700 or 8696									
	Uncertainties in $\frac{1}{I}$ From $\pm 90\text{--}110$ to $\pm 360\text{--}400$	1							

PUBLISHED

Question	Answer	Marks
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1
	Error bars in $\frac{1}{I}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top plot to bottom plot. Points must be balanced. Line must pass between (1.8, 5000) and (2.1, 5000) and between (7.2, 8500) and (7.5, 8500)	1
	Worst acceptable line drawn. Steepest or shallowest possible line that passes through all the error bars. All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$; distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$; uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and x into $y = mx + c$	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$ uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line, or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept)	1

PUBLISHED

Question	Answer	Marks
2(d)(i)	E determined using gradient and E and Z given to 2, 3 or 4 sf. $E = \frac{3}{\text{gradient}}$	1
	Z determined using y-intercept and E and Z given with SI units with correct powers of ten $Z = \frac{E \times y\text{-intercept}}{4} \text{ or } Z = \frac{3 \times y\text{-intercept}}{4 \times \text{gradient}}$ Unit of E: V or A Ω Unit of Z: Ω	1
2(d)(ii)	Percentage uncertainty in Z with method shown. $\% \text{uncertainty} = \left(\frac{\Delta \text{gradient}}{\text{gradient}} + \frac{\Delta y\text{-intercept}}{y\text{-intercept}} \right)$ or Correct substitution for max/min methods.	1
2(e)	R determined to a minimum of 2sf from (c)(iii) and (c)(iv) or (d)(i) with correct substitution and correct powers of ten. $0.1 \text{ mA} = 0.1 \times 10^{-3} \text{ A}$ and $R = \frac{1}{0.10 \times 10^{-3} - y\text{-intercept}} \text{ or }$ $R = \frac{E}{3 \times 0.10 \times 10^{-3}} - \frac{4Z}{3}$	1



Cambridge International AS & A Level

PHYSICS

9702/41

Paper 4 A Level Structured Questions

May/June 2022

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **16** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Annotations

✓	Indicates the point at which a mark has been awarded.
X	Indicates an incorrect answer or a point at which a decision is made not to award a mark.
XP	Indicates a physically incorrect equation ('incorrect physics'). No credit is given for substitution, or subsequent arithmetic, in a physically incorrect equation.

PUBLISHED

ECF	Indicates 'error carried forward'. Answers to later numerical questions can always be awarded up to full credit provided they are consistent with earlier incorrect answers. <u>Within</u> a section of a numerical question, ECF can be given after AE, TE and POT errors, but not after XP.
AE	Indicates an arithmetic error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
POT	Indicates a power of ten error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
TE	Indicates incorrect transcription of the correct data from the question, a graph, data sheet or a previous answer. For example, the value of 1.6×10^{-19} has been written down as 6.1×10^{-19} or 1.6×10^{19} . Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
SF	Indicates that the correct answer is seen in the working but the final answer is incorrect as it is expressed to too few significant figures.
BOD	Indicates that a mark is awarded where the candidate provides an answer that is not totally satisfactory, but the examiner feels that sufficient work has been done ('benefit of doubt').
CON	Indicates that a response is contradictory.
I	Indicates parts of a response that have been seen but disregarded as irrelevant.
M0	Indicates where an A category mark has not been awarded due to the M category mark upon which it depends not having previously been awarded.
^	Indicates where more is needed for a mark to be awarded (what is written is not wrong, but not enough). May also be used to annotate a response space that has been left completely blank.
SEEN	Indicates that a page has been seen.

Question	Answer	Marks
1(a)(i)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(a)(ii)	$g = F / m$	C1
	$F = GMm / r^2$ and so $g = [GMm / r^2] / m = GM / r^2$	A1
1(b)(i)	$g = (6.67 \times 10^{-11} \times 7.35 \times 10^{22}) / (1.74 \times 10^6)^2 = 1.62 \text{ N kg}^{-1}$	A1
1(b)(ii)	fields (due to Earth and the Moon) have equal magnitudes	B1
	fields (due to Earth and the Moon) are in opposite directions	B1
1(b)(iii)	distance of X from Earth = $(3.84 \times 10^8 - x)$	C1
	$(G \times) 7.35 \times 10^{22} / x^2 = (G \times) 5.98 \times 10^{24} / (3.84 \times 10^8 - x)^2$	C1
	$x = 3.8 \times 10^7 \text{ m}$	A1

Question	Answer	Marks
2(a)(i)	(vertically) downwards	B1
2(a)(ii)	magnetic force (on sphere) is perpendicular to its velocity	B1
	magnetic force perpendicular to velocity is the centripetal force or magnetic force perpendicular to velocity causes centripetal acceleration or acceleration perpendicular to velocity is centripetal (acceleration) or magnetic force does not change the speed of the sphere or magnetic force has constant magnitude	B1
2(b)	$mg = Eq$	C1
	$E = (1.6 \times 10^{-10} \times 9.81) / (0.27 \times 10^{-9})$ $= 5.8 \text{ N C}^{-1}$	A1
2(c)	centripetal force = magnetic force or $Bqv = mv^2 / r$	B1
	$B = mv / qr$	C1
	$= (1.6 \times 10^{-10} \times 0.78) / (0.27 \times 10^{-9} \times 3.4) = 0.14 \text{ T}$	A1

Question	Answer	Marks											
3(a)(i)	a gas that obeys $pV \propto T$	M1											
	where p = pressure, V = volume, T = thermodynamic temperature	A1											
3(a)(ii)	$T = (273 + 17) \text{ K}$	C1											
	$n = pV / RT$ $= (1.2 \times 10^5 \times 0.24) / [8.31 \times (273 + 17)]$ $= 12 \text{ mol}$	A1											
3(b)(i)	work done = $p\Delta V$	C1											
	$= 1.2 \times 10^5 \times (0.24 - 0.08) = 19200 \text{ J } (= 19.2 \text{ kJ})$	A1											
3(b)(ii)	AB work done correct (19.2)	A1											
	BC work done correct (0)	A1											
	CA increase in internal energy correct (0) and CA thermal energy correct (31.6)	A1											
	AB increase in internal energy calculated correctly from work done – 48.0	A1											
	BC increase in internal energy correctly calculated so the final column adds up to zero and BC thermal energy same as increase in internal energy	A1											
	(Fully correct table: <table><tr><td>AB</td><td>19.2</td><td>−48.0</td><td>−28.8</td></tr><tr><td>BC</td><td>0</td><td>28.8</td><td>28.8</td></tr><tr><td>CA</td><td>−31.6</td><td>31.6</td><td>0</td></tr></table>)	AB	19.2	−48.0	−28.8	BC	0	28.8	28.8	CA	−31.6	31.6	0
AB	19.2	−48.0	−28.8										
BC	0	28.8	28.8										
CA	−31.6	31.6	0										

Question	Answer	Marks
4(a)	straight line through origin shows that a is proportional to x	B1
	negative gradient shows that a is in opposite direction to x	B1
4(b)(i)	$a_0 = \omega^2 x_0$ or $a = -\omega^2 x$ or $\omega^2 = -\text{gradient}$	C1
	$\omega = \sqrt{(0.40 / 0.050)}$ $= 2.8 \text{ rad s}^{-1}$	A1
4(b)(ii)	$k = \omega^2 L$ $= 2.8^2 \times 1.24$	C1
	$= 9.7 \text{ m s}^{-2}$	A1
4(c)	(increasing L causes) ω to decrease or energy $(= \frac{1}{2} m \omega^2 x_0^2) = \frac{1}{2} m k x_0^2 / L$ (and L increases)	M1
	so amplitude increases	A1

Question	Answer	Marks
5(a)(i)	conversion (from a.c.) to d.c.	B1
5(a)(ii)	full-wave (rectification)	B1
5(b)(i)	P labelled – and Q labelled +	B1
5(b)(ii)	V_{OUT} scale labelled 4 and 8 on the 2 cm tick marks	B1
	$T = 2\pi / \omega$ $= 2\pi / 25\pi$ $= 0.08 \text{ s}$	C1
	t scale labelled 0.02, 0.04, 0.06, 0.08, 0.10, 0.12 on the 2 cm tick marks	A1
5(c)(i)	correct symbol used for capacitor and capacitor connected in parallel with the $1.2 \text{ k}\Omega$ resistor.	B1
5(c)(ii)	straight lines or curves, with negative decreasing gradients, drawn between adjacent peaks, from top of first peak to meet line going up to next peak	B1
	lines, from one peak to the line going up to the next peak, show a drop in p.d. of $1\frac{1}{2}$ small squares	B1
5(c)(iii)	$V = 0.90 \times 6.0 (= 5.4 \text{ V})$ or discharge time (for each cycle) = 0.034 s	C1
	$V = V_0 \exp(-t/RC)$ $5.4 = 6.0 \exp[-0.034 / (1.2 \times 10^3 \times C)]$	C1
	$C = 2.7 \times 10^{-4} \text{ F}$	A1

PUBLISHED

Question	Answer	Marks
6(a)	product of (magnetic) flux density and area	M1
	where area is perpendicular to the (magnetic) field	A1
6(b)(i)	$N\Phi = BAN$	C1
	$= 400 \times 10^{-3} \times 0.12^2 \times 8$	C1
	$= 0.046 \text{ Wb}$	A1
6(b)(ii)	(line is a) straight line	B1
6(b)(iii)	(induced) e.m.f. = rate of change of flux linkage	C1
	e.m.f. $= N\Phi / t$	A1
	$= 0.046 / 0.60$	
6(c)	$= 0.077 \text{ V}$	
	(induced e.m.f. causes) current flow (in the coil)	B1
	either	
	current (in magnetic field) causes forces to act on the coil	B1
	(opposite sides of) coil forced inwards	B1
	or	
	current causes dissipation of energy in the resistance of the coil	(B1)
	temperature of the coil rises	(B1)

Question	Answer	Marks
7(a)	quantum of energy	M1
	of electromagnetic radiation	A1
7(b)(i)	photoelectric effect	B1
7(b)(ii)	- there is a frequency below which no electrons are emitted or threshold frequency = 5.4×10^{14} Hz - work function of the metal = 3.6×10^{-19} J (or 2.2 eV) E_{MAX} increases (linearly) with (increasing) frequency - gradient of the line is the Planck constant or gradient of the line is 6.7×10^{-34} J s <i>Any three bullet points, 1 mark each</i>	B3
7(c)(i)	different threshold frequency	B1
	(line has) same gradient but different intercept	B1
7(c)(ii)	photons have same energy	B1
	line unchanged	B1

Question	Answer	Marks
8(a)(i)	energy required to separate the nucleons (in the nucleus)	M1
	to infinity	A1
8(a)(ii)	curve starting close to the origin and forming a single peak	B1
	peak shown to left of centre, with steep line on LHS of peak and shallow line on RHS of peak	B1
8(b)(i)	fusion	B1
8(b)(ii)	both particles have low A values or both particles are at left-hand end of graph	B1
	He-3 has higher binding energy (per nucleon) than H-2	B1
8(c)	$\Delta m = [(2 \times 2.014102) - (3.016029 + 1.008665)] \text{ u}$ (= 0.00351 u)	C1
	$E = \Delta mc^2$	C1
	$= 0.00351 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ (= $5.24 \times 10^{-13} \text{ J}$)	C1
	1.00 mol of deuterium forms 0.500 mol of helium-3	C1
	total energy = $0.500 \times 6.02 \times 10^{23} \times 5.24 \times 10^{-13}$ = $1.58 \times 10^{11} \text{ J}$	A1

PUBLISHED

Question	Answer	Marks
9(a)(i)	electrons are accelerated (by an applied p.d.)	B1
	electrons hit target	B1
	X-rays produced when electrons decelerate	B1
9(a)(ii)	images of the multiple sections are combined to create a 3-D image	B1
9(b)(i)	$I = I_0 \exp(-\mu x)$	C1
	$= I_0 \exp(-0.89 \times 5.6)$	A1
	$= 0.0068 I_0$	
9(b)(ii)	$I = I_0 \exp(-2.4 \times 3.4) \times \exp(-0.89 \times 3.2)$	C1
	$= 1.7 \times 10^{-5} I_0$	A1
9(c)	comparison of intensities or values in (b) leading to conclusion consistent with these values	B1

Question	Answer	Marks
10(a)	wavelength of maximum intensity is inversely proportional to (thermodynamic) temperature	B1
10(b)(i)	$\lambda_{\text{MAX}} = 0.50 \mu\text{m}$ for A and $0.65 \mu\text{m}$ for B	C1
	$T = 5800 \times (0.50 / 0.65)$ $= 4500 \text{ K}$	A1
10(b)(ii)	(star B has) greater peak / average wavelength	B1
	(star B looks) redder	B1
10(c)(i)	apparent wavelength is greater or wavelength is greater than known value	B1
	(due to) movement of star away (from observer)	B1
10(c)(ii)	by examining the (lines in the) spectrum (of light from the star)	B1
	and comparing with known spectrum	B1



Cambridge International AS & A Level

PHYSICS

9702/42

Paper 4 A Level Structured Questions

May/June 2022

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **16** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Annotations

✓	Indicates the point at which a mark has been awarded.
X	Indicates an incorrect answer or a point at which a decision is made not to award a mark.
XP	Indicates a physically incorrect equation ('incorrect physics'). No credit is given for substitution, or subsequent arithmetic, in a physically incorrect equation.

PUBLISHED

ECF	Indicates ‘error carried forward’. Answers to later numerical questions can always be awarded up to full credit provided they are consistent with earlier incorrect answers. <u>Within</u> a section of a numerical question, ECF can be given after AE, TE and POT errors, but not after XP.
AE	Indicates an arithmetic error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
POT	Indicates a power of ten error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
TE	Indicates incorrect transcription of the correct data from the question, a graph, data sheet or a previous answer. For example, the value of 1.6×10^{-19} has been written down as 6.1×10^{-19} or 1.6×10^{19} . Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
SF	Indicates that the correct answer is seen in the working but the final answer is incorrect as it is expressed to too few significant figures.
BOD	Indicates that a mark is awarded where the candidate provides an answer that is not totally satisfactory, but the examiner feels that sufficient work has been done (‘benefit of doubt’).
CON	Indicates that a response is contradictory.
I	Indicates parts of a response that have been seen but disregarded as irrelevant.
MO	Indicates where an A category mark has not been awarded due to the M category mark upon which it depends not having previously been awarded.
^	Indicates where more is needed for a mark to be awarded (what is written is not wrong, but not enough). May also be used to annotate a response space that has been left completely blank.
SEEN	Indicates that a page has been seen.

PUBLISHED

Question	Answer	Marks
1(a)(i)	work (done) per unit mass	B1
	work (done on mass) in moving mass from infinity (to the point)	B1
1(a)(ii)	$E_P = \phi m$ $E_P = (-GM/r) \times m = -GMm/r$ or $\phi = -GM/r$ and $E_P = \phi m = -GMm/r$	B1
1(b)(i)	$\Delta E_P = 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 2.20 \times 10^{14} \times [1/(6.38 \times 10^{10}) - 1/(8.44 \times 10^{11})]$	C1
	$= 4.23 \times 10^{23} \text{ J}$	A1
1(b)(ii)	(gravitational) force is attractive so decrease or (gravitational) force does work so decrease	B1
1(b)(iii)	$\Delta E_P = \frac{1}{2}m(v_2^2 - v_1^2)$	C1
	$4.23 \times 10^{23} = \frac{1}{2} \times 2.20 \times 10^{14} \times (v^2 - 34\,100^2)$	C1
	$v (= 70\,800 \text{ m s}^{-1}) = 70.8 \text{ km s}^{-1}$	A1
1(c)	both PE and KE equations include m , so path is unchanged	B1

Question	Answer	Marks
2(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
2(b)(i)	(electric) force is perpendicular to velocity (of particles)	B1
	force (perpendicular to velocity) causes centripetal acceleration or force does not change the speed of the particles or force has constant magnitude	B1
2(b)(ii)	$F = e^2 / 4\pi\epsilon_0 x^2$	C1
	$= (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (2 \times 1.59 \times 10^{-10})^2]$	A1
	$= 2.28 \times 10^{-9} \text{ N}$	
2(b)(iii)	$F = m\omega^2$ and $\omega = 2\pi / T$ or $F = mv^2 / r$ and $v = 2\pi r / T$	C1
	$F = 4\pi^2 mr / T^2$	C1
	$T = \sqrt{[4\pi^2 \times 9.11 \times 10^{-31} \times 1.59 \times 10^{-10} / (2.28 \times 10^{-9})]}$	
	$= 1.58 \times 10^{-15} \text{ s}$	A1
2(c)(i)	- electron and positron interact - positron is anti-particle of electron (pair) annihilation occurs <i>Any two points, 1 mark each</i>	B2
	mass of the electron and positron converted into photon energy	B1
2(c)(ii)	PET scanning	B1

Question	Answer	Marks
3(a)	(thermal) energy per unit mass	B1
	energy to change state between liquid and gas at constant temperature	B1
3(b)(i)	$q = mL = 0.37 \times 2.3 \times 10^6$ $= 8.5 \times 10^5 \text{ J}$	A1
3(b)(ii)	$pV = nRT$ and $T = 373 \text{ K}$	C1
	$n = 370 / 18$	C1
	$V = [(370 / 18) \times 8.31 \times 373] / (1.0 \times 10^5) = 0.64 \text{ m}^3$	A1
3(b)(iii)	$w = p\Delta V$	C1
	$= 1.0 \times 10^5 \times 0.64$	A1
	$= 6.4 \times 10^4 \text{ J}$	
3(b)(iv)	(water does work against atmosphere so) work done on water is negative	B1
	increase in internal energy $= (8.5 - 0.64) \times 10^5 = 7.9 \times 10^5 \text{ J}$	A1
3(c)	valid reasoning of how work done by water is affected	M1
	correct use of first law to draw conclusion about effect on specific latent heat that is consistent with work done	A1

PUBLISHED

Question	Answer	Marks
4(a)	oscillations (of object) at maximum amplitude	B1
	when driving frequency equals natural frequency (of object)	B1
4(b)(i)	$T = 2\pi / \omega$	C1
	$= 2\pi / 5.0\pi$	A1
	$= 0.40 \text{ s}$	
4(b)(ii)	displacement scale labelled $-1.0, -0.5, (0), 0.5, 1.0$ on the 2 cm tick marks	B1
	t scale labelled $0.2, 0.4, 0.6, 0.8, 1.0, 1.2$ on the 2 cm tick marks	B1
4(b)(iii)	$\phi = 2\pi\Delta t / T$	C1
	$= 2\pi \times 0.10 / 0.40$ or $2\pi \times 0.30 / 0.40$	
	$= 1.6 \text{ rad}$ or 4.7 rad	A1

Question	Answer	Marks
5(a)	charge / potential (difference)	M1
	charge is charge on one plate, <u>and</u> potential is p.d. across the plates	A1
5(b)	p.d. across both capacitors = E	B1
	$Q_T = Q_1 + Q_2$	B1
	$C_T E = C_1 E + C_2 E$ hence $C_T = C_1 + C_2$	B1
5(c)(i)	$[(1/22) + (1/47)]^{-1} = 15 \mu\text{F}$	A1
5(c)(ii)	energy = $\frac{1}{2}CV^2$	C1
	$= \frac{1}{2} \times 15 \times 10^{-6} \times 12^2$	A1
	$= 1.1 \times 10^{-3} \text{ J}$	
5(c)(iii)	initial p.d. (across $22 \mu\text{F}$) = $12 \times (15/22)$ $= 8.2 \text{ V}$ or final p.d. across both capacitors = $6.0 \times (22/15)$ $= 8.8 \text{ V}$	C1
	$V = V_0 \exp[-t/(2.7 \times 10^6 \times 15 \times 10^{-6})]$	C1
	$6.0 = 8.2 \exp[-t/(2.7 \times 10^6 \times 15 \times 10^{-6})]$ or $8.8 = 12 \exp[-t/(2.7 \times 10^6 \times 15 \times 10^{-6})]$ $t = 13 \text{ s}$	A1

PUBLISHED

Question	Answer	Marks
6(a)	there must be a current (in the wire)	B1
	(wire) must be at a non-zero angle to the magnetic field	B1
6(b)(i)	arrow from X pointing horizontally to the left	B1
	arrow from Y pointing diagonally upwards and to the left at about 45°	B1
	arrow from Z pointing horizontally to the right	B1
6(b)(ii)	(flux densities at W and X are approximately) equal	B1
	(flux density at) Y greater than (flux density at) Z	B1
6(c)	current in wire creates magnetic field around wire	B1
	(each) wire sits in the magnetic field created by the other	B1
	(for each wire,) current / wire is perpendicular to magnetic field (due to other wire), (so) experiences a (magnetic) force	B1

PUBLISHED

Question	Answer	Marks
7(a)	induced e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)	V_2 stepped, all at non-zero values, between $t = 0$ and $t = 0.40$ s	B1
	V_2 shown with same non-zero magnitude up to $t = 0.15$ s and after $t = 0.25$ s but with a different magnitude between these times	B1
	V_2 shown with a magnitude between $t = 0.15$ s and $t = 0.25$ s that is three times the magnitude before $t = 0.15$ s and after $t = 0.25$ s	B1
	V_2 shown with same sign up to $t = 0.15$ s and after $t = 0.25$ s, and opposite sign in between	B1
7(c)(i)	changing current in coil causes changing (magnetic) field or changing (magnetic) flux causes induced e.m.f. in ring	B1
	induced e.m.f. in ring causes current in ring	B1
	(magnetic) field due to (induced) current in ring interacts with (coil's) field to cause upwards force (on ring) or (induced) current in ring perpendicular to (coil's magnetic) field causes upwards force (on ring)	B1
7(c)(ii)	both magnetic fields reverse direction so ring still jumps up or current (in ring) and (coil's) field both reverse so ring still jumps up	B1

PUBLISHED

Question	Answer	Marks
8(a)(i)	photoelectric effect	B1
8(a)(ii)	electron diffraction	B1
8(b)(i)	$\lambda = h / p$	C1
	$p = 4 \times 1.66 \times 10^{-27} \times 6.2 \times 10^7$ (= $4.1 \times 10^{-19} \text{ N s}$)	C1
	$\lambda = 6.63 \times 10^{-34} / 4.1 \times 10^{-19}$ $= 1.6 \times 10^{-15} \text{ m}$	A1
8(b)(ii)	line with negative gradient throughout	B1
	curve asymptotic to both axes with non-zero λ at $v = 6.2 \times 10^7 \text{ m s}^{-1}$	B1
8(c)	(de Broglie) wavelength negligible compared with width of doorway	B1

PUBLISHED

Question	Answer	Marks
9(a)(i)	speed is (directly) proportional to distance	M1
	where speed is speed of recession of galaxy (from observer) and distance is distance of galaxy away from observer	A1
9(a)(ii)	wavelengths (of spectral lines) are greater (than their known values)	B1
	redshift shows stars (in distant galaxies) moving away from Earth	B1
9(b)	(all) parts of Universe moving away from each other	B1
	more distant objects are moving away faster	B1
	matter must have been close together / very dense in the past	B1

PUBLISHED

Question	Answer	Marks
10(a)	spontaneous emission of (ionising) radiation	B1
	emission from unstable nucleus	B1
10(b)(i)	curve with decreasing negative gradient passing through $(0, N_0)$	B1
	curve passing through $(T, 0.5N_0)$	B1
	curve passing through $(2T, 0.25N_0)$ and $(3T, 0.125N_0)$	B1
10(b)(ii)	line through origin with positive gradient	B1
	straight line passing through (N_0, A_0)	B1
10(c)(i)	activity	B1
10(c)(ii)	decay constant	B1
10(d)	$N = N_0 \exp(-\ln 2 \times 1.70T / T)$	C1
	$N / N_0 = 0.31$	A1



Cambridge International AS & A Level

PHYSICS

9702/43

Paper 4 A Level Structured Questions

May/June 2022

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **16** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Annotations

✓	Indicates the point at which a mark has been awarded.
X	Indicates an incorrect answer or a point at which a decision is made not to award a mark.
XP	Indicates a physically incorrect equation ('incorrect physics'). No credit is given for substitution, or subsequent arithmetic, in a physically incorrect equation.

ECF	Indicates 'error carried forward'. Answers to later numerical questions can always be awarded up to full credit provided they are consistent with earlier incorrect answers. <u>Within</u> a section of a numerical question, ECF can be given after AE, TE and POT errors, but not after XP.
AE	Indicates an arithmetic error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
POT	Indicates a power of ten error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
TE	Indicates incorrect transcription of the correct data from the question, a graph, data sheet or a previous answer. For example, the value of 1.6×10^{-19} has been written down as 6.1×10^{-19} or 1.6×10^{19} . Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
SF	Indicates that the correct answer is seen in the working but the final answer is incorrect as it is expressed to too few significant figures.
BOD	Indicates that a mark is awarded where the candidate provides an answer that is not totally satisfactory, but the examiner feels that sufficient work has been done ('benefit of doubt').
CON	Indicates that a response is contradictory.
I	Indicates parts of a response that have been seen but disregarded as irrelevant.
M0	Indicates where an A category mark has not been awarded due to the M category mark upon which it depends not having previously been awarded.
^	Indicates where more is needed for a mark to be awarded (what is written is not wrong, but not enough). May also be used to annotate a response space that has been left completely blank.
SEEN	Indicates that a page has been seen.

Question	Answer	Marks
1(a)(i)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(a)(ii)	$g = F / m$	C1
	$F = GMm / r^2$ and so $g = [GMm / r^2] / m = GM / r^2$	A1
1(b)(i)	$g = (6.67 \times 10^{-11} \times 7.35 \times 10^{22}) / (1.74 \times 10^6)^2 = 1.62 \text{ N kg}^{-1}$	A1
1(b)(ii)	fields (due to Earth and the Moon) have equal magnitudes	B1
	fields (due to Earth and the Moon) are in opposite directions	B1
1(b)(iii)	distance of X from Earth = $(3.84 \times 10^8 - x)$	C1
	$(G \times) 7.35 \times 10^{22} / x^2 = (G \times) 5.98 \times 10^{24} / (3.84 \times 10^8 - x)^2$	C1
	$x = 3.8 \times 10^7 \text{ m}$	A1

PUBLISHED

Question	Answer	Marks
2(a)(i)	(vertically) downwards	B1
2(a)(ii)	magnetic force (on sphere) is perpendicular to its velocity	B1
	magnetic force perpendicular to velocity is the centripetal force or magnetic force perpendicular to velocity causes centripetal acceleration or acceleration perpendicular to velocity is centripetal (acceleration) or magnetic force does not change the speed of the sphere or magnetic force has constant magnitude	B1
2(b)	$mg = Eq$	C1
	$E = (1.6 \times 10^{-10} \times 9.81) / (0.27 \times 10^{-9})$ $= 5.8 \text{ N C}^{-1}$	A1
2(c)	centripetal force = magnetic force or $Bqv = mv^2 / r$	B1
	$B = mv / qr$	C1
	$= (1.6 \times 10^{-10} \times 0.78) / (0.27 \times 10^{-9} \times 3.4) = 0.14 \text{ T}$	A1

Question	Answer	Marks										
3(a)(i)	a gas that obeys $pV \propto T$	M1										
	where p = pressure, V = volume, T = thermodynamic temperature	A1										
3(a)(ii)	$T = (273 + 17) \text{ K}$	C1										
	$n = pV / RT$ $= (1.2 \times 10^5 \times 0.24) / [8.31 \times (273 + 17)]$ $= 12 \text{ mol}$	A1										
3(b)(i)	work done = $p\Delta V$	C1										
	$= 1.2 \times 10^5 \times (0.24 - 0.08) = 19200 \text{ J } (= 19.2 \text{ kJ})$	A1										
3(b)(ii)	AB work done correct (19.2)	A1										
	BC work done correct (0)	A1										
	CA increase in internal energy correct (0) and CA thermal energy correct (31.6)	A1										
	AB increase in internal energy calculated correctly from work done – 48.0	A1										
	BC increase in internal energy correctly calculated so the final column adds up to zero and BC thermal energy same as increase in internal energy	A1										
	(Fully correct table: <table><tr><td>AB</td><td>19.2</td><td>–48.0</td><td>–28.8</td></tr><tr><td>BC</td><td>0</td><td>28.8</td><td>28.8</td></tr><tr><td>CA</td><td>–31.6</td><td>31.6</td><td>0</td></tr></table>)		AB	19.2	–48.0	–28.8	BC	0	28.8	28.8	CA	–31.6
AB	19.2	–48.0	–28.8									
BC	0	28.8	28.8									
CA	–31.6	31.6	0									

Question	Answer	Marks
4(a)	straight line through origin shows that a is proportional to x	B1
	negative gradient shows that a is in opposite direction to x	B1
4(b)(i)	$a_0 = \omega^2 x_0$ or $a = -\omega^2 x$ or $\omega^2 = -\text{gradient}$	C1
	$\omega = \sqrt{(0.40 / 0.050)}$ $= 2.8 \text{ rad s}^{-1}$	A1
4(b)(ii)	$k = \omega^2 L$ $= 2.8^2 \times 1.24$	C1
	$= 9.7 \text{ m s}^{-2}$	A1
4(c)	(increasing L causes) ω to decrease or energy ($= \frac{1}{2} m \omega^2 x_0^2$) $= \frac{1}{2} m k x_0^2 / L$ (and L increases)	M1
	so amplitude increases	A1

PUBLISHED

Question	Answer	Marks
5(a)(i)	conversion (from a.c.) to d.c.	B1
5(a)(ii)	full-wave (rectification)	B1
5(b)(i)	P labelled – and Q labelled +	B1
5(b)(ii)	V_{OUT} scale labelled 4 and 8 on the 2 cm tick marks	B1
	$T = 2\pi / \omega$ $= 2\pi / 25\pi$ $= 0.08 \text{ s}$	C1
	t scale labelled 0.02, 0.04, 0.06, 0.08, 0.10, 0.12 on the 2 cm tick marks	A1
5(c)(i)	correct symbol used for capacitor and capacitor connected in parallel with the $1.2 \text{ k}\Omega$ resistor.	B1
5(c)(ii)	straight lines or curves, with negative decreasing gradients, drawn between adjacent peaks, from top of first peak to meet line going up to next peak	B1
	lines, from one peak to the line going up to the next peak, show a drop in p.d. of $1\frac{1}{2}$ small squares	B1
5(c)(iii)	$V = 0.90 \times 6.0 (= 5.4 \text{ V})$ or discharge time (for each cycle) = 0.034 s	C1
	$V = V_0 \exp(-t/RC)$ $5.4 = 6.0 \exp[-0.034 / (1.2 \times 10^3 \times C)]$	C1
	$C = 2.7 \times 10^{-4} \text{ F}$	A1

PUBLISHED

Question	Answer	Marks
6(a)	product of (magnetic) flux density and area	M1
	where area is perpendicular to the (magnetic) field	A1
6(b)(i)	$N\Phi = BAN$	C1
	$= 400 \times 10^{-3} \times 0.12^2 \times 8$	C1
	$= 0.046 \text{ Wb}$	A1
6(b)(ii)	(line is a) straight line	B1
6(b)(iii)	(induced) e.m.f. = rate of change of flux linkage	C1
	e.m.f. $= N\Phi / t$	A1
	$= 0.046 / 0.60$	
6(c)	$= 0.077 \text{ V}$	
	(induced e.m.f. causes) current flow (in the coil)	B1
	either	
	current (in magnetic field) causes forces to act on the coil	B1
	(opposite sides of) coil forced inwards	B1
	or	
	current causes dissipation of energy in the resistance of the coil	(B1)
	temperature of the coil rises	(B1)

Question	Answer	Marks
7(a)	quantum of energy	M1
	of electromagnetic radiation	A1
7(b)(i)	photoelectric effect	B1
7(b)(ii)	- there is a frequency below which no electrons are emitted or threshold frequency = 5.4×10^{14} Hz - work function of the metal = 3.6×10^{-19} J (or 2.2 eV) E_{MAX} increases (linearly) with (increasing) frequency - gradient of the line is the Planck constant or gradient of the line is 6.7×10^{-34} J s <i>Any three bullet points, 1 mark each</i>	B3
7(c)(i)	different threshold frequency	B1
	(line has) same gradient but different intercept	B1
7(c)(ii)	photons have same energy	B1
	line unchanged	B1

PUBLISHED

Question	Answer	Marks
8(a)(i)	energy required to separate the nucleons (in the nucleus)	M1
	to infinity	A1
8(a)(ii)	curve starting close to the origin and forming a single peak	B1
	peak shown to left of centre, with steep line on LHS of peak and shallow line on RHS of peak	B1
8(b)(i)	fusion	B1
8(b)(ii)	both particles have low A values or both particles are at left-hand end of graph	B1
	He-3 has higher binding energy (per nucleon) than H-2	B1
8(c)	$\Delta m = [(2 \times 2.014102) - (3.016029 + 1.008665)] \text{ u}$ (= 0.00351 u)	C1
	$E = \Delta mc^2$	C1
	$= 0.00351 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ (= $5.24 \times 10^{-13} \text{ J}$)	C1
	1.00 mol of deuterium forms 0.500 mol of helium-3	C1
	total energy = $0.500 \times 6.02 \times 10^{23} \times 5.24 \times 10^{-13}$ = $1.58 \times 10^{11} \text{ J}$	A1

PUBLISHED

Question	Answer	Marks
9(a)(i)	electrons are accelerated (by an applied p.d.)	B1
	electrons hit target	B1
	X-rays produced when electrons decelerate	B1
9(a)(ii)	images of the multiple sections are combined to create a 3-D image	B1
9(b)(i)	$I = I_0 \exp(-\mu x)$	C1
	$= I_0 \exp(-0.89 \times 5.6)$	A1
	$= 0.0068 I_0$	
9(b)(ii)	$I = I_0 \exp(-2.4 \times 3.4) \times \exp(-0.89 \times 3.2)$	C1
	$= 1.7 \times 10^{-5} I_0$	A1
9(c)	comparison of intensities or values in (b) leading to conclusion consistent with these values	B1

Question	Answer	Marks
10(a)	wavelength of maximum intensity is inversely proportional to (thermodynamic) temperature	B1
10(b)(i)	$\lambda_{\text{MAX}} = 0.50 \mu\text{m}$ for A and $0.65 \mu\text{m}$ for B	C1
	$T = 5800 \times (0.50 / 0.65)$ $= 4500 \text{ K}$	A1
10(b)(ii)	(star B has) greater peak / average wavelength	B1
	(star B looks) redder	B1
10(c)(i)	apparent wavelength is greater or wavelength is greater than known value	B1
	(due to) movement of star away (from observer)	B1
10(c)(ii)	by examining the (lines in the) spectrum (of light from the star)	B1
	and comparing with known spectrum	B1



Cambridge International AS & A Level

PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

May/June 2022

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **10** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations

✓	Correct point Method of analysis marks in Question 1
✓ ₁₋₁₀	Additional detail marks in Question 1
X	Incorrect point
^	Omission
BOD	Benefit of the doubt
NBOD	No benefit of the doubt given
ECF	Error carried forward
P	Defining the problem marks in Question 1 Power of ten error in Question 2
M0	Methods of data collection marks in Question 1
SF	Incorrect number of significant figures

Question	Answer	Marks
1	Defining the problem	
	d is the independent variable and V is the dependent variable or vary d and measure V	1
	keep A or area (of overlap) of plates <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - circuit diagram with voltmeter connected in parallel with the capacitor - capacitor and voltmeter connected to the metal plates with no power supply in discharge part of the circuit - correct symbols for capacitor and voltmeter	1
	method to charge parallel plates, e.g. separate circuit diagram showing plates connected to a d.c. power supply or combined circuit with switches and d.c. power supply	1
	use calipers to measure d or use micrometer/calipers to measure thickness of spacers	1
	use rule(r) to measure lengths to determine A and $A = \text{length} \times \text{breadth}$	1
	Method of analysis	
	plot a graph of $\frac{1}{V}$ against d or equivalent (e.g. d against $\frac{1}{V}$) (Do not accept log graphs.)	1
	$K = \frac{\text{y-intercept} \times C}{\text{gradient} \times A} \text{ or } K = \frac{C}{\text{gradient} \times AW}$ (for d against $\frac{1}{V}$: $K = -\frac{\text{y-intercept} \times C}{A}$)	1

Question	Answer	Marks
1	$W = \frac{1}{\text{y-intercept}}$ (for d against $\frac{1}{V}$: $W = -\frac{\text{gradient}}{\text{y-intercept}}$ or $W = \frac{\text{gradient} \times C}{AK}$)	1
	Additional detail including safety considerations	6
	D1 use gloves <u>to prevent</u> electric shock or do not touch metal plates to avoid shocks	
	D2 keep the <u>initial</u> p.d. across plates or initial charge <u>constant</u>	
	D3 method to determine the value of C , e.g. description of an experiment to measure p.d. or current against time during discharge through a resistor	
	D4 method of operation of circuit(s) using switch(es)	
	D5 description of method to fully discharge capacitor, e.g. between experiments, short-circuit the capacitor or use of switch in parallel with capacitor	
	D6 repeat measurements of d <u>at different points across plates</u> and average	
	D7 repeat measurements of V for same d and average V	
	D8 bottom plate resting on insulating material or top plate supported by strings	
	D9 use high voltage power supply to increase charge on plates or use a very small value of capacitance to increase voltmeter reading	
	D10 relationship valid <u>if</u> a straight line is produced (not passing through the origin)	

Question	Answer		Marks														
2(a)	gradient = n y-intercept = $\lg SZ$		1														
2(b)	<table><tr><th>$\lg (M / 10^{30} \text{ kg})$</th><th>$\lg (L / 10^{28} \text{ W})$</th></tr><tr><td>0.68 or 0.681 ± 0.03 or 0.04</td><td>0.15 or 0.146</td></tr><tr><td>0.81 or 0.806 ± 0.02 or 0.03</td><td>0.49 or 0.491</td></tr><tr><td>1.08 or 1.079 ± 0.07 or 0.08</td><td>1.51 or 1.505</td></tr><tr><td>1.36 or 1.362 ± 0.04</td><td>2.54 or 2.544</td></tr><tr><td>1.63 or 1.633 ± 0.04</td><td>3.56 or 3.556</td></tr><tr><td>1.96 or 1.959 ± 0.02</td><td>4.82 or 4.820</td></tr></table>		$\lg (M / 10^{30} \text{ kg})$	$\lg (L / 10^{28} \text{ W})$	0.68 or 0.681 ± 0.03 or 0.04	0.15 or 0.146	0.81 or 0.806 ± 0.02 or 0.03	0.49 or 0.491	1.08 or 1.079 ± 0.07 or 0.08	1.51 or 1.505	1.36 or 1.362 ± 0.04	2.54 or 2.544	1.63 or 1.633 ± 0.04	3.56 or 3.556	1.96 or 1.959 ± 0.02	4.82 or 4.820	1
	$\lg (M / 10^{30} \text{ kg})$	$\lg (L / 10^{28} \text{ W})$															
	0.68 or 0.681 ± 0.03 or 0.04	0.15 or 0.146															
	0.81 or 0.806 ± 0.02 or 0.03	0.49 or 0.491															
	1.08 or 1.079 ± 0.07 or 0.08	1.51 or 1.505															
	1.36 or 1.362 ± 0.04	2.54 or 2.544															
	1.63 or 1.633 ± 0.04	3.56 or 3.556															
	1.96 or 1.959 ± 0.02	4.82 or 4.820															
Values of $\lg (M / 10^{30} \text{ kg})$ and $\lg (L / 10^{28} \text{ W})$ correct as shown above.																	
Absolute uncertainties in $\lg (M / 10^{30} \text{ kg})$ correct as shown above.		1															
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1															
	Error bars in $\lg (M / 10^{30} \text{ kg})$ plotted correctly. All error bars must be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1															
2(c)(ii)	Straight line of best fit drawn. Points must be balanced. Do not accept line from top point to bottom point. Line must pass between (0.92, 1.0) and (0.96, 1.0) and between (1.86, 4.5) and (1.90, 4.5)	1															
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1															

Question	Answer	Marks
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not allow methods using a false origin.	1

Question	Answer	Marks
2(d)	$n = \text{gradient} = \text{(c)(iii)}$ and n and Z both given to two or three significant figures.	1
	Value of Z determined using y -intercept. Correct method must be seen. $Z = \frac{10^{y\text{-intercept}} \times 10^{28}}{S} = \frac{10^{\text{(c)(iv)}} \times 10^{28}}{3.85 \times 10^{26}}$ or $Z = 10^{y\text{-intercept} - \lg S} \times 10^{28}$ or $Z = 10^{\text{(c)(iv)} - \lg 3.85 \times 10^{26}} \times 10^{28}$	1
	Absolute uncertainty in $n = \text{absolute uncertainty in gradient}$ and $\Delta Z = \frac{(10^{y\text{-intercept}} - 10^{\text{WAL } y\text{-intercept}}) \times 10^{28}}{S}$ Correct substitution of numbers must be seen.	1
2(e)	L determined from (d) or (c)(iii) and (c)(iv) with correct substitution <u>and</u> correct power of ten(s). Do not accept incorrect POT for n or Z . $L = 3.85 \times 10^{26} \times \text{(d)} \times 3.0^{\text{(c)(iii)}}$ or $\lg L = \text{(c)(iii)} \times \lg 3.0 + y\text{-intercept}$	1



Cambridge International AS & A Level

PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

May/June 2022

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **11** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations

✓	Correct point Method of analysis marks in Question 1
✓ ₁₋₁₀	Additional detail marks in Question 1
X	Incorrect point
^	Omission
BOD	Benefit of the doubt
NBOD	No benefit of the doubt given
ECF	Error carried forward
P	Defining the problem marks in Question 1 Power of ten error in Question 2
M0	Methods of data collection marks in Question 1
SF	Incorrect number of significant figures

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	L is the independent variable and I is the dependent variable or vary L and measure I	1
	keep E constant	1
	Methods of data collection	
	labelled diagram of workable experiment including: - circuit diagram with power supply connected to ends C - ammeter in series with power supply and conductors - correct symbol for ammeter and power supply	1
	circuit diagram with voltmeter correctly positioned to measure E across the power supply	1
	use a rule(r) to measure L and x	1
	use a micrometer/calipers to measure y	1

PUBLISHED

Question	Answer	Marks
1	Method of analysis	
	plot a graph of $\frac{1}{I}$ against L or equivalent (e.g. L against $\frac{1}{I}$) (Do not accept log graphs.)	1
	$P = \frac{\text{gradient} \times AE}{2}$ (for L against $\frac{1}{I}$: $P = \frac{AE}{2 \times \text{gradient}}$)	1
	$Q = \frac{y\text{-intercept} \times Ey^2}{x}$ (for L against $\frac{1}{I}$: $Q = -\frac{y\text{-intercept} \times 2Py^2}{Ax}$ or $Q = -\frac{y\text{-intercept} \times Ey^2}{\text{gradient} \times x}$)	1

Question	Answer	Marks
1	Additional detail including safety considerations	6
	D1 do not touch/use (heat resistant) gloves to avoid <u>hot</u> conductors/metal bar or use a protective resistor/small e.m.f. to reduce the <u>current</u> or switch off when not in use/when moving bar	
	D2 keep A and y <u>constant</u>	
	D3 keep x <u>constant</u>	
	D4 use of micrometer/calipers to measure <u>diameter</u> of conductor and $A = \pi d^2 / 4$.	
	D5 repeat measurements of diameter <u>along conductors/different (perpendicular) directions/different points</u> and average or repeat measurements of y in <u>different (perpendicular) directions/different points/along bar</u> and average	
	D6 method to ensure that L is the same for each conductor, e.g. check both lengths	
	D7 method to determine L e.g. measure to edge and add $y/2$ or method to determine x e.g. measure between the conductors and add diameter	
	D8 method to keep x constant <u>with reason</u> , e.g. adhesive/plasticine/blocks (one either side of each conductor) <u>to prevent cylindrical conductors from moving</u>	
	D9 method of ensuring good electrical contact, e.g. clean metal bar/cylindrical conductors or use of solder or crocodile clips to connect circuit to the conductors	
	D10 relationship valid <u>if</u> a straight line is produced (not passing through the origin)	

Question	Answer		Marks														
2(a)	gradient = a y-intercept = $\lg SK$		1														
2(b)	<table><tr><th>$\lg (T / \text{days})$</th><th>$\lg (L / 10^{30} W)$</th></tr><tr><td>1.34 or 1.342</td><td>0.46 or 0.462 ± 0.03</td></tr><tr><td>1.51 or 1.505</td><td>0.69 or 0.690 ± 0.02</td></tr><tr><td>1.62 or 1.623</td><td>0.84 or 0.839 ± 0.01</td></tr><tr><td>1.73 or 1.732</td><td>0.99 or 0.991 ± 0.01</td></tr><tr><td>1.89 or 1.892</td><td>1.20 or 1.204 ± 0.05 or 0.06</td></tr><tr><td>1.99 or 1.987</td><td>1.32 or 1.322 ± 0.04</td></tr></table> Values of $\lg (T / \text{days})$ and $\lg (L / 10^{30} W)$ correct as shown above. Absolute uncertainties in $\lg (L / 10^{30} W)$ correct as shown above.		$\lg (T / \text{days})$	$\lg (L / 10^{30} W)$	1.34 or 1.342	0.46 or 0.462 ± 0.03	1.51 or 1.505	0.69 or 0.690 ± 0.02	1.62 or 1.623	0.84 or 0.839 ± 0.01	1.73 or 1.732	0.99 or 0.991 ± 0.01	1.89 or 1.892	1.20 or 1.204 ± 0.05 or 0.06	1.99 or 1.987	1.32 or 1.322 ± 0.04	1
$\lg (T / \text{days})$	$\lg (L / 10^{30} W)$																
1.34 or 1.342	0.46 or 0.462 ± 0.03																
1.51 or 1.505	0.69 or 0.690 ± 0.02																
1.62 or 1.623	0.84 or 0.839 ± 0.01																
1.73 or 1.732	0.99 or 0.991 ± 0.01																
1.89 or 1.892	1.20 or 1.204 ± 0.05 or 0.06																
1.99 or 1.987	1.32 or 1.322 ± 0.04																
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.		1														
	Error bars in $\lg (L / 10^{30} W)$ plotted correctly. All error bars must be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.		1														
2(c)(ii)	Straight line of best fit drawn. Points must be balanced. Do not accept line from top point to bottom point. Line must pass between (1.43, 0.60) and (1.45, 0.60) and between (1.84, 1.15) and (1.86, 1.15).		1														
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.		1														

PUBLISHED

Question	Answer	Marks
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not allow methods using a false origin.	1

Question	Answer	Marks
2(d)	$a = \text{gradient} = \text{(c)(iii) and } a \text{ and } K \text{ both given to two or three significant figures.}$	1
	Value of K determined using y-intercept. Correct method must be seen. $K = \frac{10^{y\text{-intercept}} \times 10^{30}}{S} = \frac{10^{\text{(c)(iv)}} \times 10^{30}}{3.85 \times 10^{26}}$ or $K = 10^{y\text{-intercept} - \lg S} \times 10^{30}$ or $K = 10^{\text{(c)(iv)} - \lg 3.85 \times 10^{26}} \times 10^{30}$	1
	absolute uncertainty in $a = \text{absolute uncertainty in gradient}$ and $\Delta K = \frac{(10^{y\text{-intercept}} - 10^{\text{WAL } y\text{-intercept}}) \times 10^{30}}{S}$ Correct substitution of numbers must be seen.	1
2(e)	L determined from (d) or (c)(iii) and (c)(iv) with correct substitution <u>and</u> correct power of ten(s). Do not accept incorrect POT for a or K. $L = 3.85 \times 10^{26} \times \text{(d)} \times 5.0^{\text{(c)(iii)}}$ or $\lg L = \text{(c)(iii)} \times \lg 5.0 + y\text{-intercept}$	1



Cambridge International AS & A Level

PHYSICS

9702/53

Paper 5 Planning, Analysis and Evaluation

May/June 2022

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **10** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations

✓	Correct point Method of analysis marks in Question 1
✓ ₁₋₁₀	Additional detail marks in Question 1
X	Incorrect point
^	Omission
BOD	Benefit of the doubt
NBOD	No benefit of the doubt given
ECF	Error carried forward
P	Defining the problem marks in Question 1 Power of ten error in Question 2
M0	Methods of data collection marks in Question 1
SF	Incorrect number of significant figures

Question	Answer	Marks
1	Defining the problem	
	d is the independent variable and V is the dependent variable or vary d and measure V	1
	keep A or area (of overlap) of plates <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - circuit diagram with voltmeter connected in parallel with the capacitor - capacitor and voltmeter connected to the metal plates with no power supply in discharge part of the circuit - correct symbols for capacitor and voltmeter	1
	method to charge parallel plates, e.g. separate circuit diagram showing plates connected to a d.c. power supply or combined circuit with switches and d.c. power supply	1
	use calipers to measure d or use micrometer/calipers to measure thickness of spacers	1
	use rule(r) to measure lengths to determine A and $A = \text{length} \times \text{breadth}$	1
	Method of analysis	
	plot a graph of $\frac{1}{V}$ against d or equivalent (e.g. d against $\frac{1}{V}$) (Do not accept log graphs.)	1
	$K = \frac{\text{y-intercept} \times C}{\text{gradient} \times A} \text{ or } K = \frac{C}{\text{gradient} \times AW}$ (for d against $\frac{1}{V}$: $K = -\frac{\text{y-intercept} \times C}{A}$)	1

Question	Answer	Marks
1	$W = \frac{1}{\text{y-intercept}}$ (for d against $\frac{1}{V}$: $W = -\frac{\text{gradient}}{\text{y-intercept}}$ or $W = \frac{\text{gradient} \times C}{AK}$)	1
	Additional detail including safety considerations	6
	D1 use gloves <u>to prevent</u> electric shock or do not touch metal plates to avoid shocks	
	D2 keep the <u>initial</u> p.d. across plates or initial charge <u>constant</u>	
	D3 method to determine the value of C , e.g. description of an experiment to measure p.d. or current against time during discharge through a resistor	
	D4 method of operation of circuit(s) using switch(es)	
	D5 description of method to fully discharge capacitor, e.g. between experiments, short-circuit the capacitor or use of switch in parallel with capacitor	
	D6 repeat measurements of d <u>at different points across plates</u> and average	
	D7 repeat measurements of V for same d and average V	
	D8 bottom plate resting on insulating material or top plate supported by strings	
	D9 use high voltage power supply to increase charge on plates or use a very small value of capacitance to increase voltmeter reading	
	D10 relationship valid <u>if</u> a straight line is produced (not passing through the origin)	

Question	Answer		Marks														
2(a)	gradient = n y-intercept = $\lg SZ$		1														
2(b)	<table><tr><th>$\lg (M / 10^{30} \text{ kg})$</th><th>$\lg (L / 10^{28} \text{ W})$</th></tr><tr><td>0.68 or 0.681 ± 0.03 or 0.04</td><td>0.15 or 0.146</td></tr><tr><td>0.81 or 0.806 ± 0.02 or 0.03</td><td>0.49 or 0.491</td></tr><tr><td>1.08 or 1.079 ± 0.07 or 0.08</td><td>1.51 or 1.505</td></tr><tr><td>1.36 or 1.362 ± 0.04</td><td>2.54 or 2.544</td></tr><tr><td>1.63 or 1.633 ± 0.04</td><td>3.56 or 3.556</td></tr><tr><td>1.96 or 1.959 ± 0.02</td><td>4.82 or 4.820</td></tr></table>		$\lg (M / 10^{30} \text{ kg})$	$\lg (L / 10^{28} \text{ W})$	0.68 or 0.681 ± 0.03 or 0.04	0.15 or 0.146	0.81 or 0.806 ± 0.02 or 0.03	0.49 or 0.491	1.08 or 1.079 ± 0.07 or 0.08	1.51 or 1.505	1.36 or 1.362 ± 0.04	2.54 or 2.544	1.63 or 1.633 ± 0.04	3.56 or 3.556	1.96 or 1.959 ± 0.02	4.82 or 4.820	1
	$\lg (M / 10^{30} \text{ kg})$	$\lg (L / 10^{28} \text{ W})$															
	0.68 or 0.681 ± 0.03 or 0.04	0.15 or 0.146															
	0.81 or 0.806 ± 0.02 or 0.03	0.49 or 0.491															
	1.08 or 1.079 ± 0.07 or 0.08	1.51 or 1.505															
	1.36 or 1.362 ± 0.04	2.54 or 2.544															
	1.63 or 1.633 ± 0.04	3.56 or 3.556															
	1.96 or 1.959 ± 0.02	4.82 or 4.820															
Values of $\lg (M / 10^{30} \text{ kg})$ and $\lg (L / 10^{28} \text{ W})$ correct as shown above.																	
Absolute uncertainties in $\lg (M / 10^{30} \text{ kg})$ correct as shown above.		1															
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1															
	Error bars in $\lg (M / 10^{30} \text{ kg})$ plotted correctly. All error bars must be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1															
2(c)(ii)	Straight line of best fit drawn. Points must be balanced. Do not accept line from top point to bottom point. Line must pass between (0.92, 1.0) and (0.96, 1.0) and between (1.86, 4.5) and (1.90, 4.5)	1															
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1															

Question	Answer	Marks
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not allow methods using a false origin.	1

Question	Answer	Marks
2(d)	$n = \text{gradient} = \text{(c)(iii)}$ and n and Z both given to two or three significant figures.	1
	Value of Z determined using y -intercept. Correct method must be seen. $Z = \frac{10^{y\text{-intercept}} \times 10^{28}}{S} = \frac{10^{\text{(c)(iv)}} \times 10^{28}}{3.85 \times 10^{26}}$ or $Z = 10^{y\text{-intercept} - \lg S} \times 10^{28}$ or $Z = 10^{\text{(c)(iv)} - \lg 3.85 \times 10^{26}} \times 10^{28}$	1
	Absolute uncertainty in $n = \text{absolute uncertainty in gradient}$ and $\Delta Z = \frac{(10^{y\text{-intercept}} - 10^{\text{WAL } y\text{-intercept}}) \times 10^{28}}{S}$ Correct substitution of numbers must be seen.	1
2(e)	L determined from (d) or (c)(iii) and (c)(iv) with correct substitution <u>and</u> correct power of ten(s). Do not accept incorrect POT for n or Z . $L = 3.85 \times 10^{26} \times \text{(d)} \times 3.0^{\text{(c)(iii)}}$ or $\lg L = \text{(c)(iii)} \times \lg 3.0 + y\text{-intercept}$	1



Cambridge International AS & A Level

PHYSICS

9702/41

Paper 4 A Level Structured Questions

May/June 2023

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Question	Answer	Marks
1(a)(i)	force per unit mass	B1
1(a)(ii)	force per unit positive charge	B1
1(a)(iii)	similarity: • inversely proportional to distance (from point) • points of equal potential lie on concentric spheres • zero at infinite distance <i>Any point, 1 mark</i>	B1
	difference: • gravitational potential is (always) negative • electric potential can be positive or negative <i>Any point, 1 mark</i>	B1
1(b)(i)	$g = GM / r^2$	M1
	$E = Q / 4\pi\epsilon_0 r^2$	M1
	algebra showing the elimination of r leading to $M / Q = (1 / 4\pi G\epsilon_0) (g / E)$	A1
1(b)(ii)	$\alpha = 1 / (4\pi \times 6.67 \times 10^{-11} \times 8.85 \times 10^{-12}) = 1.35 \times 10^{20} \text{ (kg}^2 \text{ C}^{-2}\text{)}$ or $\alpha = (8.99 \times 10^9) / (6.67 \times 10^{-11}) = 1.35 \times 10^{20} \text{ (kg}^2 \text{ C}^{-2}\text{)}$	A1
1(c)(i)	$E = \alpha g Q / M$ $= (1.35 \times 10^{20} \times 9.81 \times 4.80 \times 10^5) / (5.98 \times 10^{24})$	C1
	$= 106 \text{ N C}^{-1}$ or 106 V m^{-1}	A1
1(c)(ii)	same (direction)	B1

PUBLISHED

Question	Answer	Marks
2(a)	horizontal force on sphere causes centripetal acceleration	B1
	weight of sphere is (now) equal to vertical component of tension or horizontal and vertical components (of force) (now) combine to give greater tension (in spring)	B1
	greater tension in spring so greater extension of spring	B1
2(b)(i)	$r = 10.8 \times \sin 27^\circ = 4.9 \text{ cm}$	A1
2(b)(ii)	$T \cos \theta = mg$ or $T \cos \theta = W$ and $W = mg$	C1
	$T \cos 27^\circ = 0.29 \times 9.81$ leading to $T = 3.2 \text{ N}$	A1
2(b)(iii)	$\Delta T = 3.2 - (0.29 \times 9.81)$	C1
	$k = \Delta T / \Delta x$ $= [3.2 - (0.29 \times 9.81)] / [10.8 - 8.5]$ $= 0.15 \text{ N cm}^{-1}$	A1
2(c)(i)	centripetal acceleration $= (T \sin \theta) / m$ $= (3.2 \times \sin 27^\circ) / 0.29$	C1
	$= 5.0 \text{ m s}^{-2}$	A1

Question	Answer	Marks
2(c)(ii)	$a = r\omega^2$ and $\omega = 2\pi / T$ or $a = v^2 / r$ and $v = 2\pi r / T$	C1
	$T = 2\pi \times \sqrt{(0.049 / 5.0)}$ $= 0.62 \text{ s}$	A1

Question	Answer	Marks
3(a)	no <u>net</u> thermal energy is transferred (between them)	B1
3(b)(i)	variation (of density with temperature) is linear or each temperature has a unique value of density	B1
3(b)(ii)	- variation (of density with temperature) is not linear - region where the density does not vary with temperature - different temperatures have the same density <i>Any two points, 1 mark each</i>	B2
3(c)(i)	boiling point = 80 °C	A1
3(c)(ii)	$Q = Pt$ and $t = 21 \text{ s}$ (thermal energy supplied = $810 \times 21 = 17000 \text{ J}$)	C1
	$c = Q / m\Delta\theta$	C1
	thermal energy absorbed by beaker = $42 \times 0.84 \times (80 - 25)$ (= 1940 J)	C1
	s.h.c. of liquid = $[(810 \times 21) - (42 \times 0.84 \times (80 - 25))] / [120 \times (80 - 25)]$ = $2.3 \text{ J g}^{-1} \text{ K}^{-1}$	A1
3(d)	sketch: straight diagonal line from 25 °C to 100 °C and then horizontal at 100 °C	B1
	straight diagonal line starting at 25 °C with gradient approximately half that of the original line	B1

Question	Answer	Marks															
4(a)	- particles are in (continuous) random motion - particles have negligible volume (compared with the gas) - negligible forces between particles (except during collisions) (all) collisions (perfectly) elastic - time of collision negligible (in comparison with time between collisions) <i>Any two points, 1 mark each</i>	B2															
4(b)(i)	(general starting equation) $pV = nRT$	C1															
	$T = (2pV / nR)$ where R is the (molar) gas constant	A1															
4(b)(ii)	sketch: straight vertical line XY from $(V, 2p)$ to (V, p)	B1															
	straight horizontal line YZ from (V, p) to $(2V, p)$	B1															
	curve with gradient increasing from Z to X from $(2V, p)$ to $(V, 2p)$	B1															
4(b)(iii)	XY work done on gas correct (= 0)	B1															
	ZX increase in internal energy correct (= 0)	B1															
	YZ work done on gas correct (= $-pV$)	B1															
	XY increase in internal energy such that the increase in internal energy column adds up to zero	B1															
	all three thermal energies transferred such that $\Delta U = q + w$ in each row (completely correct answer: <table><tr><td>change</td><td>ΔU</td><td>Q</td><td>w</td></tr><tr><td>X to Y</td><td>$-U$</td><td>$-U$</td><td>0</td></tr><tr><td>Y to Z</td><td>[$+U$]</td><td>$U + pV$</td><td>$-pV$</td></tr><tr><td>Z to X</td><td>0</td><td>$-W$</td><td>[$+W$]</td></tr></table>)	change	ΔU	Q	w	X to Y	$-U$	$-U$	0	Y to Z	[$+U$]	$U + pV$	$-pV$	Z to X	0	$-W$	[$+W$]
change	ΔU	Q	w														
X to Y	$-U$	$-U$	0														
Y to Z	[$+U$]	$U + pV$	$-pV$														
Z to X	0	$-W$	[$+W$]														

PUBLISHED

Question	Answer	Marks
5(a)(i)	correct circuit symbol for a diode shown correctly connected in series with the wires leading into and out of the dotted box	B1
5(a)(ii)	smoothing / V_{OUT} is smoothed	B1
5(b)(i)	frequency = $1 / 0.04$ = 25 Hz	A1
5(b)(ii)	$V = V_0 \exp(-t / RC)$ and $\tau = RC$ or $V = V_0 \exp(-t / \tau)$	C1
	$3.25 = 5.50 \exp(-0.020 / \tau)$ leading to $\tau = 0.038$ s	A1
5(b)(iii)	$\tau = RC$	C1
	capacitance = $0.038 / 14000$ = 2.7×10^{-6} F	A1
5(c)	V_{IN} has constant magnitude in both positive and negative directions	B1
	(so) V_{OUT} is (now) constant / V_{OUT} does not vary with time	B1

PUBLISHED

Question	Answer	Marks
6(a)	a region where a force acts on	M1
	a current-carrying conductor or a <u>moving</u> charge or a magnetic material / magnetic pole	A1
6(b)	concentric circles around the wire	B1
	spacing between circles increases with distance from wire	B1
	arrows showing direction of field is clockwise	B1
6(c)(i)	$F = BIL$	C1
	force per unit length = BI $= 2.6 \times 10^{-3} \times 5.0$ $= 0.013 \text{ N m}^{-1}$	A1
6(c)(ii)	to the right	B1
6(c)(iii)	force (per unit length) has the same magnitude due to Newton's 3rd law	B1
	$0.013 = 1.5 \times 10^{-3} \times I$ current = 8.7 A	A1

PUBLISHED

Question	Answer	Marks
7(a)	wavelength associated with a <u>moving</u> particle	B1
7(b)(i)	(electron) diffraction	B1
7(b)(ii)	beam spreads out indicating diffraction or light and dark regions indicate an interference pattern	B1
	electron beam is behaving as a wave	B1
7(c)(i)	central blob and concentric rings	B1
	rings closer together (than previously)	B1
7(c)(ii)	(greater p.d. so) electrons to have greater momentum	B1
	greater momentum so decrease in (de Broglie) wavelength	B1
	lower (de Broglie) wavelength (for same grating spacing in crystal) causes: smaller diffraction angle or smaller angle of intensity maxima (for each order) or decrease in fringe spacing in diffraction pattern	B1

PUBLISHED

Question	Answer	Marks
8(a)(i)	specific acoustic impedance = $1200 \times 1400 = 1.68 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$	A1
8(a)(ii)	density of air shown in table as 1.29	A1
	speed of sound in tissue shown in table as 1540	A1
8(b)(i)	intensity reflection coefficient = $(Z_1 - Z_2)^2 / (Z_1 + Z_2)^2$ $= (1680000 - 440)^2 / (1680000 + 440)^2$	C1
	$= 0.999$	A1
8(b)(ii)	intensity reflection coefficient = $(Z_1 - Z_2)^2 / (Z_1 + Z_2)^2$ $= (1680000 - 1680000)^2 / (1680000 + 1680000)^2$ $= 0$	A1
8(c)	without gel, (almost) all of the (incident) ultrasound is reflected (from skin)	B1
	with gel, (almost) all of the (incident) ultrasound is transmitted (into the body)	B1

PUBLISHED

Question	Answer	Marks
9(a)	time for activity (of sample) to halve	B1
9(b)	sketch: line with positive gradient starting at (0,0) and extending to $t = 80$ min	B1
	exponential curve, extending from $t = 0$ to $t = 80$ min, with gradient of steadily decreasing magnitude	B1
	line passing through (0,0), (20, $0.5N_0$) and (40, $0.75 N_0$)	B1
9(c)(i)	every (undecayed) nucleus has the same probability of decay	M1
	fewer (undecayed) nuclei remaining (with time), so fewer will decay (in a given time interval)	A1
9(c)(ii)	- sample emits in all directions but detector only captures emissions in one direction - some emissions are absorbed before reaching detector - some emissions are scattered within the sample - simultaneous arrival of multiple particles only registers once - some particles may reach detector but not cause ionisation <i>Any two points, 1 mark each</i>	B2
	measured count rate is less than the activity	B1

PUBLISHED

Question	Answer	Marks
10(a)	speed is (directly) proportional to distance	M1
	speed is speed of recession of galaxy from an observer, and distance is the distance of the galaxy from the observer	A1
10(b)	$F = L / (4\pi d^2)$	C1
	$= (3.8 \times 10^{31}) / [4\pi \times (1.8 \times 10^{24})^2]$	A1
	$= 9.3 \times 10^{-19} \text{ W m}^{-2}$	
10(c)(i)	galaxy is moving away (from the Earth)	B1
	wavelength (of light from the galaxy) increased by the Doppler effect / due to redshift	B1
10(c)(ii)	$\Delta\lambda / \lambda = v / c$ $v = [(492 - 486) \times 3.00 \times 10^8] / 486$ $(v = 3.7 \times 10^6 \text{ m s}^{-1})$	C1
	$H_0 = v / d$	C1
	$= (3.7 \times 10^6) / (1.8 \times 10^{24})$ $= 2.1 \times 10^{-18} \text{ s}^{-1}$	A1



Cambridge International AS & A Level

PHYSICS

9702/42

Paper 4 A Level Structured Questions

May/June 2023

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED**Abbreviations**

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Question	Answer	Marks
1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)	$GMm / R^2 = mR\omega^2$	M1
	$\omega = 2\pi / T$ <u>and</u> algebra leading to $4\pi^2 R^3 = GMT^2$	A1
	or	
	$GMm / R^2 = mv^2 / R$	(M1)
	$v = 2\pi R / T$ <u>and</u> algebra leading to $4\pi^2 R^3 = GMT^2$	(A1)
1(c)	$4\pi^2 \times R^3 = 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times (24 \times 60 \times 60)^2$ $(R = 4.22 \times 10^7 \text{ m})$	C1
	$h = R - (6.37 \times 10^6)$	C1
	$h = (4.22 \times 10^7) - (6.37 \times 10^6)$ $= 3.6 \times 10^7 \text{ m}$	A1
1(d)(i)	$\omega = 2\pi / T$	C1
	$= 2\pi / (24 \times 60 \times 60)$	A1
	$= 7.3 \times 10^{-5} \text{ rad s}^{-1}$	
1(d)(ii)	orbit is from east to west	B1
	orbit is not equatorial / orbit is polar	B1

Question	Answer	Marks
2(a)(i)	(gas that obeys) $pV \propto T$ (for all values of p, V and T)	M1
	where T is thermodynamic temperature	A1
2(a)(ii)	temperature = $-273.15\text{ }^{\circ}\text{C}$	A1
2(b)(i)	$pV = NkT$	C1
	$N = (1.37 \times 10^5 \times 0.640) / (1.38 \times 10^{-23} \times (227 + 273))$	C1
	$= 1.27 \times 10^{25}$	A1
2(b)(ii)	mass = $0.0424 / (1.27 \times 10^{25})$ $= 3.34 \times 10^{-27}\text{ kg}$	A1
2(b)(iii)	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$	C1
	$3.34 \times 10^{-27} \times v^2 = 3 \times 1.38 \times 10^{-23} \times 500$	C1
	$v = 2490\text{ m s}^{-1}$	A1
	or	
	$pV = \frac{1}{3}(Nm) \langle c^2 \rangle$ <u>and</u> Nm = mass of gas	(C1)
	$0.0424 \times v^2 = 3 \times 1.37 \times 10^5 \times 0.640$	(C1)
	$v = 2490\text{ m s}^{-1}$	(A1)
2(c)	sketch: line from (0, 0) to (500, v)	B1
	line with decreasing positive gradient throughout	B1

PUBLISHED

Question	Answer	Marks
3(a)	<u>change</u> in internal energy = work done + energy transfer by heating	C1
	<u>increase</u> in internal energy = work done <u>on</u> system + energy transferred <u>to</u> the system by heating	A1
3(b)(i)	AB change in internal energy: decrease	B1
	AB work done on gas: positive	B1
	BC change in internal energy: increase	B1
	BC work done on gas: zero	B1
3(b)(ii)	more work done by gas in CD than is done on gas in AB or (no work done on gas in BC and DA so) (overall) gas does work	B1
	(overall) change in internal energy is zero	B1
	(must be an overall) input of thermal energy	B1

PUBLISHED

Question	Answer	Marks
4(a)(i)	$\omega = 2\pi f$	C1
	$f = 9.7 / 2\pi$	A1
	$= 1.5 \text{ Hz}$	
4(a)(ii)	amplitude = $\sqrt{11.6} = 3.4 \text{ cm}$	A1
4(a)(iii)	$a_0 = \omega^2 x_0$	C1
	$= 9.7^2 \times 3.4 \times 10^{-2}$	A1
	$= 3.2 \text{ m s}^{-2}$	
4(b)	sketch: straight line through the origin with negative gradient	B1
	line with negative gradient passing through $(+3.4, -a_0)$ and $(-3.4, +a_0)$	B1
	line with ends at $x = \pm 3.4 \text{ cm}$ and $a = \pm a_0$	B1
4(c)	sum of potential energy and kinetic energy is constant	B1
	at maximum displacement, kinetic energy is zero or at maximum displacement, potential energy is maximum	B1
	at zero displacement, kinetic energy is maximum or at zero displacement, potential energy is minimum	B1

Question	Answer	Marks
5(a)(i)	$Q_A = CV$	A1
5(a)(ii)	$E_A = \frac{1}{2}CV^2$	A1
5(b)(i)	some of the charge transfers to (the plates of) capacitor B	B1
	transfer is because the p.d.s across the capacitors are not equal or transfer stops when the p.d.s across the capacitors become equal	B1
5(b)(ii)	$V_A = V_B$	M1
	charge on A + charge on B = CV	M1
	$CV_B + 3CV_B = CV$ leading to $V_B = V/4$	A1
	or	
	$C_T = 4C$	(M1)
	$Q_T = CV$	(M1)
	$V_B = CV/4C = V/4$	(A1)
5(b)(iii)	$\Delta E = \frac{1}{2}CV^2 - nCV^2$, where n is a multiple that is less than $\frac{1}{2}$ or total final energy = $\frac{1}{2} \times 4C \times (V/4)^2$ $= \frac{1}{8}CV^2$	C1
	$\Delta E = \frac{1}{2}CV^2 - \frac{1}{8}CV^2$ $= \frac{3}{8}CV^2$	A1

PUBLISHED

Question	Answer	Marks
6(a)(i)	product of (magnetic) flux density and area	M1
	area perpendicular to the (magnetic) field	A1
6(a)(ii)	flux = $B \times \pi r^2$ = $0.17 \times \pi \times 0.36^2$	C1
	= $6.9 \times 10^{-2} \text{ Wb}$	A1
6(b)	time for one revolution = $1 / 25 \text{ s}$	C1
	e.m.f. = rate of cutting flux or $\Delta \Phi / \Delta t$	C1
	= 0.069×25 = 1.7 V	A1
6(c)	current (in disc) is perpendicular to magnetic field or current causes force to act on disc	B1
	force opposes rotation of disc	B1
	left-hand rule indicates current is from rim to axle	B1

Question	Answer	Marks
7(a)(i)	full-wave (rectification)	B1
7(a)(ii)	lower left diode shown pointing left	B1
	lower right and upper left diodes shown pointing left	B1
7(a)(iii)	arrow indicating current direction in resistor to the right	B1
7(b)(i)	sketch: periodic line showing minimum $V_{OUT} = 0$ and maximum $V_{OUT} = +V_0$	B1
	line showing peak V_{OUT} at $t = 0, 0.5T, 1.0T, 1.5T$ and $2.0T$, with V_{OUT} going to zero half-way in between each peak	B1
	line showing correct modulated sine shape	B1
7(b)(ii)	sketch: sinusoidal curve with troughs sitting on the time axis	B1
	<u>peak</u> power at $t = 0, 0.5T, 1.0T, 1.5T$ and $2.0T$ and zero power half-way in between each peak	B1
7(b)(iii)	same power-time graph with or without rectification, so same V_{rms} or V^2 -time graph is same for both V_{OUT} and V_{IN} , so same V_{rms} or power does not depend on sign of V , so same V_{rms}	B1

PUBLISHED

Question	Answer	Marks
8(a)	transition (emits) (one) photon with energy equal to the difference in energy between the two levels	B1
	frequency of radiation corresponds to energy of photon	B1
8(b)(i)	line to the left of the pair in Fig. 8.2, labelled A	B1
	larger gap between line A and the nearest of the pair in Fig. 8.2 than between the lines in the pair	B1
8(b)(ii)	line to the left of both the pair in Fig. 8.2 and line A, labelled B	B1
	larger gap between line B and line A than between line A and the nearest one of the pair in Fig. 8.2	B1
8(c)	$E = hf$	C1
	$E_3 = E_1 + h(f_A + f_B)$	A1

PUBLISHED

Question	Answer	Marks
9(a)	difference between mass of nucleus and (total) mass of nucleons	M1
	when infinitely separated	A1
9(b)(i)	neutron	B1
9(b)(ii)	$E = \Delta m c^2$	C1
	$\Delta m = (0.030377 - 0.002388 - 0.009105)u$ (= 0.018884u)	C1
	energy release = $(0.030377 - 0.002388 - 0.009105) \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 = 2.8 \times 10^{-12} \text{ J}$	A1
9(c)(i)	number of atoms per unit time = $(1.4 \times 10^{28}) / (2.8 \times 10^{-12})$ (= $5.0 \times 10^{39} \text{ s}^{-1}$)	C1
	mass of one atom = $4 \times 1.66 \times 10^{-27}$ or $(4 \times 10^{-3}) / (6.02 \times 10^{23})$ (= $6.64 \times 10^{-27} \text{ kg}$)	C1
	mass per unit time = $6.64 \times 10^{-27} \times 5.0 \times 10^{39}$ = $3.3 \times 10^{13} \text{ kg s}^{-1}$	A1
9(c)(ii)	$L = 4\pi\sigma r^2 T^4$	C1
	$1.4 \times 10^{28} = 4\pi \times 5.67 \times 10^{-8} \times (2.3 \times 10^9)^2 \times T^4$	
	$T = 7800 \text{ K}$	A1

PUBLISHED

Question	Answer	Marks
10(a)(i)	electrons	B1
10(a)(ii)	electrons are decelerated / stopped on impact with the target	B1
	(kinetic) energy lost by electrons emitted as (X-ray) photons	B1
10(a)(iii)	$eV = hc / \lambda$	C1
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (1.60 \times 10^{-19} \times 5800)$	C1
	$= 2.14 \times 10^{-10} \text{ m}$	A1
10(b)	$I = I_0 \exp(-\mu x)$	C1
	$I_T / I_0 = \exp(-(1.4 \times 2.8))$ $= 0.020$	C1
	% absorbed $= (1.000 - 0.0198) \times 100$ $= 98\%$	A1



Cambridge International AS & A Level

PHYSICS

9702/43

Paper 4 A Level Structured Questions

May/June 2023

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Question	Answer	Marks
1(a)(i)	force per unit mass	B1
1(a)(ii)	force per unit positive charge	B1
1(a)(iii)	similarity: • inversely proportional to distance (from point) • points of equal potential lie on concentric spheres • zero at infinite distance <i>Any point, 1 mark</i>	B1
	difference: • gravitational potential is (always) negative • electric potential can be positive or negative <i>Any point, 1 mark</i>	B1
1(b)(i)	$g = GM / r^2$	M1
	$E = Q / 4\pi\epsilon_0 r^2$	M1
	algebra showing the elimination of r leading to $M / Q = (1 / 4\pi G\epsilon_0) (g / E)$	A1
1(b)(ii)	$\alpha = 1 / (4\pi \times 6.67 \times 10^{-11} \times 8.85 \times 10^{-12}) = 1.35 \times 10^{20} \text{ (kg}^2 \text{ C}^{-2}\text{)}$ or $\alpha = (8.99 \times 10^9) / (6.67 \times 10^{-11}) = 1.35 \times 10^{20} \text{ (kg}^2 \text{ C}^{-2}\text{)}$	A1
1(c)(i)	$E = \alpha g Q / M$ $= (1.35 \times 10^{20} \times 9.81 \times 4.80 \times 10^5) / (5.98 \times 10^{24})$	C1
	$= 106 \text{ N C}^{-1}$ or 106 V m^{-1}	A1
1(c)(ii)	same (direction)	B1

PUBLISHED

Question	Answer	Marks
2(a)	horizontal force on sphere causes centripetal acceleration	B1
	weight of sphere is (now) equal to vertical component of tension or horizontal and vertical components (of force) (now) combine to give greater tension (in spring)	B1
	greater tension in spring so greater extension of spring	B1
2(b)(i)	$r = 10.8 \times \sin 27^\circ = 4.9 \text{ cm}$	A1
2(b)(ii)	$T \cos \theta = mg$ or $T \cos \theta = W$ and $W = mg$	C1
	$T \cos 27^\circ = 0.29 \times 9.81$ leading to $T = 3.2 \text{ N}$	A1
2(b)(iii)	$\Delta T = 3.2 - (0.29 \times 9.81)$	C1
	$k = \Delta T / \Delta x$ $= [3.2 - (0.29 \times 9.81)] / [10.8 - 8.5]$ $= 0.15 \text{ N cm}^{-1}$	A1
2(c)(i)	centripetal acceleration $= (T \sin \theta) / m$ $= (3.2 \times \sin 27^\circ) / 0.29$	C1
	$= 5.0 \text{ m s}^{-2}$	A1

PUBLISHED

Question	Answer	Marks
2(c)(ii)	$a = r\omega^2$ and $\omega = 2\pi / T$ or $a = v^2 / r$ and $v = 2\pi r / T$	C1
	$T = 2\pi \times \sqrt{(0.049 / 5.0)}$ $= 0.62 \text{ s}$	A1

Question	Answer	Marks
3(a)	no <u>net</u> thermal energy is transferred (between them)	B1
3(b)(i)	variation (of density with temperature) is linear or each temperature has a unique value of density	B1
3(b)(ii)	- variation (of density with temperature) is not linear - region where the density does not vary with temperature - different temperatures have the same density <i>Any two points, 1 mark each</i>	B2
3(c)(i)	boiling point = 80 °C	A1
3(c)(ii)	$Q = Pt$ and $t = 21 \text{ s}$ (thermal energy supplied = $810 \times 21 = 17000 \text{ J}$)	C1
	$c = Q / m\Delta\theta$	C1
	thermal energy absorbed by beaker = $42 \times 0.84 \times (80 - 25)$ (= 1940 J)	C1
	s.h.c. of liquid = $[(810 \times 21) - (42 \times 0.84 \times (80 - 25))] / [120 \times (80 - 25)]$ = $2.3 \text{ J g}^{-1} \text{ K}^{-1}$	A1
3(d)	sketch: straight diagonal line from 25 °C to 100 °C and then horizontal at 100 °C	B1
	straight diagonal line starting at 25 °C with gradient approximately half that of the original line	B1

Question	Answer	Marks															
4(a)	- particles are in (continuous) random motion - particles have negligible volume (compared with the gas) - negligible forces between particles (except during collisions) (all) collisions (perfectly) elastic - time of collision negligible (in comparison with time between collisions) <i>Any two points, 1 mark each</i>	B2															
4(b)(i)	(general starting equation) $pV = nRT$	C1															
	$T = (2pV / nR)$ where R is the (molar) gas constant	A1															
4(b)(ii)	sketch: straight vertical line XY from $(V, 2p)$ to (V, p)	B1															
	straight horizontal line YZ from (V, p) to $(2V, p)$	B1															
	curve with gradient increasing from Z to X from $(2V, p)$ to $(V, 2p)$	B1															
4(b)(iii)	XY work done on gas correct (= 0)	B1															
	ZX increase in internal energy correct (= 0)	B1															
	YZ work done on gas correct (= $-pV$)	B1															
	XY increase in internal energy such that the increase in internal energy column adds up to zero	B1															
	all three thermal energies transferred such that $\Delta U = q + w$ in each row (completely correct answer: <table><tr><td>change</td><td>ΔU</td><td>Q</td><td>w</td></tr><tr><td>X to Y</td><td>$-U$</td><td>$-U$</td><td>0</td></tr><tr><td>Y to Z</td><td>[$+U$]</td><td>$U + pV$</td><td>$-pV$</td></tr><tr><td>Z to X</td><td>0</td><td>$-W$</td><td>[$+W$]</td></tr></table>)	change	ΔU	Q	w	X to Y	$-U$	$-U$	0	Y to Z	[$+U$]	$U + pV$	$-pV$	Z to X	0	$-W$	[$+W$]
change	ΔU	Q	w														
X to Y	$-U$	$-U$	0														
Y to Z	[$+U$]	$U + pV$	$-pV$														
Z to X	0	$-W$	[$+W$]														

PUBLISHED

Question	Answer	Marks
5(a)(i)	correct circuit symbol for a diode shown correctly connected in series with the wires leading into and out of the dotted box	B1
5(a)(ii)	smoothing / V_{OUT} is smoothed	B1
5(b)(i)	frequency = $1 / 0.04$ = 25 Hz	A1
5(b)(ii)	$V = V_0 \exp(-t / RC)$ and $\tau = RC$ or $V = V_0 \exp(-t / \tau)$	C1
	$3.25 = 5.50 \exp(-0.020 / \tau)$ leading to $\tau = 0.038$ s	A1
5(b)(iii)	$\tau = RC$	C1
	capacitance = $0.038 / 14000$ = 2.7×10^{-6} F	A1
5(c)	V_{IN} has constant magnitude in both positive and negative directions	B1
	(so) V_{OUT} is (now) constant / V_{OUT} does not vary with time	B1

PUBLISHED

Question	Answer	Marks
6(a)	a region where a force acts on	M1
	a current-carrying conductor or a <u>moving</u> charge or a magnetic material / magnetic pole	A1
6(b)	concentric circles around the wire	B1
	spacing between circles increases with distance from wire	B1
	arrows showing direction of field is clockwise	B1
6(c)(i)	$F = BIL$	C1
	force per unit length = BI $= 2.6 \times 10^{-3} \times 5.0$ $= 0.013 \text{ N m}^{-1}$	A1
6(c)(ii)	to the right	B1
6(c)(iii)	force (per unit length) has the same magnitude due to Newton's 3rd law	B1
	$0.013 = 1.5 \times 10^{-3} \times I$ current = 8.7 A	A1

PUBLISHED

Question	Answer	Marks
7(a)	wavelength associated with a <u>moving</u> particle	B1
7(b)(i)	(electron) diffraction	B1
7(b)(ii)	beam spreads out indicating diffraction or light and dark regions indicate an interference pattern	B1
	electron beam is behaving as a wave	B1
7(c)(i)	central blob and concentric rings	B1
	rings closer together (than previously)	B1
7(c)(ii)	(greater p.d. so) electrons to have greater momentum	B1
	greater momentum so decrease in (de Broglie) wavelength	B1
	lower (de Broglie) wavelength (for same grating spacing in crystal) causes: smaller diffraction angle or smaller angle of intensity maxima (for each order) or decrease in fringe spacing in diffraction pattern	B1

PUBLISHED

Question	Answer	Marks
8(a)(i)	specific acoustic impedance = $1200 \times 1400 = 1.68 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$	A1
8(a)(ii)	density of air shown in table as 1.29	A1
	speed of sound in tissue shown in table as 1540	A1
8(b)(i)	intensity reflection coefficient = $(Z_1 - Z_2)^2 / (Z_1 + Z_2)^2$ $= (1680000 - 440)^2 / (1680000 + 440)^2$	C1
	$= 0.999$	A1
8(b)(ii)	intensity reflection coefficient = $(Z_1 - Z_2)^2 / (Z_1 + Z_2)^2$ $= (1680000 - 1680000)^2 / (1680000 + 1680000)^2$ $= 0$	A1
8(c)	without gel, (almost) all of the (incident) ultrasound is reflected (from skin)	B1
	with gel, (almost) all of the (incident) ultrasound is transmitted (into the body)	B1

PUBLISHED

Question	Answer	Marks
9(a)	time for activity (of sample) to halve	B1
9(b)	sketch: line with positive gradient starting at (0,0) and extending to $t = 80$ min	B1
	exponential curve, extending from $t = 0$ to $t = 80$ min, with gradient of steadily decreasing magnitude	B1
	line passing through (0,0), (20, $0.5N_0$) and (40, $0.75 N_0$)	B1
9(c)(i)	every (undecayed) nucleus has the same probability of decay	M1
	fewer (undecayed) nuclei remaining (with time), so fewer will decay (in a given time interval)	A1
9(c)(ii)	- sample emits in all directions but detector only captures emissions in one direction - some emissions are absorbed before reaching detector - some emissions are scattered within the sample - simultaneous arrival of multiple particles only registers once - some particles may reach detector but not cause ionisation <i>Any two points, 1 mark each</i>	B2
	measured count rate is less than the activity	B1

PUBLISHED

Question	Answer	Marks
10(a)	speed is (directly) proportional to distance	M1
	speed is speed of recession of galaxy from an observer, and distance is the distance of the galaxy from the observer	A1
10(b)	$F = L / (4\pi d^2)$	C1
	$= (3.8 \times 10^{31}) / [4\pi \times (1.8 \times 10^{24})^2]$	A1
	$= 9.3 \times 10^{-19} \text{ W m}^{-2}$	
10(c)(i)	galaxy is moving away (from the Earth)	B1
	wavelength (of light from the galaxy) increased by the Doppler effect / due to redshift	B1
10(c)(ii)	$\Delta\lambda / \lambda = v / c$ $v = [(492 - 486) \times 3.00 \times 10^8] / 486$ ($v = 3.7 \times 10^6 \text{ m s}^{-1}$)	C1
	$H_0 = v / d$	C1
	$= (3.7 \times 10^6) / (1.8 \times 10^{24})$ $= 2.1 \times 10^{-18} \text{ s}^{-1}$	A1



Cambridge International AS & A Level

PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

May/June 2023

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

PUBLISHED**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1	Defining the problem	
	θ is the independent variable and t is the dependent variable or vary θ and measure t	1
	keep d <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - plane supported by stand / support - stand/support on bench/floor/horizontal surface - minimum of two labels from cube, cylinder, (inclined) plane, method of support, bench/floor/horizontal surface, pulley, string	1
	diagram showing method to measure d , e.g. <u>clamped</u> vertical rule near cylinder or drawn rule on plane used to measure d or distance d marked on plane and rule used to determine d	1
	use a protractor to measure θ or use a rule(r) to measure appropriate lengths for a trigonometric calculation	1
	use a timer/stop-watch to measure t or light gates connected to a timer to measure t	1

PUBLISHED

Question	Answer	Marks
1	Method of Analysis	
	plot a graph of $\frac{1}{t^2}$ against $\sin \theta$ or equivalent (e.g. $\sin \theta$ against $\frac{1}{t^2}$) Do not accept logarithms.	1
	$H = -\frac{2d(A+B) \times \text{gradient}}{A}$ (for $\sin \theta$ against $\frac{1}{t^2}$: $H = -\frac{2d(A+B)}{A \times \text{gradient}}$)	1
	$K = -\frac{2d(A+B) \times y\text{-intercept}}{A}$ (for $\sin \theta$ against $\frac{1}{t^2}$: $K = -H \times y\text{-intercept}$)	1

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	Safety precaution linked to <u>falling</u> cylinder, e.g. use of cushion/sand box to collect cylinder/prevent damage to cylinder/floor/bench/injury	
D2	protractor correctly positioned on diagram or appropriate trigonometric relationship for marked lengths	
D3	keep A and B <u>constant</u>	
D4	use a (top-pan) balance to measure A and B	
D5	correct positioning of light gates to determine t , e.g. two light gates either end of distance d , connected to a timer or correct <u>position</u> of video camera with <u>timer in frame</u> of the video to <u>determine</u> t	
D6	method to release cylinder/cube, e.g. cube held by set square, set square moved to release cube.	
D7	reasoned method to keep d constant <u>as θ changes</u> , e.g. (when measuring d by position of cylinder) adjust the length of the string or adjust the position of vertical marks or adjust the position of the vertical rule or initial position of the cube or (when measuring d by position of cube) use fixed marks on the plane or ruler placed on the plane with d measured between the marks	
D8	method to increase t for cylinder to fall, e.g. use large d to increase t	
D9	repeat measurements of t for the same θ and average t	
D10	relationship valid <u>if</u> a straight line is produced (not passing through the origin) Do not accept straight line passing through the origin.	

Question	Answer	Marks														
2(a)	$\text{gradient} = -R \ln \left(\frac{V}{V_0} \right)$	1														
2(b)	<table><tr><th>$C / 10^{-4} \text{ F}$</th><th>T / s</th></tr><tr><td>0.89 or 0.892</td><td>13.7 ± 0.8</td></tr><tr><td>1.3 or 1.32</td><td>20.4 ± 0.7</td></tr><tr><td>1.6 or 1.58</td><td>24.3 ± 0.6</td></tr><tr><td>1.0 or 1.03</td><td>16.1 ± 0.8</td></tr><tr><td>1.2 or 1.18</td><td>18.3 ± 0.7</td></tr><tr><td>2.1 or 2.08</td><td>31.5 ± 0.6</td></tr></table> Values of C and T correct as shown above.	$C / 10^{-4} \text{ F}$	T / s	0.89 or 0.892	13.7 ± 0.8	1.3 or 1.32	20.4 ± 0.7	1.6 or 1.58	24.3 ± 0.6	1.0 or 1.03	16.1 ± 0.8	1.2 or 1.18	18.3 ± 0.7	2.1 or 2.08	31.5 ± 0.6	1
$C / 10^{-4} \text{ F}$	T / s															
0.89 or 0.892	13.7 ± 0.8															
1.3 or 1.32	20.4 ± 0.7															
1.6 or 1.58	24.3 ± 0.6															
1.0 or 1.03	16.1 ± 0.8															
1.2 or 1.18	18.3 ± 0.7															
2.1 or 2.08	31.5 ± 0.6															
	Absolute uncertainties in T correct as shown above.	1														
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1														
	Error bars in T plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1														

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (1.42, 22.0) and (1.45, 22.0) and between (1.95, 30.0) and (2.00, 30.0)	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(d)	– 0.69 or – 0.693 and ± 0.06	1
2(e)(i)	R determined using gradient and R given to 2 or 3 significant figures. $R = - \frac{\text{gradient}}{\ln\left(\frac{V}{V_0}\right)} = \frac{\text{(c)(iii)}}{\text{(d)}}$	1
	R correctly determined using gradient and SI unit with correct power of ten for R (e.g. Ω).	1

Question	Answer	Marks
2(e)(ii)	Percentage uncertainty in R with method shown. $\text{percentage uncertainty} = \left(\frac{\Delta \left(\ln \left(\frac{V}{V_0} \right) \right)}{\ln \left(\frac{V}{V_0} \right)} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or Correct substitution for max/min methods.	1

Question	Answer	Marks
2(f)	C determined to a minimum of 2 significant figures from (c)(iii) or (d) and (e)(i) with correct substitution. $C = \frac{T}{\text{gradient}} = \frac{60.0}{\text{gradient}}$ or $C = \frac{-T}{-R \ln\left(\frac{V}{V_0}\right)} = \frac{-60.0}{\mathbf{(e)(i)} \times \mathbf{(d)}}$	1
	Absolute uncertainty in C determined with <u>correct method</u> used: Using gradient to determine C: $\Delta C = \left(\frac{\Delta \text{gradient}}{\text{gradient}} \right) \times C$ Allow using R to determine C: $\Delta C = \left(\frac{\Delta \left(\ln\left(\frac{V}{V_0}\right) \right)}{\ln\left(\frac{V}{V_0}\right)} + \frac{\mathbf{(e)(ii)}}{100} \right) \times C$	1



Cambridge International AS & A Level

PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

May/June 2023

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

PUBLISHED**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	R is the independent variable and E is the dependent variable or vary R and measure E	1
	keep V <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - coil P placed close to coil Q - separate workable circuit for coil Q (a.c.) voltmeter or oscilloscope connected across coil Q (Do not accept a power supply connected to coil Q.)	1
	a.c. power supply/signal generator connected to resistor and coil P in series	1
	workable circuit with power supply <u>and</u> (a.c.) voltmeter/oscilloscope in parallel with resistor <u>and</u> coil P or across terminals of power supply/signal generator	1
	method to determine R , e.g. measure current in R and p.d. across R and use $R = V_R/I$ or measure R using an ohmmeter	1

PUBLISHED

Question	Answer	Marks
1	Method of Analysis	
	plot a graph of $\frac{1}{E}$ against R or equivalent (e.g. R against $\frac{1}{E}$) Do not accept logarithms.	1
	$M = \frac{1}{2\pi fV \times \text{gradient}}$ (for R against $\frac{1}{E}$: $M = \frac{\text{gradient}}{2\pi fV}$)	1
	$k = 2\pi fVM \times y\text{-intercept}$ or $k = \frac{y\text{-intercept}}{\text{gradient}}$ (for R against $\frac{1}{E}$: $k = -y\text{-intercept}$)	1

Question	Answer	Marks
1	Additional detail including safety considerations	6
	D1 precaution linked to <u>hot coil</u> (<i>P</i>) / <u>hot resistor</u> , e.g. use of (heat-proof) gloves, wait until circuit cools down or precaution linked to <u>shocks</u> from <u>high voltages</u> e.g. use of (insulating) gloves or switch off supply before touching the circuit (to change <i>R</i>)	
	D2 keep the number of turns on (both) coils <u>constant</u>	
	D3 keep <i>f</i> <u>constant</u>	
	D4 keep distance between the coils <u>constant</u>	
	D5 method to keep distance between the coils constant, e.g. fix/clamp coils to bench	
	D6 method to measure <i>f</i> , e.g. read from signal generator or use of oscilloscope	
	D7 method to determine <i>f</i> from oscilloscope, e.g. period from oscilloscope $T = \text{time-base} \times \text{horizontal distance}$ and $f = 1/T$	
	D8 method to determine <i>V</i> or <i>E</i> from oscilloscope, e.g. $V = y\text{-gain} \times \text{vertical distance}$	
	D9 method to increase <i>E</i> e.g. use iron core/more turns on <u>coil Q</u> /high frequency/high p.d. (across <i>R</i> and coil <i>P</i>)	
	D10 relationship valid <u>if</u> a straight line is produced (not passing through the origin) Do not accept straight line passing through the origin.	

PUBLISHED

Question	Answer		Marks														
2(a)	gradient = $\frac{Yk}{p}$ y-intercept = $\frac{YkZ}{p}$		1														
2(b)	<table><tr><th>$V / 10^{-5} \text{ m}^3$</th><th>absolute uncertainty</th></tr><tr><td>3.81 or 3.815</td><td>± 0.03</td></tr><tr><td>3.99 or 3.986</td><td>± 0.03</td></tr><tr><td>4.16 or 4.163</td><td>± 0.04</td></tr><tr><td>4.33 or 4.335</td><td>± 0.04</td></tr><tr><td>4.48 or 4.481</td><td>± 0.04</td></tr><tr><td>4.65 or 4.652</td><td>± 0.04</td></tr></table>		$V / 10^{-5} \text{ m}^3$	absolute uncertainty	3.81 or 3.815	± 0.03	3.99 or 3.986	± 0.03	4.16 or 4.163	± 0.04	4.33 or 4.335	± 0.04	4.48 or 4.481	± 0.04	4.65 or 4.652	± 0.04	1
	$V / 10^{-5} \text{ m}^3$	absolute uncertainty															
	3.81 or 3.815	± 0.03															
3.99 or 3.986	± 0.03																
4.16 or 4.163	± 0.04																
4.33 or 4.335	± 0.04																
4.48 or 4.481	± 0.04																
4.65 or 4.652	± 0.04																
	Values of V correct as shown above.																
	Absolute uncertainties in V correct as shown above.		1														
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.		1														
	Error bars in V plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.		1														

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (27.5, 3.90) and (29.5, 3.90) and between (82.0, 4.60) and (84.0, 4.60).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and y into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

PUBLISHED

Question	Answer	Marks
2(d)(i)	Y determined using gradient and Y and Z given to 2 or 3 significant figures. $Y = \frac{p \times \text{gradient}}{k} = 7.3188 \times 10^{27} \times \text{gradient}$	1
	Z determined using y-intercept and <u>Y and</u> Z given with SI units. $Z = \frac{p \times y\text{-intercept}}{Yk} \text{ or } Z = \frac{y\text{-intercept}}{\text{gradient}}$ Units: Y: no unit Z: °C	1
2(d)(ii)	Percentage uncertainty in Y with method shown. $\text{percentage uncertainty} = \left(\frac{\Delta p}{p} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or Correct substitution for max/min methods.	1

PUBLISHED

Question	Answer	Marks
2(e)	θ determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with <u>correct substitution</u> and <u>correct powers of ten</u>. $V = \frac{\pi \times (0.0279)^2 \times 0.0600}{4} = 3.67 \times 10^{-5}$ and $\theta = \frac{pV}{Yk} - Z \text{ or } \theta = \frac{V}{\text{gradient}} - Z \text{ or } \theta = \frac{V - y\text{-intercept}}{\text{gradient}}$ or using h directly: $\theta = \frac{p\pi d^2 h}{4Yk} - Z \text{ or } \theta = \frac{\pi d^2 h}{4 \times \text{gradient}} - Z \text{ or } \theta = \frac{\frac{\pi d^2 h}{4} - y\text{-intercept}}{\text{gradient}}$	1



Cambridge International AS & A Level

PHYSICS

9702/53

Paper 5 Planning, Analysis and Evaluation

May/June 2023

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

PUBLISHED**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1	Defining the problem	
	θ is the independent variable and t is the dependent variable or vary θ and measure t	1
	keep d <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - plane supported by stand / support - stand/support on bench/floor/horizontal surface - minimum of two labels from cube, cylinder, (inclined) plane, method of support, bench/floor/horizontal surface, pulley, string	1
	diagram showing method to measure d , e.g. <u>clamped</u> vertical rule near cylinder or drawn rule on plane used to measure d or distance d marked on plane and rule used to determine d	1
	use a protractor to measure θ or use a rule(r) to measure appropriate lengths for a trigonometric calculation	1
	use a timer/stop-watch to measure t or light gates connected to a timer to measure t	1

PUBLISHED

Question	Answer	Marks
1	Method of Analysis	
	plot a graph of $\frac{1}{t^2}$ against $\sin \theta$ or equivalent (e.g. $\sin \theta$ against $\frac{1}{t^2}$) Do not accept logarithms.	1
	$H = -\frac{2d(A+B) \times \text{gradient}}{A}$ (for $\sin \theta$ against $\frac{1}{t^2}$: $H = -\frac{2d(A+B)}{A \times \text{gradient}}$)	1
	$K = -\frac{2d(A+B) \times y\text{-intercept}}{A}$ (for $\sin \theta$ against $\frac{1}{t^2}$: $K = -H \times y\text{-intercept}$)	1

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	Safety precaution linked to <u>falling</u> cylinder, e.g. use of cushion/sand box to collect cylinder/prevent damage to cylinder/floor/bench/injury	
D2	protractor correctly positioned on diagram or appropriate trigonometric relationship for marked lengths	
D3	keep A and B <u>constant</u>	
D4	use a (top-pan) balance to measure A and B	
D5	correct positioning of light gates to determine t , e.g. two light gates either end of distance d , connected to a timer or correct <u>position</u> of video camera with <u>timer in frame</u> of the video to <u>determine</u> t	
D6	method to release cylinder/cube, e.g. cube held by set square, set square moved to release cube.	
D7	reasoned method to keep d constant <u>as θ changes</u> , e.g. (when measuring d by position of cylinder) adjust the length of the string or adjust the position of vertical marks or adjust the position of the vertical rule or initial position of the cube or (when measuring d by position of cube) use fixed marks on the plane or ruler placed on the plane with d measured between the marks	
D8	method to increase t for cylinder to fall, e.g. use large d to increase t	
D9	repeat measurements of t for the same θ and average t	
D10	relationship valid <u>if</u> a straight line is produced (not passing through the origin) Do not accept straight line passing through the origin.	

Question	Answer	Marks														
2(a)	$\text{gradient} = -R \ln \left(\frac{V}{V_0} \right)$	1														
2(b)	<table><tr><th>$C / 10^{-4} \text{ F}$</th><th>T / s</th></tr><tr><td>0.89 or 0.892</td><td>13.7 ± 0.8</td></tr><tr><td>1.3 or 1.32</td><td>20.4 ± 0.7</td></tr><tr><td>1.6 or 1.58</td><td>24.3 ± 0.6</td></tr><tr><td>1.0 or 1.03</td><td>16.1 ± 0.8</td></tr><tr><td>1.2 or 1.18</td><td>18.3 ± 0.7</td></tr><tr><td>2.1 or 2.08</td><td>31.5 ± 0.6</td></tr></table> Values of C and T correct as shown above.	$C / 10^{-4} \text{ F}$	T / s	0.89 or 0.892	13.7 ± 0.8	1.3 or 1.32	20.4 ± 0.7	1.6 or 1.58	24.3 ± 0.6	1.0 or 1.03	16.1 ± 0.8	1.2 or 1.18	18.3 ± 0.7	2.1 or 2.08	31.5 ± 0.6	1
$C / 10^{-4} \text{ F}$	T / s															
0.89 or 0.892	13.7 ± 0.8															
1.3 or 1.32	20.4 ± 0.7															
1.6 or 1.58	24.3 ± 0.6															
1.0 or 1.03	16.1 ± 0.8															
1.2 or 1.18	18.3 ± 0.7															
2.1 or 2.08	31.5 ± 0.6															
	Absolute uncertainties in T correct as shown above.	1														
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1														
	Error bars in T plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1														

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (1.42, 22.0) and (1.45, 22.0) and between (1.95, 30.0) and (2.00, 30.0)	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(d)	– 0.69 or – 0.693 and ± 0.06	1
2(e)(i)	R determined using gradient and R given to 2 or 3 significant figures. $R = -\frac{\text{gradient}}{\ln\left(\frac{V}{V_0}\right)} = \frac{\text{(c)(iii)}}{\text{(d)}}$	1
	R correctly determined using gradient and SI unit with correct power of ten for R (e.g. Ω).	1

Question	Answer	Marks
2(e)(ii)	Percentage uncertainty in R with method shown. $\text{percentage uncertainty} = \left(\frac{\Delta \left(\ln \left(\frac{V}{V_0} \right) \right)}{\ln \left(\frac{V}{V_0} \right)} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or Correct substitution for max/min methods.	1

Question	Answer	Marks
2(f)	C determined to a minimum of 2 significant figures from (c)(iii) or (d) and (e)(i) with correct substitution. $C = \frac{T}{\text{gradient}} = \frac{60.0}{\text{gradient}}$ or $C = \frac{-T}{-R \ln\left(\frac{V}{V_0}\right)} = \frac{-60.0}{\mathbf{(e)(i)} \times \mathbf{(d)}}$	1
	Absolute uncertainty in C determined with <u>correct method</u> used: Using gradient to determine C: $\Delta C = \left(\frac{\Delta \text{gradient}}{\text{gradient}} \right) \times C$ Allow using R to determine C: $\Delta C = \left(\frac{\Delta \left(\ln\left(\frac{V}{V_0}\right) \right)}{\ln\left(\frac{V}{V_0}\right)} + \frac{\mathbf{(e)(ii)}}{100} \right) \times C$	1

Cambridge International AS & A Level

PHYSICS**9702/41**

Paper 4 A Level Structured Questions

May/June 2024

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **15** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Question	Answer	Marks
1(a)	work done per unit mass	B1
	work done moving mass from infinity (to the point)	B1
1(b)(i)	potential is zero at infinity	B1
	work is done by (two) masses in moving them closer together or work is done on (two) masses in moving them apart	B1
1(b)(ii)	magnitude of potential shown as 4ϕ	B1
	potential negative and shown as a multiple of $-\phi$ [potential = -4ϕ if fully correct]	B1
1(b)(iii)	field strength at X: $\phi/4R$	A1
	field strength at Y: $4\phi/R$	A1
	potential energy at X: $-M\phi$	A1
	potential energy at Y: $-8M\phi$	A1

PUBLISHED

Question	Answer	Marks
2(a)(i)	0 K	B1
2(a)(ii)	(measurement) depends on properties of the liquid	B1
2(b)(i)	- resistivity varies with temperature - variation with temperature is linear - unique value of resistivity for each (different value of) temperature <i>Any two points, 1 mark each</i>	B2
2(b)(ii)	thermometer has high heat capacity/specific heat capacity or energy transfer needed for thermometer to reach correct temperature or thermometer takes time to reach the correct temperature	B1
2(b)(iii)	thermocouple	B1
2(c)	(variation is) inverse or (variation is) non-linear	B1

Question	Answer	Marks
3(a)(i)	gas for which $pV \propto T$	M1
	where T is thermodynamic temperature	A1
3(a)(ii)	no intermolecular forces	B1
	(so) potential energy is zero	B1
3(b)(i)	$pV = NkT$	C1
	$N = (2.0 \times 10^5 \times 0.26) / (1.38 \times 10^{-23} \times 290)$	A1
	$= 1.3 \times 10^{25}$	
3(b)(ii)	$E_K = (3/2) kT$	C1
	$E_K = (3/2) \times 1.38 \times 10^{-23} \times 290$	A1
	$= 6.0 \times 10^{-21} \text{ J}$	
3(b)(iii)	internal energy = total KE + PE of molecules or PE = 0 so internal energy = total KE of molecules	B1
	internal energy = $1.3 \times 10^{25} \times 6.0 \times 10^{-21}$	A1
	$= 7.8 \times 10^4 \text{ J}$	
3(c)	straight line with positive gradient	B1
	line passing through the origin	B1

PUBLISHED

Question	Answer	Marks
4(a)	oscillation (of object) at maximum amplitude	B1
	when driving frequency = natural frequency (of system)	B1
4(b)(i)	light damping	B1
4(b)(ii)	<u>oscillations</u> (of ball) lose energy	B1
	(due to) resistive forces (acting on ball)	B1
4(b)(iii)	frequency = $1 / 0.25$ = 4.0 Hz	A1
4(c)	curve showing a maximum amplitude at a single non-zero frequency	B1
	single maximum amplitude shown at 4.0 Hz	B1

Question	Answer	Marks
5(a)	force per unit charge	B1
	force on positive charge	B1
5(b)(i)	four straight vertical parallel lines, approximately evenly spaced	B1
	arrows downwards	B1
5(b)(ii)	$E = V / d$	C1
	$E = 430 / 0.067$ $= 6.4 \times 10^3 \text{ N C}^{-1}$	A1
5(b)(iii)	smooth curve within plates and straight lines outside plates	B1
	direction of deflection shown as upwards	B1
5(c)(i)	into the page	B1
5(c)(ii)	forces are in opposite directions	B1
	(undeviated) when (magnitudes of) forces are equal	B1
5(c)(iii)	$Eq = Bqv$	C1
	$B = E / v = (6.4 \times 10^3) / (2.6 \times 10^7)$ $= 2.5 \times 10^{-4} \text{ T}$	A1

Question	Answer	Marks
6(a)	- p.d. across capacitor proportional to charge on capacitor - p.d. across capacitor = p.d. across resistor - current in resistor proportional to p.d. across resistor - current in resistor = rate of decrease of charge on capacitor <i>Any two points, 1 mark each</i>	B2
	charge proportional to current so rate of decrease of current decreases as current decreases (therefore exponential shape)	B1
6(b)(i)	$R = V / I$ $= 12 / (0.13 \times 10^{-3})$	C1
	$= 9.2 \times 10^4 \Omega$	A1
6(b)(ii)	correct read-off of at least one pair of values for I and t	C1
	attempted read-off of t when $I = 0.048$ mA or substitution of a correct pair of values of I and t into $I = 0.13 \exp(-t / \tau)$	C1
	$\tau = 4.3$ s	A1
6(c)	$\tau = RC$	C1
	$C = \tau / R = 4.3 / (9.2 \times 10^4)$ $= 4.7 \times 10^{-5}$ F	A1

PUBLISHED

Question	Answer	Marks
7(a)	rectification (of the input voltage)	M1
	full-wave	A1
7(b)(i)	$P = V^2 / R$ or maximum $V = 9.0 \text{ V}$	C1
	$P_{\text{MAX}} = 9.0^2 / 370 = 0.22 \text{ W}$	A1
7(b)(ii)	sinusoidal shape with minima sitting on the time axis	B1
	correct frequency and phase, with minima at 0, 0.02, 0.04, 0.06 and 0.08 s and maxima at 0.01, 0.03, 0.05 and 0.07 s	B1
	all maxima shown at 0.22 W	B1
7(b)(iii)	mean power = peak power / 2 = $0.22 / 2$ = 0.11 W	A1
7(c)	power–time graph is identical	B1
	(so) mean powers are equal	B1

Question	Answer	Marks
8(a)	packet / quantum of <u>energy</u>	M1
	of electromagnetic radiation	A1
8(b)(i)	electron(s)	B1
8(b)(ii)	X labelled – and Y labelled +	B1
8(c)(i)	0.032 MeV	A1
8(c)(ii)	momentum = E / c	C1
	momentum = $(0.032 \times 1.60 \times 10^{-13}) / (3.00 \times 10^8)$ $= 1.7 \times 10^{-23} \text{ N s}$	A1
8(c)(iii)	$E = hf$ and $\lambda = c / f$	C1
	$\lambda = hc / E$ $= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (0.032 \times 1.60 \times 10^{-13})$	C1
	$\lambda = 3.9 \times 10^{-11} \text{ m}$	A1
8(d)	discussion of bone and soft tissue	B1
	discussion of different attenuation (coefficients) or discussion differences in penetration / transmission / absorption	B1
	<u>transmitted intensities</u> (by bone and tissue) are very different (leading to good contrast images)	B1

PUBLISHED

Question	Answer	Marks
9(a)	time for activity (of sample) to halve	B1
9(b)(i)	activity (of X at time t)	B1
9(b)(ii)	- Y is a stable isotope - total number of nuclei is constant - half-life (of X) is 13.6 s - decay constant (of X) is 0.051 s^{-1} - amount (of X) at $t = 0$ is 0.066 mol - activity (of X) at $t = 0$ is $2.0 \times 10^{21} \text{ Bq}$ <i>Any three points, 1 mark each</i>	B3
9(c)	mass of 1 nucleus = $(7.3 \times 10^{-4}) / (4.0 \times 10^{22})$	C1
	nucleon number = mass of nucleus / (1.66×10^{-27})	C1
	$= (7.3 \times 10^{-4}) / (4.0 \times 10^{22} \times 1.66 \times 10^{-27})$ $= 11 \text{ and given as an integer}$	A1

PUBLISHED

Question	Answer	Marks
10(a)(i)	<u>total</u> power	B1
	power radiated (by the star)	B1
10(a)(ii)	standard candle has known luminosity	B1
	radiant flux intensity measured by observer	B1
	(distance calculated using) $F = L / 4\pi d^2$	B1
10(b)(i)	luminosity $= 4\pi\sigma r^2 T^4$ $= 4\pi \times 5.67 \times 10^{-8} \times (6.96 \times 10^8)^2 \times 5780^4$	C1
	$= 3.85 \times 10^{26} \text{ W}$	A1
10(b)(ii)	$\lambda_{\text{MAX}} T = \text{constant}$	C1
	temperature $= (5780 \times 501) / 624$ $= 4640 \text{ K}$	A1

Cambridge International AS & A Level

PHYSICS**9702/42**

Paper 4 A Level Structured Questions

May/June 2024

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **16** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	angle (subtended at the centre of a circle) when arc (length) = radius	B1
1(b)(i)	arrow, labelled V, pointing in NE direction	B1
1(b)(ii)	arrow, labelled A, pointing in NW direction	B1
1(c)(i)	$v = r\omega$	C1
	$\omega = 0.68 / (0.093 - 0.012)$ $= 8.4 \text{ rad s}^{-1}$	A1
1(c)(ii)	$a = v^2 / r$ or $a = r\omega^2$	C1
	$a = 0.68^2 / (0.093 - 0.012)$ or $(0.093 - 0.012) \times 8.4^2$ $= 5.7 \text{ m s}^{-2}$	A1
1(d)	angular speed: same for both pieces	B1
	linear speed: less for second piece than first piece	B1
	acceleration: less for second piece than first piece	B1

Question	Answer	Marks
2(a)	(if in thermal contact) no <u>net transfer</u> of (thermal) energy (between them)	B1
2(b)(i)	$pV = nRT$	C1
	$T = (1.20 \times 10^5 \times 0.0260) / (0.740 \times 8.31)$ (= 507 K)	M1
	temperature = 507 – 273 = 234 °C	A1
2(b)(ii)	thermal equilibrium <u>so</u> temperatures (of X and Y) are equal	B1
	$pV = NkT$	C1
	$N = (2.90 \times 10^5 \times 0.0430) / (1.38 \times 10^{-23} \times 507)$ = 1.78×10^{24}	A1
2(b)(iii)	(molecular) kinetic energy is proportional to temperature or kinetic energy (of molecules) is same in both cylinders - kinetic energy proportional to mass $\times$ mean-square speed or temperature proportional to mass $\times$ mean-square speed or r.m.s. speed proportional to $\sqrt{(\text{temperature} / \text{mass})}$ - mean-square speed inversely proportional to mass or r.m.s. speed inversely proportional to $\sqrt{(\text{mass})}$ <i>Any two bulleted points, 1 mark each</i>	B2
	r.m.s. speed (of molecules) in X is half r.m.s. speed (of molecules) in Y	B1

PUBLISHED

Question	Answer	Marks
3(a)	sum of potential energy and kinetic energy	B1
	(total) energy of random motion of particles	B1
3(b)(i)	no change in separation so no change in (molecular) potential energy	B1
	temperature increases so kinetic energy (of molecules) increases	B1
	kinetic energy increases and potential energy unchanged, so internal energy increases	B1
3(b)(ii)	temperature constant so no change in (molecular) kinetic energy	B1
	separation increases so potential energy (of molecules) increases	B1
	potential energy increases and kinetic energy unchanged, so internal energy increases	B1

Question	Answer	Marks
4(a)	straight line through the origin shows that a is proportional to x	B1
	negative gradient shows that a and x are (always) in opposite <u>directions</u>	B1
4(b)(i)	$a = -\omega^2 x$ $\omega = \sqrt{(2A / 3Y)}$	A1
4(b)(ii)	$v_0 = \omega x_0$	C1
	$= 3Y \times \sqrt{(2A / 3Y)}$ $= \sqrt{(6AY)}$	A1
4(b)(iii)	$E = \frac{1}{2} m \omega^2 x_0^2$	C1
	$= \frac{1}{2} m \times (2A / 3Y) \times (3Y)^2$ $= 3mAY$	A1
4(c)	$\omega = 2\pi / T$ $(= 2\pi / 0.75)$	C1
	$x = 1.8 \sin (8.4 t)$	A1

Question	Answer	Marks
5(a)	work done per unit charge	B1
	work (done) moving positive charge from infinity (to the point)	B1
5(b)	<i>Any three points from:</i> <i>Up to 2 points from:</i> • radius of sphere X is 0.30 m • radius of sphere Y is 0.10 m • radius of X is treble the radius of Y <i>Up to 2 points from:</i> • charge on X is positive • charge on Y is positive • spheres X and Y carry charges of the same sign <i>Up to 1 point from:</i> • (magnitudes of) charges on the spheres are equal • charges on the spheres have the same magnitude	B3
5(c)	proton remains at rest (in the position of release)	M1
	potential <u>energy</u> of proton is (already) at its minimum or (electric) forces (from spheres) on proton are equal and opposite or no resultant (electric) force on proton or resultant electric field strength (at proton) is zero	A1

Question	Answer	Marks
6(a)	equal charge on both capacitors	B1
	$V_X + V_Y = V$	M1
	$(Q / C_X) + (Q / C_Y) = (Q / C_T)$ <u>leading to</u> $(1 / C_X) + (1 / C_Y) = (1 / C_T)$ or $(V_X / Q) + (V_Y / Q) = (V / Q)$ <u>leading to</u> $(1 / C_X) + (1 / C_Y) = (1 / C_T)$	A1
6(b)(i)	$E = \frac{1}{2}CV^2$	C1
	$V = \sqrt{[(2 \times 2.5 \times 10^{-3}) / (200 \times 10^{-6})]} = 5.0 \text{ V}$	A1
6(b)(ii)	total capacitance = 600 μF	C1
	$E = \frac{1}{2} \times 600 \times 10^{-6} \times 5.0^2$ $(= 7.5 \times 10^{-3} \text{ J})$	C1
	$= 7.5 \text{ mJ}$	A1
6(b)(iii)	line with positive gradient starting at (0, 2.5)	B1
	straight line passing through (400, 7.5)	B1

Question	Answer	Marks
7(a)	(induced) e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)(i)	$\Phi = BA$	C1
	$= 7.2 \times 10^{-3} \times 3.2 \times 10^{-4}$	A1
	$= 2.3 \times 10^{-6} \text{ Wb}$	
7(b)(ii)	tangent drawn at steepest point on Fig. 7.2	C1
	evidence of multiplication by 340	C1
	maximum rate of change of flux = 0.82 Wb s^{-1}	A1
7(b)(iii)	$V_0 = 0.82 \text{ V}$ or V_0 given as identical numerical answer to the answer in (b)(ii)	A1
7(b)(iv)	sinusoidal curve of period 2.0 ms from $t = 0$ to $t = 6.0 \text{ ms}$	B1
	all peaks at $+V_0$ and all troughs at $-V_0$	B1
	line showing $V = 0$ at (and only at) $t = 0, 1.0, 2.0, 3.0, 4.0, 5.0$ and 6.0 ms	B1
7(b)(v)	$A = 0.82 \text{ V}$ or A has same numerical value as answer in (b)(iii) , with unit V	A1
	$B = 2\pi / (2.0 \times 10^{-3})$	C1
	$= 3100 \text{ rad s}^{-1}$	A1

Question	Answer	Marks
8(a)(i)	movement of star causes change in (observed) frequency or movement of star causes redshift	B1
	observed frequency is lower (than emitted frequency)	B1
8(a)(ii)	all three lines shown to left of corresponding printed lines	B1
	distance between drawn line and corresponding printed line approximately the same for all three lines	B1
8(b)(i)	$E = hf$ and $\lambda = c / f$	C1
	$E = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (488 \times 10^{-9})$ $= 4.08 \times 10^{-19} \text{ J}$	A1
8(b)(ii)	photon energy = $(4.08 \times 10^{-19}) / (1.60 \times 10^{-19})$ $= 2.55 \text{ eV}$	C1
	energy level = $-3.40 + 2.55$ $= -0.85 \text{ eV}$	A1
8(b)(iii)	$\Delta\lambda = \lambda \times (v / c)$ $= (488 \times 6.2 \times 10^6) / (3.00 \times 10^8)$ (= 10 nm)	C1
	observed wavelength = $488 + \Delta\lambda = 488 + 10$ $= 498 \text{ nm}$	A1

PUBLISHED

Question	Answer	Marks
8(c)	$v = H_0 d$	C1
	$d = (6.2 \times 10^6) / (2.3 \times 10^{-18})$ $= 2.7 \times 10^{24} \text{ m}$	A1

Question	Answer	Marks
9(a)	energy required to separate (all) the nucleons (in the nucleus)	M1
	to infinity	A1
9(b)(i)	$\Delta m = \{[(84 \times 1.007276) + (128 \times 1.008665)] - 211.942749\} \text{ (u)}$ $(= 1.778 \text{ u})$	C1
	$= 1.778 \times 1.66 \times 10^{-27} \text{ (kg)}$	C1
	$= 2.95 \times 10^{-27} \text{ kg}$	A1
9(b)(ii)	$E = (\Delta)mc^2$	C1
	binding energy $= 2.95 \times 10^{-27} \times (3.00 \times 10^8)^2$ $= 2.66 \times 10^{-10} \text{ J}$	A1
9(b)(iii)	binding energy per nucleon $= (2.66 \times 10^{-10}) / 212$ $= 1.25 \times 10^{-12} \text{ J}$	A1
9(c)(i)	line rising to a single peak that is to the left of the '9' in the Fig. 9.1 label and then continually decreasing	B1
	steep positive gradient on the left of the peak and shallow negative gradient on the right	B1
9(c)(ii)	X shown on the line at a value of A that is to the right of the left-hand edge of the 'A' in the axis label, and to the left of '2' in the 250 label	B1
9(c)(iii)	nucleus formed (as a result of the decay) has a lower nucleon number	B1
	(nucleus formed has a) greater binding energy per nucleon	B1

PUBLISHED

Question	Answer	Marks
10(a)	time gives information about depth (of boundary)	B1
	intensity gives information about <u>nature</u> of <u>boundary</u>	B1
10(b)(i)	product of density and speed	M1
	speed of ultrasound in medium (and density of medium)	A1
10(b)(ii)	$Z_{\text{water}} = 1000 \times 1420 (= 1.42 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1})$ and $Z_{\text{glass}} = 2500 \times 4560 (= 11.4 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1})$	C1
	intensity reflection coefficient = $(11.4 - 1.42)^2 / (11.4 + 1.42)^2$	C1
	= 0.61	A1

Cambridge International AS & A Level

PHYSICS**9702/43**

Paper 4 A Level Structured Questions

May/June 2024

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **15** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	work done per unit mass	B1
	work done moving mass from infinity (to the point)	B1
1(b)(i)	potential is zero at infinity	B1
	work is done by (two) masses in moving them closer together or work is done on (two) masses in moving them apart	B1
1(b)(ii)	magnitude of potential shown as 4ϕ	B1
	potential negative and shown as a multiple of $-\phi$ [potential = -4ϕ if fully correct]	B1
1(b)(iii)	field strength at X: $\phi/4R$	A1
	field strength at Y: $4\phi/R$	A1
	potential energy at X: $-M\phi$	A1
	potential energy at Y: $-8M\phi$	A1

PUBLISHED

Question	Answer	Marks
2(a)(i)	0 K	B1
2(a)(ii)	(measurement) depends on properties of the liquid	B1
2(b)(i)	- resistivity varies with temperature - variation with temperature is linear - unique value of resistivity for each (different value of) temperature <i>Any two points, 1 mark each</i>	B2
2(b)(ii)	thermometer has high heat capacity/specific heat capacity or energy transfer needed for thermometer to reach correct temperature or thermometer takes time to reach the correct temperature	B1
2(b)(iii)	thermocouple	B1
2(c)	(variation is) inverse or (variation is) non-linear	B1

Question	Answer	Marks
3(a)(i)	gas for which $pV \propto T$	M1
	where T is thermodynamic temperature	A1
3(a)(ii)	no intermolecular forces	B1
	(so) potential energy is zero	B1
3(b)(i)	$pV = NkT$	C1
	$N = (2.0 \times 10^5 \times 0.26) / (1.38 \times 10^{-23} \times 290)$	A1
	$= 1.3 \times 10^{25}$	
3(b)(ii)	$E_K = (3/2) kT$	C1
	$E_K = (3/2) \times 1.38 \times 10^{-23} \times 290$	A1
	$= 6.0 \times 10^{-21} \text{ J}$	
3(b)(iii)	internal energy = total KE + PE of molecules or PE = 0 so internal energy = total KE of molecules	B1
	internal energy = $1.3 \times 10^{25} \times 6.0 \times 10^{-21}$	A1
	$= 7.8 \times 10^4 \text{ J}$	
3(c)	straight line with positive gradient	B1
	line passing through the origin	B1

PUBLISHED

Question	Answer	Marks
4(a)	oscillation (of object) at maximum amplitude	B1
	when driving frequency = natural frequency (of system)	B1
4(b)(i)	light damping	B1
4(b)(ii)	<u>oscillations</u> (of ball) lose energy	B1
	(due to) resistive forces (acting on ball)	B1
4(b)(iii)	frequency = $1 / 0.25$ = 4.0 Hz	A1
4(c)	curve showing a maximum amplitude at a single non-zero frequency	B1
	single maximum amplitude shown at 4.0 Hz	B1

PUBLISHED

Question	Answer	Marks
5(a)	force per unit charge	B1
	force on positive charge	B1
5(b)(i)	four straight vertical parallel lines, approximately evenly spaced	B1
	arrows downwards	B1
5(b)(ii)	$E = V / d$	C1
	$E = 430 / 0.067$ $= 6.4 \times 10^3 \text{ N C}^{-1}$	A1
5(b)(iii)	smooth curve within plates and straight lines outside plates	B1
	direction of deflection shown as upwards	B1
5(c)(i)	into the page	B1
5(c)(ii)	forces are in opposite directions	B1
	(undeviated) when (magnitudes of) forces are equal	B1
5(c)(iii)	$Eq = Bqv$	C1
	$B = E / v = (6.4 \times 10^3) / (2.6 \times 10^7)$ $= 2.5 \times 10^{-4} \text{ T}$	A1

Question	Answer	Marks
6(a)	- p.d. across capacitor proportional to charge on capacitor - p.d. across capacitor = p.d. across resistor - current in resistor proportional to p.d. across resistor - current in resistor = rate of decrease of charge on capacitor <i>Any two points, 1 mark each</i>	B2
	charge proportional to current so rate of decrease of current decreases as current decreases (therefore exponential shape)	B1
6(b)(i)	$R = V / I$ $= 12 / (0.13 \times 10^{-3})$	C1
	$= 9.2 \times 10^4 \Omega$	A1
6(b)(ii)	correct read-off of at least one pair of values for I and t	C1
	attempted read-off of t when $I = 0.048$ mA or substitution of a correct pair of values of I and t into $I = 0.13 \exp(-t / \tau)$	C1
	$\tau = 4.3$ s	A1
6(c)	$\tau = RC$	C1
	$C = \tau / R = 4.3 / (9.2 \times 10^4)$ $= 4.7 \times 10^{-5}$ F	A1

PUBLISHED

Question	Answer	Marks
7(a)	rectification (of the input voltage)	M1
	full-wave	A1
7(b)(i)	$P = V^2 / R$ or maximum $V = 9.0 \text{ V}$	C1
	$P_{\text{MAX}} = 9.0^2 / 370 = 0.22 \text{ W}$	A1
7(b)(ii)	sinusoidal shape with minima sitting on the time axis	B1
	correct frequency and phase, with minima at 0, 0.02, 0.04, 0.06 and 0.08 s and maxima at 0.01, 0.03, 0.05 and 0.07 s	B1
	all maxima shown at 0.22 W	B1
7(b)(iii)	mean power = peak power / 2 = $0.22 / 2$ = 0.11 W	A1
7(c)	power–time graph is identical	B1
	(so) mean powers are equal	B1

Question	Answer	Marks
8(a)	packet / quantum of <u>energy</u>	M1
	of electromagnetic radiation	A1
8(b)(i)	electron(s)	B1
8(b)(ii)	X labelled – and Y labelled +	B1
8(c)(i)	0.032 MeV	A1
8(c)(ii)	momentum = E / c	C1
	momentum = $(0.032 \times 1.60 \times 10^{-13}) / (3.00 \times 10^8)$ $= 1.7 \times 10^{-23} \text{ N s}$	A1
8(c)(iii)	$E = hf$ and $\lambda = c / f$	C1
	$\lambda = hc / E$ $= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (0.032 \times 1.60 \times 10^{-13})$	C1
	$\lambda = 3.9 \times 10^{-11} \text{ m}$	A1
8(d)	discussion of bone and soft tissue	B1
	discussion of different attenuation (coefficients) or discussion differences in penetration / transmission / absorption	B1
	<u>transmitted intensities</u> (by bone and tissue) are very different (leading to good contrast images)	B1

PUBLISHED

Question	Answer	Marks
9(a)	time for activity (of sample) to halve	B1
9(b)(i)	activity (of X at time t)	B1
9(b)(ii)	- Y is a stable isotope - total number of nuclei is constant - half-life (of X) is 13.6 s - decay constant (of X) is 0.051 s^{-1} - amount (of X) at $t = 0$ is 0.066 mol - activity (of X) at $t = 0$ is $2.0 \times 10^{21} \text{ Bq}$ <i>Any three points, 1 mark each</i>	B3
9(c)	mass of 1 nucleus = $(7.3 \times 10^{-4}) / (4.0 \times 10^{22})$	C1
	nucleon number = mass of nucleus / (1.66×10^{-27})	C1
	$= (7.3 \times 10^{-4}) / (4.0 \times 10^{22} \times 1.66 \times 10^{-27})$ $= 11 \text{ and given as an integer}$	A1

PUBLISHED

Question	Answer	Marks
10(a)(i)	<u>total</u> power	B1
	power radiated (by the star)	B1
10(a)(ii)	standard candle has known luminosity	B1
	radiant flux intensity measured by observer	B1
	(distance calculated using) $F = L / 4\pi d^2$	B1
10(b)(i)	luminosity $= 4\pi\sigma r^2 T^4$ $= 4\pi \times 5.67 \times 10^{-8} \times (6.96 \times 10^8)^2 \times 5780^4$	C1
	$= 3.85 \times 10^{26} \text{ W}$	A1
10(b)(ii)	$\lambda_{\text{MAX}} T = \text{constant}$	C1
	temperature $= (5780 \times 501) / 624$ $= 4640 \text{ K}$	A1

Cambridge International AS & A Level

PHYSICS**9702/51**

Paper 5 Planning, Analysis and Evaluation

May/June 2024

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1	Defining the problem	
	t is the independent variable and T_C is the dependent variable or vary t and measure T_C	1
	keep T_R <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • solid cylinder cooling • insulation surrounding all of the cylinder • thermometer touching cylinder inside insulation • insulation and thermometer labelled	1
	method to heat the cylinder uniformly, e.g. place in oven/immerse in hot water or diagram showing cylinder in oven or hot water	1
	method to determine time t , e.g. stopwatch or temperature sensor connected to a data logger	1
	method to measure L e.g. use a ruler/calipers/micrometer and method to measure d e.g. use calipers/micrometer	1
	Method of Analysis	
	plot a graph of $\ln (T_C - T_R)$ against t or equivalent	1
	$U = -\frac{mc \times \text{gradient}}{A}$	1
	$Z = e^{\text{y-intercept}}$	1

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	precaution to <u>prevent burns</u> or use of <u>hot cylinder</u> / <u>oven</u> / <u>hot water</u> e.g. use of gloves, use of tongs	
D2	keep thickness of the insulating material constant (for each T_c)	
D3	method to measure m , e.g. use a (top-pan) balance	
D4	for water bath/oven methods, wait for initial temperature of the cylinder to become <u>uniform</u> or <u>constant throughout the cylinder</u>	
D5	(surface) $A = \pi dL + \frac{\pi d^2}{2}$ or $\pi dL + 2\left(\frac{\pi d^2}{4}\right)$	
D6	repeat measurements of d <u>along the length</u> of the cylinder / in <u>different directions</u> and determine the average value of d	
D7	description of how c is determined from a separate experiment by heating the cylinder using electrical heater and $c = \frac{\Delta E}{m\Delta\theta}$	
D8	method of determining energy supplied to electrical heater to determine c , e.g. use of joulemeter for ΔE or electrical method using ammeter and voltmeter to determine IVt	
D9	use several temperature sensors and determine the average T_c	
D10	relationship valid <u>if</u> a straight line is produced (with y-intercept = $\ln Z$) Do not accept line passing through the origin.	

Question	Answer		Marks														
2(a)	gradient = $-\frac{1}{kf_s}$ y-intercept = $\frac{1}{f_s}$		1														
2(b)	<table><tr><th>v/ms^{-1}</th><th>$\frac{1}{f}/10^{-3}\text{Hz}^{-1}$</th></tr><tr><td>$3.5 \pm 0.4$</td><td>1.118 or 1.1183</td></tr><tr><td>6.3 ± 0.4</td><td>1.110 or 1.1096</td></tr><tr><td>8.7 ± 0.5</td><td>1.101 or 1.1013</td></tr><tr><td>11.4 ± 0.5</td><td>1.092 or 1.0919</td></tr><tr><td>13.9 ± 0.6</td><td>1.083 or 1.0827</td></tr><tr><td>16.2 ± 0.6</td><td>1.074 or 1.0739</td></tr></table>		v/ms^{-1}	$\frac{1}{f}/10^{-3}\text{Hz}^{-1}$	3.5 ± 0.4	1.118 or 1.1183	6.3 ± 0.4	1.110 or 1.1096	8.7 ± 0.5	1.101 or 1.1013	11.4 ± 0.5	1.092 or 1.0919	13.9 ± 0.6	1.083 or 1.0827	16.2 ± 0.6	1.074 or 1.0739	1
	v/ms^{-1}	$\frac{1}{f}/10^{-3}\text{Hz}^{-1}$															
	3.5 ± 0.4	1.118 or 1.1183															
6.3 ± 0.4	1.110 or 1.1096																
8.7 ± 0.5	1.101 or 1.1013																
11.4 ± 0.5	1.092 or 1.0919																
13.9 ± 0.6	1.083 or 1.0827																
16.2 ± 0.6	1.074 or 1.0739																
	Values of v and $\frac{1}{f}$ correct as shown above.																
	Uncertainties in v correct as shown above.		1														
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.		1														
	Error bars in v plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.		1														

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Line must pass between (14.5, 1.080) and (14.9, 1.080) and between (4.5, 1.115) and (4.8, 1.115).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Gradient must be negative. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and y into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

Question	Answer	Marks
2(d)(i)	f_s determined using y-intercept and f_s given to 2, 3 or 4 significant figures and k given to 2 or 3 significant figures. $f_s = \frac{1}{y\text{-intercept}}$	1
	k determined using gradient with method shown and f_s <u>and</u> k given with SI units with appropriate powers of ten. $k = -\frac{y\text{-intercept}}{\text{gradient}} \text{ or } k = -\frac{1}{\text{gradient} \times f_s}$ Units of f_s : Hz Units of k : m s^{-1}	1
2(d)(ii)	Percentage uncertainty in k with method shown. $\text{percentage uncertainty} = \left(\frac{\Delta y\text{-intercept}}{y\text{-intercept}} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or correct substitution for max/min methods.	1
2(e)	v determined (non-zero) to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution. $v = \frac{\frac{1}{f} - y\text{-intercept}}{\text{gradient}}$ or $v = k - \frac{k f_s}{f}$	1

Cambridge International AS & A Level

PHYSICS**9702/52**

Paper 5 Planning, Analysis and Evaluation

May/June 2024**MARK SCHEME**Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **10** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	t is the independent variable and s is the dependent variable or vary t and measure s	1
	keep k <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • spring connected to magnet • vertical rule parallel to spring to determine s • rule held in position by a stand • stand resting on the bench • rule labelled and at least one other label from stand, clamp, card, (magnetic) sheet, (cylindrical) magnet, spring	1
	$s = (\text{new}) \text{ length/position of spring} - \text{original length/position of spring}$	1
	use a micrometer to measure t	1
	measure B using a (calibrated) Hall probe and rotate probe until <u>maximum</u> value or measure B using Hall probe first in one direction, then in the opposite direction and average	1

PUBLISHED

Question	Answer	Marks
1	Method of Analysis	
	plot a graph of s against $\frac{1}{t}$ or equivalent (allow $\lg s$ against $\lg t$)	1
	relationship valid <u>if</u> a straight line that passes through the origin is produced (for $\lg s$ against $\lg t$: relationship valid <u>if</u> a straight line with gradient -1)	1
	$Z = \frac{k}{ALB} \times \text{gradient}$ (for $\lg s$ against $\lg t$: $Z = \frac{k \times 10^{\text{y-intercept}}}{ALB}$)	1

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	precaution related to <u>spring</u> and/or <u>magnet hitting eyes</u> , e.g. use of goggles/use of safety screen around experiment	
D2	keep A , L and B <u>constant</u>	
D3	use a rule to measure L	
D4	micrometer/calipers to measure <u>diameter</u> d of the magnet and $A = \pi d^2 / 4$	
D5	description of method to determine k , e.g. add mass to spring and $k = mg / \text{extension}$ or use newton meter to measure force applied to spring and $k = \text{force} / \text{extension}$ or take several readings of force and extension, plot a force–extension graph and $k = \text{gradient}$	
D6	(magnetic) sheet clamped to bench	
D7	use pointer(s)/marker(s) on the spring to read off values from the rule	
D8	method to use video recorder and <u>replay</u> to determine maximum length of the spring or increase s or force <u>gradually/slowly</u> until magnet (just) leaves the card	
D9	repeat measurements of t in <u>different positions</u> on the card and average t or repeat measurements of s for each value of t and average s	
D10	method to check that the spring has not exceeded the elastic limit	
D11	use of non-magnetic stand or named non-magnetic material for stand, e.g. wood	

Question	Answer		Marks														
2(a)	gradient = n y-intercept = $\lg \frac{2}{C}$		1														
2(b)	<table><tr><th>$\lg (L / \text{cm})$</th><th>$\lg (T / 10^{-5} \text{ s})$</th></tr><tr><td>1.73 or 1.732</td><td>1.38 or 1.380 ± 0.02</td></tr><tr><td>1.85 or 1.845</td><td>1.51 or 1.505 ± 0.01</td></tr><tr><td>1.93 or 1.934</td><td>1.59 or 1.591 ± 0.01</td></tr><tr><td>2.033 or 2.0334</td><td>1.69 or 1.690 ± 0.02</td></tr><tr><td>2.146 or 2.1461</td><td>1.81 or 1.806 ± 0.01</td></tr><tr><td>2.223 or 2.2227</td><td>1.87 or 1.869 ± 0.01</td></tr></table>		$\lg (L / \text{cm})$	$\lg (T / 10^{-5} \text{ s})$	1.73 or 1.732	1.38 or 1.380 ± 0.02	1.85 or 1.845	1.51 or 1.505 ± 0.01	1.93 or 1.934	1.59 or 1.591 ± 0.01	2.033 or 2.0334	1.69 or 1.690 ± 0.02	2.146 or 2.1461	1.81 or 1.806 ± 0.01	2.223 or 2.2227	1.87 or 1.869 ± 0.01	1
	$\lg (L / \text{cm})$	$\lg (T / 10^{-5} \text{ s})$															
	1.73 or 1.732	1.38 or 1.380 ± 0.02															
	1.85 or 1.845	1.51 or 1.505 ± 0.01															
1.93 or 1.934	1.59 or 1.591 ± 0.01																
2.033 or 2.0334	1.69 or 1.690 ± 0.02																
2.146 or 2.1461	1.81 or 1.806 ± 0.01																
2.223 or 2.2227	1.87 or 1.869 ± 0.01																
	Values of $\lg (L / \text{cm})$ and $\lg (T / 10^{-5} \text{ s})$ correct as shown above.																
	Uncertainties in $\lg (T / 10^{-5} \text{ s})$ correct as shown above.		1														
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.		1														
	Error bars in $\lg T$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.		1														

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Line must pass between (1.780, 1.45) and (1.800, 1.45) and between (2.085, 1.75) and (2.100, 1.75)	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and x into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

Question	Answer	Marks
2(d)	Value of n determined using gradient ($n = \text{gradient}$) and C given to 2 or 3 significant figures.	1
	Value of C determined using y -intercept with method shown. $C = \frac{2}{10^{y\text{-intercept}}}$	1
	Absolute uncertainties in n and C . uncertainty in $n = \text{uncertainty in gradient}$ and $\Delta C = C - \frac{2}{10^{\text{worst } y\text{-intercept}}} \text{ or } \Delta C = \frac{\frac{2}{10^{\text{min worst } y\text{-intercept}}} - \frac{2}{10^{\text{max worst } y\text{-intercept}}}}{2}$ Clear method must be shown with ΔC correctly evaluated.	1
2(e)	Value of L determined (non-zero) to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d) with correct substitution and correct power of ten. Units of T and either C or y -intercept must be consistent. $\log L = \frac{\log T - \log \frac{2}{C}}{n} = \frac{\log 10 - y\text{-intercept}}{n}$ $L = 10^{\frac{\log 10 - y\text{-intercept}}{n}}$ or $L = \sqrt[n]{\frac{TC}{2}}$	1

Cambridge International AS & A Level

PHYSICS**9702/53**

Paper 5 Planning, Analysis and Evaluation

May/June 2024**MARK SCHEME**Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1	Defining the problem	
	t is the independent variable and T_C is the dependent variable or vary t and measure T_C	1
	keep T_R <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • solid cylinder cooling • insulation surrounding all of the cylinder • thermometer touching cylinder inside insulation • insulation and thermometer labelled	1
	method to heat the cylinder uniformly, e.g. place in oven/immerse in hot water or diagram showing cylinder in oven or hot water	1
	method to determine time t , e.g. stopwatch or temperature sensor connected to a data logger	1
	method to measure L e.g. use a ruler/calipers/micrometer and method to measure d e.g. use calipers/micrometer	1
	Method of Analysis	
	plot a graph of $\ln (T_C - T_R)$ against t or equivalent	1
	$U = -\frac{mc \times \text{gradient}}{A}$	1
	$Z = e^{\text{y-intercept}}$	1

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	precaution to <u>prevent burns</u> or use of <u>hot cylinder</u> / <u>oven</u> / <u>hot water</u> e.g. use of gloves, use of tongs	
D2	keep thickness of the insulating material constant (for each T_c)	
D3	method to measure m , e.g. use a (top-pan) balance	
D4	for water bath/oven methods, wait for initial temperature of the cylinder to become <u>uniform</u> or <u>constant throughout the cylinder</u>	
D5	(surface) $A = \pi dL + \frac{\pi d^2}{2}$ or $\pi dL + 2\left(\frac{\pi d^2}{4}\right)$	
D6	repeat measurements of d <u>along the length</u> of the cylinder / in <u>different directions</u> and determine the average value of d	
D7	description of how c is determined from a separate experiment by heating the cylinder using electrical heater and $c = \frac{\Delta E}{m\Delta\theta}$	
D8	method of determining energy supplied to electrical heater to determine c , e.g. use of joulemeter for ΔE or electrical method using ammeter and voltmeter to determine IVt	
D9	use several temperature sensors and determine the average T_c	
D10	relationship valid <u>if</u> a straight line is produced (with y-intercept = $\ln Z$) Do not accept line passing through the origin.	

PUBLISHED

Question	Answer	Marks														
2(a)	$\text{gradient} = -\frac{1}{kf_s}$ $y\text{-intercept} = \frac{1}{f_s}$	1														
2(b)	<table><tr><th>v/ms^{-1}</th><th>$\frac{1}{f}/10^{-3}\text{Hz}^{-1}$</th></tr><tr><td>$3.5 \pm 0.4$</td><td>1.118 or 1.1183</td></tr><tr><td>6.3 ± 0.4</td><td>1.110 or 1.1096</td></tr><tr><td>8.7 ± 0.5</td><td>1.101 or 1.1013</td></tr><tr><td>11.4 ± 0.5</td><td>1.092 or 1.0919</td></tr><tr><td>13.9 ± 0.6</td><td>1.083 or 1.0827</td></tr><tr><td>16.2 ± 0.6</td><td>1.074 or 1.0739</td></tr></table> Values of v and $\frac{1}{f}$ correct as shown above.	v/ms^{-1}	$\frac{1}{f}/10^{-3}\text{Hz}^{-1}$	3.5 ± 0.4	1.118 or 1.1183	6.3 ± 0.4	1.110 or 1.1096	8.7 ± 0.5	1.101 or 1.1013	11.4 ± 0.5	1.092 or 1.0919	13.9 ± 0.6	1.083 or 1.0827	16.2 ± 0.6	1.074 or 1.0739	1
v/ms^{-1}	$\frac{1}{f}/10^{-3}\text{Hz}^{-1}$															
3.5 ± 0.4	1.118 or 1.1183															
6.3 ± 0.4	1.110 or 1.1096															
8.7 ± 0.5	1.101 or 1.1013															
11.4 ± 0.5	1.092 or 1.0919															
13.9 ± 0.6	1.083 or 1.0827															
16.2 ± 0.6	1.074 or 1.0739															
	Uncertainties in v correct as shown above.	1														
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1														
	Error bars in v plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1														

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Line must pass between (14.5, 1.080) and (14.9, 1.080) and between (4.5, 1.115) and (4.8, 1.115).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Gradient must be negative. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and y into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

Question	Answer	Marks
2(d)(i)	f_s determined using y-intercept and f_s given to 2, 3 or 4 significant figures and k given to 2 or 3 significant figures. $f_s = \frac{1}{y\text{-intercept}}$	1
	k determined using gradient with method shown and f_s <u>and</u> k given with SI units with appropriate powers of ten. $k = -\frac{y\text{-intercept}}{\text{gradient}} \text{ or } k = -\frac{1}{\text{gradient} \times f_s}$ Units of f_s : Hz Units of k : m s^{-1}	1
2(d)(ii)	Percentage uncertainty in k with method shown. percentage uncertainty = $\left(\frac{\Delta y\text{-intercept}}{y\text{-intercept}} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or correct substitution for max/min methods.	1
2(e)	v determined (non-zero) to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution. $v = \frac{\frac{1}{f} - y\text{-intercept}}{\text{gradient}}$ or $v = k - \frac{k f_s}{f}$	1

Cambridge International AS & A Level

PHYSICS**9702/41**

Paper 4 A Level Structured Questions

May/June 2025

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **19** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	arithmetic error
	benefit of the doubt given
	contradiction in response, mark not awarded
	correct point or mark awarded
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect physics
	incorrect point or mark not awarded
	information missing or insufficient for credit

PUBLISHED

Annotation	Meaning
MO	mandatory mark not awarded
SEEN	point has been noted, but no credit has been given or blank page seen
POT	power of ten error
TE	transcription error

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Question	Answer	Marks
1(a)	work done per unit mass	B1
	work (done in) moving mass from infinity (to the point)	B1
1(b)(i)	evidence of addition of 3.4×10^6 to 1.7×10^6 or 6.8×10^6	C1
	$GM \times 122 / (5.1 \times 10^6)$ or $GM \times 122 / (10.2 \times 10^6)$	C1
	$6.67 \times 10^{-11} \times M \times 122 \times [(5.1 \times 10^6)^{-1} - (10.2 \times 10^6)^{-1}] = 5.1 \times 10^8$	A1
	leading to $M = 6.4 \times 10^{23}$ kg	
1(b)(ii)	$\phi = (-) (6.67 \times 10^{-11} \times 6.4 \times 10^{23}) / (3.4 \times 10^6)$	C1
	$= -1.3 \times 10^7 \text{ J kg}^{-1}$	A1
1(c)(i)	Mars takes (just under) 25 hours to rotate once on its axis	B1
1(c)(ii)	orbit is equatorial or orbit is in same direction as direction of rotation of Mars	B1

Question	Answer	Marks
2(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
2(b)(i)	charge = $(+)2e$	A1
2(b)(ii)	$F = 2 \times (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (170 \times 10^{-12})^2] = 1.6 \times 10^{-8} \text{ N}$	A1
2(c)(i)	$F = mv^2 / r$	C1
	$v = [(1.6 \times 10^{-8} \times 170 \times 10^{-12}) / (9.11 \times 10^{-31})]^{1/2}$ $= 1.7 \times 10^6 \text{ m s}^{-1}$	A1
2(c)(ii)	$F = mr\omega^2$ and $\omega = 2\pi / T$ $F = 4\pi^2 mr / T^2$	C1
	$T = [(4\pi^2 \times 9.11 \times 10^{-31} \times 170 \times 10^{-12}) / (1.6 \times 10^{-8})]^{1/2}$ $= 6.2 \times 10^{-16} \text{ s}$	A1
	or	
	$v = 2\pi r / T$	(C1)
	$T = (2\pi \times 170 \times 10^{-12}) / (1.73 \times 10^6)$ $= 6.2 \times 10^{-16} \text{ s}$	(A1)

Question	Answer	Marks
2(d)(i)	$E \propto Q / r^2$	C1
	ratio = $[1.60 \times 10^{-19} \times (170 \times 10^{-12})^2] / [3.2 \times 10^{-19} \times (340 \times 10^{-12})^2]$ = 0.13	A1
2(d)(ii)	resultant force slightly less (than 1.6×10^{-8} N) so speed lower or resultant force slightly less (than 1.6×10^{-8} N) so period greater	B1

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause state change)	B1
	(thermal) energy to change state at constant temperature	B1
3(b)	(for vaporisation): involves greater change in volume (of substance) or involves greater increase in separation of molecules	B1
	more work has to be done by molecules (to separate) or greater increase in potential energy of molecules	M1
	kinetic energy of molecules unchanged, so more thermal energy needed	A1
3(c)	$Q = mc\Delta\theta$ and $Q = mL$	C1
	$\Delta\theta$ for the water = 26.4 – 10.3	C1
	$(37.0 \times L) + (37.0 \times 4.18 \times 10.3) = (208 \times 4.18 \times 16.1)$	C1
	$L = 335 \text{ J g}^{-1}$	A1

Question	Answer	Marks								
4(a)(i)	sum of potential energy and kinetic energy	B1								
	(total) energy of random motion of particles	B1								
4(a)(ii)	potential energy (of molecules) (in an ideal gas) is zero, so the internal energy of the gas is equal to the total kinetic energy (of molecules)	B1								
	kinetic energy of molecules is proportional to (thermodynamic) temperature (so internal energy is proportional to (thermodynamic) temperature))	B1								
4(b)	cooling work done = 0	B1								
	compression increase in internal energy = $+2U$	B1								
	cooling change in internal energy = $-U$	B1								
	both rows: thermal energy adds to work to give increase in internal energy in terms of U and/or W (if fully correct, thermal energy for compression = $2U - W$ and thermal energy for cooling = $-U$: <table border="1"><tr><td>compression</td><td>$+W$</td><td>$2U - W$</td><td>$+2U$</td></tr><tr><td>cooling</td><td>0</td><td>$-U$</td><td>$-U$</td></tr></table>)	compression	$+W$	$2U - W$	$+2U$	cooling	0	$-U$	$-U$	B1
	compression	$+W$	$2U - W$	$+2U$						
cooling	0	$-U$	$-U$							

Question	Answer	Marks
5(a)(i)	amplitude = 0.60 m	A1
5(a)(ii)	oscillations are simple harmonic	B1
5(b)	<i>Any three points from:</i> • mean / equilibrium position is at $h = 1.4$ m • total energy of oscillations = 9.0 J • angular frequency of oscillations = 1.2 rad s^{-1} or period of oscillations = 5.1 s or frequency of oscillation = 0.19 Hz • maximum speed of block = 0.73 m s^{-1} • mass of block = 33 kg	B3
5(c)	U-shaped curve resting on h axis (with minimum at $E_P = 0$)	B1
	curve from $h = 0.8$ m to $h = 2.0$ m, with minimum E_P shown at $h = 1.4$ m	B1
	both end-points of curve shown at $E_P = 9.0$ J	B1

Question	Answer	Marks
6(a)(i)	conversion from a.c. to d.c.	B1
6(a)(ii)	smoothing	B1
6(b)(i)	$A = 12 \text{ V}$	A1
	$B = 2\pi / (20 \times 10^{-3})$ $= 310 \text{ rad s}^{-1}$	A1
6(b)(ii)	full-wave (rectification)	B1
6(b)(iii)	four diodes shown, with correct circuit symbols	B1
	four diodes correctly connected to form a bridge rectifier	B1
6(b)(iv)	$V = V_0 \exp(-t / \tau)$ or $V = V_0 \exp(-t / RC)$ and $\tau = RC$	C1
	$8.0 = 12 \exp(-7.3 \times 10^{-3} / \tau)$	C1
	$\tau = 0.018 \text{ s}$	A1
6(c)	time constant = RC	C1
	$R = (0.018 / 570 \times 10^{-6})$ $= 32 \Omega$	A1

Question	Answer	Marks
7(a)	• force per unit length • force per unit current • length / current perpendicular to field <i>1 mark for any two points, 2 marks for all three points.</i>	B2
7(b)(i)	$F = BQv$	B1
7(b)(ii)	arrow at Y pointing vertically upwards	B1
7(b)(iii)	upwards deflection showing circular path	B1
7(c)(i)	electric field applied vertically downwards (may be shown on a labelled diagram)	B1
	electric force on particle in opposite direction to magnetic force (may be shown on a labelled diagram)	B1
	particle undeflected when magnitudes of electric and magnetic forces are equal	B1
7(c)(ii)	$EQ = BQv$	B1
	$v = E / B$	A1

Question	Answer	Marks
8(a)	wavelength associated with a moving particle	B1
8(b)	$\lambda = h / p$	C1
	$= (6.63 \times 10^{-34}) / (9.11 \times 10^{-31} \times 4.9 \times 10^7)$ $= 1.5 \times 10^{-11} \text{ m}$	A1
8(c)	similarity: <i>any one point from:</i> • same mass • same <u>magnitude</u> of charge • both leptons	B1
	difference: <i>any one point from:</i> • electron has negative charge, positron has positive charge • positron is anti-particle of electron • electron is a particle, positron is an anti-particle	B1
8(d)(i)	(pair) annihilation	B1
8(d)(ii)	their mass gets converted into energy	B1
	(their mass–energy) becomes the energy of the gamma photons	B1
8(d)(iii)	they travel in opposite directions to conserve momentum	B1
8(d)(iv)	kinetic energy = $\frac{1}{2} \times 9.11 \times 10^{-31} \times (4.9 \times 10^7)^2 = 1.1 \times 10^{-15} \text{ J}$	A1

Question	Answer	Marks
8(d)(v)	$E = mc^2$	C1
	$E = hc / \lambda$ or $E = hf$ and $c = f\lambda$	C1
	$(1.1 \times 10^{-15}) + (9.11 \times 10^{-31} \times (3.00 \times 10^8)^2) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$ $\lambda = 2.39 \times 10^{-12} \text{ m}$	A1

PUBLISHED

Question	Answer	Marks
9(a)	number of nuclear disintegrations per unit time	B1
9(b)	activity is proportional to the number of undecayed nuclei	B1
	activity = $(-)$ rate of change of number of undecayed nuclei	B1
	N is proportional to the rate of change of N (so exponential variation)	B1
9(c)(i)	$120 = 180 \exp(-\lambda \times 8.4)$	C1
	$\lambda = 0.048 \text{ min}^{-1}$	A1
9(c)(ii)	half-life = $\ln 2 / 0.048$ = 14 min	A1
9(c)(iii)	line with negative gradient throughout, starting at (0, 180)	B1
	curve with negative gradient passing through (8.4, 120)	B1
	curve with decreasing negative gradient, from $t = 0$ to $t = 24$ min, passing through (14, 90)	B1

PUBLISHED

Question	Answer	Marks
10(a)	speed is (directly) proportional to distance	M1
	speed is speed of recession of galaxy from an observer, and distance is the distance of the galaxy from the observer	A1
10(b)(i)	galaxy is receding from the Earth	B1
	observed wavelength is redshifted from emitted wavelength	B1
10(b)(ii)	$\Delta\lambda / \lambda = v / c$	C1
	$(4.91 - 4.62) / 4.62 = v / (3.00 \times 10^8)$	
	$v = 1.9 \times 10^7 \text{ m s}^{-1}$	A1
10(b)(iii)	wavelength (of maximum intensity) is inversely proportional to temperature	B1
	observed wavelength too high, so determined temperature too low	B1
10(c)	$v = H_0 d$	C1
	$d = (1.9 \times 10^7) / (2.3 \times 10^{-18})$	A1
	$= 8.3 \times 10^{24} \text{ m}$	

Cambridge International AS & A Level

PHYSICS**9702/42**

Paper 4 A Level Structured Questions

May/June 2025

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **18** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	arithmetic error
	benefit of the doubt given
	contradiction in response, mark not awarded
	correct point or mark awarded
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect physics
	incorrect point or mark not awarded
	information missing or insufficient for credit

PUBLISHED

Annotation	Meaning
MO	mandatory mark not awarded
SEEN	point has been noted, but no credit has been given or blank page seen
POT	power of ten error
TE	transcription error

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Question	Answer	Marks
1(a)	angle (subtended at centre of a circle) when arc (length) = radius	B1
1(b)(i)	$v = r\omega$	C1
	$\omega = 17 / 0.46$ $= 37 \text{ rad s}^{-1}$	A1
1(b)(ii)	$T = 2\pi r / v$ or $T = 2\pi / \omega$	C1
	$= 2\pi \times 0.46 / 17$ or $2\pi / 37$ $= 0.17 \text{ s}$	A1
1(b)(iii)	distance = $2\pi \times 0.038 = 0.24 \text{ m}$	A1
1(b)(iv)	angle = arc length / radius	C1
	$= 0.24 / 0.15$ $= 1.6 \text{ rad}$	A1
1(c)	point X moves through a smaller distance in the same time or (linear) speed of movement of point X / chain decreases or (linear) speed of (circumference of) both cogs decreases	B1
	angular speed (of pedals) decreases	B1

Question	Answer	Marks
2(a)(i)	direction of force	B1
	force acting on a (test) mass	B1
2(a)(ii)	at least four radial lines from the Earth's surface, equally spaced around the surface	B1
	arrows indicating direction towards Earth	B1
2(b)(i)	top pole labelled S and bottom pole labelled N	B1
2(b)(ii)	solenoid field pattern at the poles: at least two field lines either side of both poles, close to the poles, clustered closely together, leaving the surface approximately perpendicularly to the surface and curving away from the axis of the poles as their distance from the surface increases	B1
	solenoid field pattern above the equator: at least one field line either side of the Earth connecting two points on the surface that are on the same side of the poles, one north of the equator and one south of it, passing above the surface near the magnetic equator approximately parallel to the surface	B1
2(c)(i)	(around the surface) lines are evenly spaced	B1
	all lines perpendicular to surface or pointing down towards surface (at all points around the surface)	B1

PUBLISHED

Question	Answer	Marks
2(c)(ii)	<i>Any three bulleted points from:</i> • strongest at the poles • weakest near the Equator <i>Up to two points from:</i> • perpendicular to surface at the poles • parallel to the surface near the Equator • angle to surface increases from Equator to poles	B3

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause temperature change)	B1
	(thermal) energy per unit change in temperature	B1
3(b)(i)	density = mass / volume	C1
	$\text{mass} = 2.700 \times 10^3 \times 3.612 \times 10^{-3}$ $= 9.752 \text{ kg}$	A1
3(b)(ii)	$\text{volume} = 3.612 \times 10^{-3} \times (2.700 / 2.620) = 3.722 \times 10^{-3} \text{ m}^3$ or $\text{volume} = 9.752 / (2.620 \times 10^3) = 3.722 \times 10^{-3} \text{ m}^3$	A1

PUBLISHED

Question	Answer	Marks
3(b)(iii)	$W = p\Delta V$	C1
	$= 1.01 \times 10^5 \times (3.722 - 3.612) \times 10^{-3}$	A1
	$= 11.1 \text{ J}$	
3(b)(iv)	volume (of block) increases	B1
	work is done against the atmosphere so work done (on block) is negative	B1
3(b)(v)	thermal energy $= (4.38 \times 10^6) + 11.1$	B1
	specific heat capacity $= (4.38 \times 10^6) / (9.75 \times 500)$	C1
	$= 898 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$	A1
3(c)	work done is negligible compared with (change in) internal energy so (answer in (b)(v) would be) unchanged	B1

PUBLISHED

Question	Answer	Marks
4(a)(i)	thermodynamic temperature	B1
4(a)(ii)	molar gas constant	B1
4(b)(i)	m : mass of one molecule (of the gas)	B1
	$\langle c^2 \rangle$: mean-square speed (of molecules)	B1
4(b)(ii)	$NBT / A = \frac{1}{3} Nm \langle c^2 \rangle$	M1
	clear use of $E_K = \frac{1}{2} m \langle c^2 \rangle$ leading to $E_K = 3BT / 2A$	A1
4(c)	line with positive gradient passing through the origin	B1
	smooth curve with decreasing positive gradient	B1

Question	Answer	Marks
5(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which) acceleration is (always) in the opposite <u>direction</u> to displacement or acceleration is (always) directed towards a fixed point	B1
5(b)(i)	period = $2\pi / 16$ = 0.39 s	A1
5(b)(ii)	$v_0 = \omega x_0$ or $v_0 = \omega \sqrt{x_0^2 - 0^2}$	C1
	$x_0 = 0.56 / 16$ = 0.035 m	A1
5(b)(iii)	$v = \pm 16 \sqrt{0.0352 - x^2}$	A1
5(b)(iv)	closed loop surrounding the origin	B1
	loop crosses $v = 0$ at maximum values of x at $x = \pm 3.5$ cm	B1
	loop crosses $x = 0$ at maximum values of v at $v = \pm 0.56$ m s ⁻¹	B1

Question	Answer	Marks
6(a)	plate X marked as negative and plate Y marked as positive	B1
6(b)(i)	$E = V / x$	C1
	$= (58 \times 10^3) / 0.041$	A1
	$= 1.4 \times 10^6 \text{ N C}^{-1}$	
6(b)(ii)	$ma = eE$	C1
	$a = (1.60 \times 10^{-19} \times 1.41 \times 10^6) / (9.11 \times 10^{-31})$	A1
	$= 2.5 \times 10^{17} \text{ m s}^{-2}$	
6(c)(i)	$eV = hc / \lambda$ or $eV = hf$ and $f = c / \lambda$	C1
	$(1.60 \times 10^{-19} \times 58 \times 10^3) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$	M1
	clear conversion from m to pm leading to $\lambda = 21 \text{ pm}$	A1
6(c)(ii)	X-rays	B1
6(c)(iii)	<i>Any two points from:</i> • waves are passed into structure and transmitted waves detected • different parts of the structure absorb different fractions of energy • difference in detected / transmitted <u>intensities</u> used (to form image)	B2

Question	Answer	Marks
7(a)	$V_C = V_R$	B1
7(b)(i)	$C = Q / V$	C1
	$= (7.2 \times 10^{-3}) / 12 \quad (= 6.0 \times 10^{-4} \text{ F})$ $= 600 \mu\text{F}$	A1
7(b)(ii)	$R = V / I$	C1
	$= 12 / (1.5 \times 10^{-3}) \quad (= 8000 \Omega)$ $= 8.0 \text{ k}\Omega$	A1
7(b)(iii)	$\tau = RC$	C1
	$= 8000 \times 6.0 \times 10^{-4}$ $= 4.8 \text{ s}$	A1
7(c)	Any two points from: - charge and current are both (directly) proportional to voltage - charge is (directly) proportional to current - current is the rate of change of charge	B2
	Q is proportional to the rate of change of Q (so exponential variation)	B1

Question	Answer	Marks
8(a)(i)	half-wave (rectification)	B1
8(a)(ii)	$V_0 = 6.0 \times \sqrt{2}$ $= 8.5 \text{ V}$	A1
8(b)(i)	$P = V^2 / R$	C1
	$P_0 = 8.5^2 / 45$ $= 1.6 \text{ W}$	A1
8(b)(ii)	two humps of width $0.5T$ and two sections of zero power of width $0.5T$	B1
	all humps drawn have width $0.5T$, minima at $P = 0$ and peaks at $P = P_0$	B1
	correct sinusoidal shape, with smooth troughs sitting on t -axis at $P = 0$	B1
8(b)(iii)	mean power within each hump is $\frac{1}{2}P_0$ from the symmetry of the curve	B1
	additional half factor from removal of half of the power in each cycle	B1
8(b)(iv)	$\langle P \rangle = V_{\text{r.m.s.}}^2 / R$ $V_{\text{r.m.s.}} = \sqrt{[(1.6 / 4) \times 45]}$ $= 4.2 \text{ V}$	A1

PUBLISHED

Question	Answer	Marks
9(a)	emission of electrons (from a metal surface)	B1
	when electromagnetic radiation is incident (on surface / electrons)	B1
9(b)(i)	current falls to zero when applied voltage equals energy per unit charge of emitted electrons	B1
	energy of photon depends on frequency	B1
	maximum energy of electron depends on energy of photon	B1
9(b)(ii)	<i>Any three points from:</i> • threshold frequency = 1.5×10^{15} Hz • threshold wavelength = 2.0×10^{-7} m • work function = 6.2 eV (or 9.9×10^{-19} J) • Planck constant = 6.6×10^{-34} J s (not 6.63×10^{-34} J s) • number per unit time (of photons / electrons) = $1.7 \times 10^{16} \text{ s}^{-1}$ (in stage 1) • power of incident radiation = 0.028 W (in stage 1)	B3

Question	Answer	Marks															
10(a)	(decay is) not affected by external / environmental factors	B1															
10(b)(i)	half-life of X = $2T$ and half-life of Y = $3T$	B1															
	both samples show decay constant, in terms of $1 / T$, equal to $\ln 2 / \text{half-life}$ (decay constant of X = $\ln 2 / 2T$ and decay constant of Y = $\ln 2 / 3T$ if both half-lives correct)	B1															
	both samples show N_0 , in terms of AT , equal to initial activity / decay constant (N_0 for X = $8AT / \ln 2$ and N_0 for Y = $3AT / \ln 2$ if both decay constants correct) (Fully correct table: <table><tr><td></td><td>half-life</td><td>decay constant</td><td>A_0</td><td>N_0</td></tr><tr><td>X</td><td>$2T$</td><td>$\ln 2 / 2T$</td><td>$4A$</td><td>$8AT / \ln 2$</td></tr><tr><td>Y</td><td>$3T$</td><td>$\ln 2 / 3T$</td><td>A</td><td>$3AT / \ln 2$</td></tr></table>)		half-life	decay constant	A_0	N_0	X	$2T$	$\ln 2 / 2T$	$4A$	$8AT / \ln 2$	Y	$3T$	$\ln 2 / 3T$	A	$3AT / \ln 2$	B1
		half-life	decay constant	A_0	N_0												
X	$2T$	$\ln 2 / 2T$	$4A$	$8AT / \ln 2$													
Y	$3T$	$\ln 2 / 3T$	A	$3AT / \ln 2$													
10(b)(ii)	correct substitution of A_0 and λ into $A_0 \exp (-\lambda t)$ for sample X or sample Y	C1															
	$4A \exp (-t \ln 2 / 2T) = A \exp (-t \ln 2 / 3T)$	C1															
	$t = 12T$	A1															
10(c)	<i>Any two points from:</i> (radiation) emitted in all directions, not just in direction of detector - some radiation absorbed by air / sample / window of detector - some radiation may not register even though it reaches detector	B2															

Cambridge International AS & A Level

PHYSICS**9702/43**

Paper 4 A Level Structured Questions

May/June 2025

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **19** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	arithmetic error
	benefit of the doubt given
	contradiction in response, mark not awarded
	correct point or mark awarded
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect physics
	incorrect point or mark not awarded
	information missing or insufficient for credit

PUBLISHED

Annotation	Meaning
MO	mandatory mark not awarded
SEEN	point has been noted, but no credit has been given or blank page seen
POT	power of ten error
TE	transcription error

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Question	Answer	Marks
1(a)	work done per unit mass	B1
	work (done in) moving mass from infinity (to the point)	B1
1(b)(i)	evidence of addition of 3.4×10^6 to 1.7×10^6 or 6.8×10^6	C1
	$GM \times 122 / (5.1 \times 10^6)$ or $GM \times 122 / (10.2 \times 10^6)$	C1
	$6.67 \times 10^{-11} \times M \times 122 \times [(5.1 \times 10^6)^{-1} - (10.2 \times 10^6)^{-1}] = 5.1 \times 10^8$	A1
	leading to $M = 6.4 \times 10^{23}$ kg	
1(b)(ii)	$\phi = (-) (6.67 \times 10^{-11} \times 6.4 \times 10^{23}) / (3.4 \times 10^6)$	C1
	$= -1.3 \times 10^7 \text{ J kg}^{-1}$	A1
1(c)(i)	Mars takes (just under) 25 hours to rotate once on its axis	B1
1(c)(ii)	orbit is equatorial or orbit is in same direction as direction of rotation of Mars	B1

Question	Answer	Marks
2(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
2(b)(i)	charge = $(+)2e$	A1
2(b)(ii)	$F = 2 \times (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (170 \times 10^{-12})^2] = 1.6 \times 10^{-8} \text{ N}$	A1
2(c)(i)	$F = mv^2 / r$	C1
	$v = [(1.6 \times 10^{-8} \times 170 \times 10^{-12}) / (9.11 \times 10^{-31})]^{1/2}$ $= 1.7 \times 10^6 \text{ m s}^{-1}$	A1
2(c)(ii)	$F = mr\omega^2$ and $\omega = 2\pi / T$ $F = 4\pi^2 mr / T^2$	C1
	$T = [(4\pi^2 \times 9.11 \times 10^{-31} \times 170 \times 10^{-12}) / (1.6 \times 10^{-8})]^{1/2}$ $= 6.2 \times 10^{-16} \text{ s}$	A1
	or	
	$v = 2\pi r / T$	(C1)
	$T = (2\pi \times 170 \times 10^{-12}) / (1.73 \times 10^6)$ $= 6.2 \times 10^{-16} \text{ s}$	(A1)

Question	Answer	Marks
2(d)(i)	$E \propto Q / r^2$	C1
	ratio = $[1.60 \times 10^{-19} \times (170 \times 10^{-12})^2] / [3.2 \times 10^{-19} \times (340 \times 10^{-12})^2]$ = 0.13	A1
2(d)(ii)	resultant force slightly less (than 1.6×10^{-8} N) so speed lower or resultant force slightly less (than 1.6×10^{-8} N) so period greater	B1

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause state change)	B1
	(thermal) energy to change state at constant temperature	B1
3(b)	(for vaporisation): involves greater change in volume (of substance) or involves greater increase in separation of molecules	B1
	more work has to be done by molecules (to separate) or greater increase in potential energy of molecules	M1
	kinetic energy of molecules unchanged, so more thermal energy needed	A1
3(c)	$Q = mc\Delta\theta$ and $Q = mL$	C1
	$\Delta\theta$ for the water = 26.4 – 10.3	C1
	$(37.0 \times L) + (37.0 \times 4.18 \times 10.3) = (208 \times 4.18 \times 16.1)$	C1
	$L = 335 \text{ J g}^{-1}$	A1

PUBLISHED

Question	Answer	Marks								
4(a)(i)	sum of potential energy and kinetic energy	B1								
	(total) energy of random motion of particles	B1								
4(a)(ii)	potential energy (of molecules) (in an ideal gas) is zero, so the internal energy of the gas is equal to the total kinetic energy (of molecules)	B1								
	kinetic energy of molecules is proportional to (thermodynamic) temperature (so internal energy is proportional to (thermodynamic) temperature))	B1								
4(b)	cooling work done = 0	B1								
	compression increase in internal energy = $+2U$	B1								
	cooling change in internal energy = $-U$	B1								
	both rows: thermal energy adds to work to give increase in internal energy in terms of U and/or W (if fully correct, thermal energy for compression = $2U - W$ and thermal energy for cooling = $-U$: <table border="1"><tr><td>compression</td><td>$+W$</td><td>$2U - W$</td><td>$+2U$</td></tr><tr><td>cooling</td><td>0</td><td>$-U$</td><td>$-U$</td></tr></table>)	compression	$+W$	$2U - W$	$+2U$	cooling	0	$-U$	$-U$	B1
	compression	$+W$	$2U - W$	$+2U$						
cooling	0	$-U$	$-U$							

PUBLISHED

Question	Answer	Marks
5(a)(i)	amplitude = 0.60 m	A1
5(a)(ii)	oscillations are simple harmonic	B1
5(b)	<i>Any three points from:</i> - mean / equilibrium position is at $h = 1.4$ m - total energy of oscillations = 9.0 J - angular frequency of oscillations = 1.2 rad s^{-1} or period of oscillations = 5.1 s or frequency of oscillation = 0.19 Hz - maximum speed of block = 0.73 m s^{-1} - mass of block = 33 kg	B3
5(c)	U-shaped curve resting on h axis (with minimum at $E_P = 0$)	B1
	curve from $h = 0.8$ m to $h = 2.0$ m, with minimum E_P shown at $h = 1.4$ m	B1
	both end-points of curve shown at $E_P = 9.0$ J	B1

Question	Answer	Marks
6(a)(i)	conversion from a.c. to d.c.	B1
6(a)(ii)	smoothing	B1
6(b)(i)	$A = 12 \text{ V}$	A1
	$B = 2\pi / (20 \times 10^{-3})$ $= 310 \text{ rad s}^{-1}$	A1
6(b)(ii)	full-wave (rectification)	B1
6(b)(iii)	four diodes shown, with correct circuit symbols	B1
	four diodes correctly connected to form a bridge rectifier	B1
6(b)(iv)	$V = V_0 \exp(-t / \tau)$ or $V = V_0 \exp(-t / RC)$ and $\tau = RC$	C1
	$8.0 = 12 \exp(-7.3 \times 10^{-3} / \tau)$	C1
	$\tau = 0.018 \text{ s}$	A1
6(c)	time constant = RC	C1
	$R = (0.018 / 570 \times 10^{-6})$ $= 32 \Omega$	A1

Question	Answer	Marks
7(a)	• force per unit length • force per unit current • length / current perpendicular to field <i>1 mark for any two points, 2 marks for all three points.</i>	B2
7(b)(i)	$F = BQv$	B1
7(b)(ii)	arrow at Y pointing vertically upwards	B1
7(b)(iii)	upwards deflection showing circular path	B1
7(c)(i)	electric field applied vertically downwards (may be shown on a labelled diagram)	B1
	electric force on particle in opposite direction to magnetic force (may be shown on a labelled diagram)	B1
	particle undeflected when magnitudes of electric and magnetic forces are equal	B1
7(c)(ii)	$EQ = BQv$	B1
	$v = E / B$	A1

Question	Answer	Marks
8(a)	wavelength associated with a moving particle	B1
8(b)	$\lambda = h / p$	C1
	$= (6.63 \times 10^{-34}) / (9.11 \times 10^{-31} \times 4.9 \times 10^7)$ $= 1.5 \times 10^{-11} \text{ m}$	A1
8(c)	similarity: <i>any one point from:</i> • same mass • same <u>magnitude</u> of charge • both leptons	B1
	difference: <i>any one point from:</i> • electron has negative charge, positron has positive charge • positron is anti-particle of electron • electron is a particle, positron is an anti-particle	B1
8(d)(i)	(pair) annihilation	B1
8(d)(ii)	their mass gets converted into energy	B1
	(their mass–energy) becomes the energy of the gamma photons	B1
8(d)(iii)	they travel in opposite directions to conserve momentum	B1
8(d)(iv)	kinetic energy = $\frac{1}{2} \times 9.11 \times 10^{-31} \times (4.9 \times 10^7)^2 = 1.1 \times 10^{-15} \text{ J}$	A1

Question	Answer	Marks
8(d)(v)	$E = mc^2$	C1
	$E = hc / \lambda$ or $E = hf$ and $c = f\lambda$	C1
	$(1.1 \times 10^{-15}) + (9.11 \times 10^{-31} \times (3.00 \times 10^8)^2) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$ $\lambda = 2.39 \times 10^{-12} \text{ m}$	A1

PUBLISHED

Question	Answer	Marks
9(a)	number of nuclear disintegrations per unit time	B1
9(b)	activity is proportional to the number of undecayed nuclei	B1
	activity = $(-)$ rate of change of number of undecayed nuclei	B1
	N is proportional to the rate of change of N (so exponential variation)	B1
9(c)(i)	$120 = 180 \exp(-\lambda \times 8.4)$	C1
	$\lambda = 0.048 \text{ min}^{-1}$	A1
9(c)(ii)	half-life = $\ln 2 / 0.048$ = 14 min	A1
9(c)(iii)	line with negative gradient throughout, starting at (0, 180)	B1
	curve with negative gradient passing through (8.4, 120)	B1
	curve with decreasing negative gradient, from $t = 0$ to $t = 24$ min, passing through (14, 90)	B1

PUBLISHED

Question	Answer	Marks
10(a)	speed is (directly) proportional to distance	M1
	speed is speed of recession of galaxy from an observer, and distance is the distance of the galaxy from the observer	A1
10(b)(i)	galaxy is receding from the Earth	B1
	observed wavelength is redshifted from emitted wavelength	B1
10(b)(ii)	$\Delta\lambda / \lambda = v / c$ $(4.91 - 4.62) / 4.62 = v / (3.00 \times 10^8)$	C1
	$v = 1.9 \times 10^7 \text{ m s}^{-1}$	A1
10(b)(iii)	wavelength (of maximum intensity) is inversely proportional to temperature	B1
	observed wavelength too high, so determined temperature too low	B1
10(c)	$v = H_0 d$	C1
	$d = (1.9 \times 10^7) / (2.3 \times 10^{-18})$ $= 8.3 \times 10^{24} \text{ m}$	A1

Cambridge International AS & A Level

PHYSICS**9702/51**

Paper 5 Planning, Analysis and Evaluation

May/June 2025

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	benefit of the doubt given
	correct awarding one mark from additional detail 1. similar numbered ticks are used for additional detail 2, 3, 4 etc.
	correct point or mark awarded
	defining the problem mark
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect point or mark not awarded
	incorrect unit
	information missing or insufficient for credit

Annotation	Meaning
MO	methods of data collection mark
SEEN	point has been noted, but no credit has been given or blank page seen
R	repeat of point previously awarded mark

Question	Answer	Marks
1	Defining the problem	
	vary f and measure V or f is the independent variable and V is the dependent variable	1
	keep E <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - circuit with a.c. supply - oscilloscope connected in parallel with the resistor - workable circuit - oscilloscope and a.c. supply labelled	1
	labelled signal generator or <u>variable frequency</u> power supply connected across the terminals	1
	method to determine V or E from oscilloscope, e.g. multiply amplitude / height of wave by y -gain on oscilloscope	1
	method to determine f from oscilloscope, e.g. determine period T by multiplying number of divisions in 1 cycle or horizontal distance in 1 cycle by the time base and $f = 1/T$	1
	Method of Analysis	
	plot a graph of $\frac{1}{V}$ against f or equivalent, e.g. f against $\frac{1}{V}$ Allow logarithms e.g. $\lg V$ against $\lg f$.	1
	relationship valid <u>if</u> a straight line is produced <u>passing through the origin</u> (for $\lg V$ against $\lg f$: relationship valid <u>if</u> a straight line is produced with <u>gradient = -1</u>)	1

Question	Answer	Marks				
1	<table><tr><td>$\frac{1}{V}$ against f</td><td>f against $\frac{1}{V}$</td></tr><tr><td>$K = \frac{lES}{AN^2} \times \text{gradient}$</td><td>$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$</td></tr></table> (for $\lg V$ against $\lg f$: $K = \frac{ElS}{AN^2} \times 10^{-y\text{-intercept}}$).	$\frac{1}{V}$ against f	f against $\frac{1}{V}$	$K = \frac{lES}{AN^2} \times \text{gradient}$	$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$	1
	$\frac{1}{V}$ against f	f against $\frac{1}{V}$				
	$K = \frac{lES}{AN^2} \times \text{gradient}$	$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$				
	Additional detail including safety considerations	6				
	D1 precaution linked to <u>hot coil</u> or <u>hot resistor</u> or prevention of <u>burns</u> from <u>coil</u> or <u>resistor</u> , e.g. use gloves / switch off power supply when not measuring V to prevent burns from coil / resistor					
	D2 keep N and A and l and S <u>constant</u>					
	D3 method to keep S constant, e.g. switch off power supply between readings to prevent heating of resistor or to allow resistor to cool					
	D4 method to determine A , e.g. use calipers / micrometer to measure diameter (of coil) / d and $A = \frac{\pi d^2}{4}$					
	D5 repeat measurements of diameter d along the length of the coil / in different directions and determine the average value of d					
	D6 method to determine the value of S , e.g. separate circuit diagram showing resistor connected to ohmmeter, or circuit diagram showing resistor connected to a power supply with an ammeter and voltmeter and $S = V / I$					
D7 measure l with a ruler / calipers						
D8 oscilloscope drawn connected across terminals / across signal generator and description to determine E						
D9 adjust y -gain for maximum amplitude or adjust time base for length of one wave or measure n waves and divide measured time by n						

Question	Answer	Marks
1	D10 method to keep E constant, e.g. check p.d. and alter supply or method to keep l constant, e.g. tape coil or method to keep A constant, e.g. wind wire on a cylinder	

Question	Answer	Marks							
2(a)	gradient = $\frac{R}{E}$ y-intercept = $\frac{Z}{E}$	1							
2(b)	<table><tr><td>$\frac{1}{I} / 10^3 \text{ A}^{-1}$</td></tr><tr><td>2.20 or 2.198</td></tr><tr><td>1.90 or 1.905</td></tr><tr><td>1.72 or 1.724</td></tr><tr><td>1.57 or 1.575</td></tr><tr><td>1.46 or 1.460</td></tr><tr><td>1.31 or 1.307</td></tr></table> Values of $\frac{1}{I} / 10^3 \text{ A}^{-1}$ correct as shown above.	$\frac{1}{I} / 10^3 \text{ A}^{-1}$	2.20 or 2.198	1.90 or 1.905	1.72 or 1.724	1.57 or 1.575	1.46 or 1.460	1.31 or 1.307	1
$\frac{1}{I} / 10^3 \text{ A}^{-1}$									
2.20 or 2.198									
1.90 or 1.905									
1.72 or 1.724									
1.57 or 1.575									
1.46 or 1.460									
1.31 or 1.307									

Question	Answer	Marks
2(b)	Uncertainties in $\frac{1}{I} / 10^3 \text{ A}^{-1}$ from ± 0.02 or ± 0.03 decreasing to ± 0.01 .	1
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1
	Error bars in $\frac{1}{I}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1
2(c)(ii)	Straight line of best fit drawn. Thickness of the line must be less than half a small square. Do not accept line from top point to bottom point. Line must pass between (0.101, 1.40) and (0.104, 1.40) and between (0.189, 2.10) and (0.194, 2.10)	1
	Worst acceptable straight line drawn (steepest or shallowest possible line that passes through all the error bars). Thickness of the line must be less than half a small square. All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1

Question	Answer	Marks
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and y into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1
2(d)(i)	R determined using gradient and R and Z given to 2 or 3 significant figures. $R = \text{gradient} \times 5.8$	1
	Z determined using y-intercept and R and Z given with units with appropriate powers of ten. $Z = \text{y-intercept} \times 5.8$ unit of R: Ω or V A^{-1} unit of Z: Ω or V A^{-1}	1
2(d)(ii)	Percentage uncertainty determined using $\Delta E = 0.2$ (V) with method shown. $\Delta R\% = \left(\frac{\Delta E}{E} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or $\Delta R\% = \left(\frac{0.2}{5.8} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$	1

Question	Answer	Marks
2(e)	<i>I</i> determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution. $I = \frac{1}{\frac{\text{gradient}}{20} + y\text{-intercept}}$ or $I = \frac{E}{\left(\frac{R}{20} + Z\right)}$	1

Cambridge International AS & A Level

PHYSICS**9702/52**

Paper 5 Planning, Analysis and Evaluation

May/June 2025

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	benefit of the doubt given
	correct awarding one mark from additional detail 1. similar numbered ticks are used for additional detail 2, 3, 4 etc.
	correct point or mark awarded
	defining the problem mark
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect point or mark not awarded
	incorrect unit
	information missing or insufficient for credit

Annotation	Meaning
MO	methods of data collection mark
SEEN	point has been noted, but no credit has been given or blank page seen
R	repeat of point previously awarded mark

Question	Answer	Marks
1	Defining the problem	
	vary m and measure v or m is the independent variable and v is the dependent variable	1
	keep h <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • axle resting on support(s) (on stands) • supports placed on bench • light gate (connected to timer) positioned at a distance h • light gate labelled and h indicated	1
	vertical metre rule <u>clamped to a stand</u> in a position close to block to measure h	1
	method to determine v using an interrupt length, e.g. $v = \text{length of block} / \text{time recorded by the timer}$	1
	method to measure m , e.g. use a (top-pan) balance	1
	Method of analysis	
	plot a graph of $\frac{1}{v^2}$ against $\frac{1}{m}$ or equivalent Do not accept logarithms.	1

Question	Answer		Marks
1	$\frac{1}{v^2}$ against $\frac{1}{m}$	$\frac{1}{m}$ against $\frac{1}{v^2}$	1
	$P = \frac{1}{h \times y\text{-intercept}}$	$P = -\frac{\text{gradient}}{y\text{-intercept} \times h}$ or $P = \frac{\pi r^2 z \times \text{gradient}}{2Qh}$	
	$\frac{1}{v^2}$ against $\frac{1}{m}$	$\frac{1}{m}$ against $\frac{1}{v^2}$	1
	$Q = \frac{\pi r^2 z}{2Ph \times \text{gradient}}$ or $Q = \frac{\pi r^2 z \times y\text{-intercept}}{2 \times \text{gradient}}$	$Q = -\frac{\pi r^2 z \times y\text{-intercept}}{2}$	
	Additional detail including safety considerations		6
	D1 precaution linked to falling block resulting in damage to block / bench, e.g. use a cushion / sand tray to prevent damage to bench or precaution linked to stands falling, e.g. clamp stand(s) to the bench prevent stand falling		
	D2 keep r and z <u>constant</u>		
	D3 method to determine r e.g. use calipers / ruler to measure diameter d and r = d / 2		
	D4 measure z with a micrometer / calipers		

PUBLISHED

Question	Answer	Marks
1	D5 set square correctly positioned between rule and bench to ensure that rule to measure h is vertical	
	D6 method to keep h constant by identifying constant initial position of the bottom of the block, e.g. clamped pin / rod to indicate the starting point each time or (fiducial) marker on rule	
	D7 description of method to ensure that axle can rotate, e.g. axle is lubricated at the supports to enable axle to rotate, axle is not fixed at the supports so axle can rotate	
	D8 use a large length of block to reduce (percentage) uncertainty in interrupt time or increase the time light gate is interrupted	
	D9 repeat experiment for the same value of m and determine the average v	
	D10 relationship valid <u>if</u> a straight line is produced (with y -intercept = $\frac{1}{hP}$). Do not accept line passing through the origin.	

Question	Answer	Marks														
2(a)	$\text{gradient} = \frac{C}{EA}$ $\text{y-intercept} = \frac{1}{E}$	1														
2(b)	<table><tr><th>V/V</th><th>$\frac{1}{V} / V^{-1}$</th></tr><tr><td>4.25</td><td>0.235 or 0.2353</td></tr><tr><td>3.70</td><td>0.270 or 0.2703</td></tr><tr><td>3.25</td><td>0.308 or 0.3077</td></tr><tr><td>2.90</td><td>0.345 or 0.3448</td></tr><tr><td>2.60</td><td>0.385 or 0.3846</td></tr><tr><td>2.35</td><td>0.426 or 0.4255</td></tr></table> Values of V/V and $\frac{1}{V} / V^{-1}$ correct as shown above.	V/V	$\frac{1}{V} / V^{-1}$	4.25	0.235 or 0.2353	3.70	0.270 or 0.2703	3.25	0.308 or 0.3077	2.90	0.345 or 0.3448	2.60	0.385 or 0.3846	2.35	0.426 or 0.4255	1
V/V	$\frac{1}{V} / V^{-1}$															
4.25	0.235 or 0.2353															
3.70	0.270 or 0.2703															
3.25	0.308 or 0.3077															
2.90	0.345 or 0.3448															
2.60	0.385 or 0.3846															
2.35	0.426 or 0.4255															
	Uncertainties in V all ± 0.05 and uncertainties in $\frac{1}{V}$ from ± 0.002 or ± 0.003 increasing to ± 0.009.	1														

Question	Answer	Marks
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1
	Error bars in $\frac{1}{V}$ plotted correctly. All error bars must be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1
2(c)(ii)	Straight line of best fit drawn. Thickness of the line must be less than half a small square. Do not accept line from top point to bottom point. Line must pass between (2.65, 0.26) and (2.80, 0.26) and between (6.30, 0.40) and (6.50, 0.40).	1
	Worst acceptable straight line drawn (steepest or shallowest possible line that passes through all the error bars). Thickness of the line must be less than half a small square. All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

Question	Answer	Marks
2(d)(i)	E determined using y-intercept and E and C given to 2 or 3 significant figures. $E = \frac{1}{y\text{-intercept}}$	1
	C determined using gradient and E and C given with correct units with appropriate powers of ten. $C = \frac{A \times \text{gradient}}{y\text{-intercept}}$ or $C = A \times E \times \text{gradient}$ unit of E : V unit of C : F	1
2(d)(ii)	Percentage uncertainty determined with method shown. $\Delta C\% = \left(\frac{\Delta A}{A} + \frac{\Delta \text{gradient}}{\text{gradient}} + \frac{\Delta y\text{-intercept}}{y\text{-intercept}} \right) \times 100$ or $\Delta C\% = \left(\frac{\Delta A}{A} + \frac{\Delta \text{gradient}}{\text{gradient}} + \frac{\Delta E}{E} \right) \times 100$ with method to determine ΔE shown	1
2(e)	V determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution. $V = \frac{1}{10 \times \text{gradient} + y\text{-intercept}}$ or $V = \frac{EA}{10C + A}$	1

Cambridge International AS & A Level

PHYSICS**9702/53**

Paper 5 Planning, Analysis and Evaluation

May/June 2025

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	benefit of the doubt given
	correct awarding one mark from additional detail 1. similar numbered ticks are used for additional detail 2, 3, 4 etc.
	correct point or mark awarded
	defining the problem mark
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect point or mark not awarded
	incorrect unit
	information missing or insufficient for credit

Annotation	Meaning
MO	methods of data collection mark
SEEN	point has been noted, but no credit has been given or blank page seen
R	repeat of point previously awarded mark

Question	Answer	Marks
1	Defining the problem	
	vary f and measure V or f is the independent variable and V is the dependent variable	1
	keep E <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - circuit with a.c. supply - oscilloscope connected in parallel with the resistor - workable circuit - oscilloscope and a.c. supply labelled	1
	labelled signal generator or <u>variable frequency</u> power supply connected across the terminals	1
	method to determine V or E from oscilloscope, e.g. multiply amplitude / height of wave by y -gain on oscilloscope	1
	method to determine f from oscilloscope, e.g. determine period T by multiplying number of divisions in 1 cycle or horizontal distance in 1 cycle by the time base and $f = 1/T$	1
	Method of Analysis	
	plot a graph of $\frac{1}{V}$ against f or equivalent, e.g. f against $\frac{1}{V}$ Allow logarithms e.g. $\lg V$ against $\lg f$.	1
	relationship valid <u>if</u> a straight line is produced <u>passing through the origin</u> (for $\lg V$ against $\lg f$: relationship valid <u>if</u> a straight line is produced with <u>gradient = -1</u>)	1

Question	Answer	Marks				
1	<table><tr><td>$\frac{1}{V}$ against f</td><td>f against $\frac{1}{V}$</td></tr><tr><td>$K = \frac{lES}{AN^2} \times \text{gradient}$</td><td>$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$</td></tr></table> (for $\lg V$ against $\lg f$: $K = \frac{ElS}{AN^2} \times 10^{-y\text{-intercept}}$).	$\frac{1}{V}$ against f	f against $\frac{1}{V}$	$K = \frac{lES}{AN^2} \times \text{gradient}$	$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$	1
	$\frac{1}{V}$ against f	f against $\frac{1}{V}$				
	$K = \frac{lES}{AN^2} \times \text{gradient}$	$K = \frac{lES}{AN^2} \times \frac{1}{\text{gradient}}$				
	Additional detail including safety considerations	6				
	D1 precaution linked to <u>hot coil</u> or <u>hot resistor</u> or prevention of <u>burns</u> from <u>coil</u> or <u>resistor</u> , e.g. use gloves / switch off power supply when not measuring V to prevent burns from coil / resistor					
	D2 keep N and A and l and S <u>constant</u>					
	D3 method to keep S constant, e.g. switch off power supply between readings to prevent heating of resistor or to allow resistor to cool					
	D4 method to determine A , e.g. use calipers / micrometer to measure diameter (of coil) / d and $A = \frac{\pi d^2}{4}$					
	D5 repeat measurements of diameter d along the length of the coil / in different directions and determine the average value of d					
	D6 method to determine the value of S , e.g. separate circuit diagram showing resistor connected to ohmmeter, or circuit diagram showing resistor connected to a power supply with an ammeter and voltmeter and $S = V / I$					
D7 measure l with a ruler / calipers						
D8 oscilloscope drawn connected across terminals / across signal generator and description to determine E						
D9 adjust y -gain for maximum amplitude or adjust time base for length of one wave or measure n waves and divide measured time by n						

PUBLISHED

Question	Answer	Marks
1	D10 method to keep E constant, e.g. check p.d. and alter supply or method to keep l constant, e.g. tape coil or method to keep A constant, e.g. wind wire on a cylinder	

Question	Answer	Marks							
2(a)	gradient = $\frac{R}{E}$ y-intercept = $\frac{Z}{E}$	1							
2(b)	<table><tr><td>$\frac{1}{I} / 10^3 \text{ A}^{-1}$</td></tr><tr><td>2.20 or 2.198</td></tr><tr><td>1.90 or 1.905</td></tr><tr><td>1.72 or 1.724</td></tr><tr><td>1.57 or 1.575</td></tr><tr><td>1.46 or 1.460</td></tr><tr><td>1.31 or 1.307</td></tr></table> Values of $\frac{1}{I} / 10^3 \text{ A}^{-1}$ correct as shown above.	$\frac{1}{I} / 10^3 \text{ A}^{-1}$	2.20 or 2.198	1.90 or 1.905	1.72 or 1.724	1.57 or 1.575	1.46 or 1.460	1.31 or 1.307	1
$\frac{1}{I} / 10^3 \text{ A}^{-1}$									
2.20 or 2.198									
1.90 or 1.905									
1.72 or 1.724									
1.57 or 1.575									
1.46 or 1.460									
1.31 or 1.307									

Question	Answer	Marks
2(b)	Uncertainties in $\frac{1}{I} / 10^3 \text{ A}^{-1}$ from ± 0.02 or ± 0.03 decreasing to ± 0.01 .	1
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1
	Error bars in $\frac{1}{I}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1
2(c)(ii)	Straight line of best fit drawn. Thickness of the line must be less than half a small square. Do not accept line from top point to bottom point. Line must pass between (0.101, 1.40) and (0.104, 1.40) and between (0.189, 2.10) and (0.194, 2.10)	1
	Worst acceptable straight line drawn (steepest or shallowest possible line that passes through all the error bars). Thickness of the line must be less than half a small square. All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1

Question	Answer	Marks
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and y into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1
2(d)(i)	R determined using gradient and R and Z given to 2 or 3 significant figures. $R = \text{gradient} \times 5.8$	1
	Z determined using y-intercept and R and Z given with units with appropriate powers of ten. $Z = \text{y-intercept} \times 5.8$ unit of R: Ω or V A^{-1} unit of Z: Ω or V A^{-1}	1
2(d)(ii)	Percentage uncertainty determined using $\Delta E = 0.2$ (V) with method shown. $\Delta R\% = \left(\frac{\Delta E}{E} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or $\Delta R\% = \left(\frac{0.2}{5.8} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$	1

Question	Answer	Marks
2(e)	<i>I</i> determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution. $I = \frac{1}{\frac{\text{gradient}}{20} + y\text{-intercept}}$ or $I = \frac{E}{\left(\frac{R}{20} + Z\right)}$	1



Cambridge International AS & A Level

PHYSICS

9702/41

Paper 4 A Level Structured Questions

October/November 2022

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **15** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

PUBLISHED**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance
 For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	$F = (Gm_1m_2) / r^2$	M1
	where G is the gravitational constant	A1
1(b)	gravitational force provides the centripetal force	B1
	$mR\omega^2 = GMm / R^2$ and $\omega = 2\pi / T$ or $mv^2 / R = GMm / R^2$ and $v = 2\pi R / T$ or $4\pi^2 mR / T^2 = GMm / R^2$	M1
	correct completion of algebra to get $T^2 = (4\pi^2 / GM) R^3$, with identification of $(4\pi^2 / GM)$ as k	A1
1(c)(i)	$(24 \times 3600)^2 = (4\pi^2 \times R^3) / (6.67 \times 10^{-11} \times 6.0 \times 10^{24})$	C1
	$R = 4.2 \times 10^7 \text{ m}$	A1
1(c)(ii)	(orbit) must be above the Equator	B1
	(direction) must be from west to east	B1

PUBLISHED

Question	Answer	Marks
2(a)	• resistance of a metal • volume of a gas at constant pressure • e.m.f. of a thermocouple <i>Any two points, 1 mark each</i>	B2
2(b)(i)	$Q = mc\Delta T$	C1
	evidence of realisation that Q lost by water = Q gained by mercury	C1
	$18.7 \times 4.18 \times (37.4 - T) = 6.94 \times 0.140 \times (T - 23.0)$	C1
	$T = 37.2\text{ }^{\circ}\text{C}$	A1
2(b)(ii)	use a liquid with a lower (specific) heat capacity (than mercury) or use a smaller mass of mercury	B1
2(c)(i)	depends on properties of a real substance	B1
	0 °C is not absolute zero	B1
2(c)(ii)	ideal gas	B1

PUBLISHED

Question	Answer	Marks
3(a)	$a = -\omega^2 x$	M1
	a = acceleration, x = displacement from equilibrium position and ω = angular frequency	A1
3(b)(i)	$x_0 = 0.12 \text{ m}$	A1
3(b)(ii)	$v = \omega\sqrt{(x_0^2 - x^2)}$ two (x, v) pairs correctly read from Fig. 3.2 (one may be $(x_0, 0)$ or value of x_0 from (i))	C1
	e.g. $0.20 = \omega\sqrt{(0.12^2 - 0)}$ leading to $\omega = 1.7 \text{ rad s}^{-1}$	A1
3(b)(iii)	$E = \frac{1}{2}M\omega^2 x_0^2$	C1
	$0.050 = \frac{1}{2} \times M \times 1.67^2 \times 0.12^2$ $M = 2.5 \text{ kg}$	A1
	or	
	$(E_K)_{\max} = \frac{1}{2}Mv_0^2$	(C1)
	$0.050 = \frac{1}{2} M \times 0.20^2$ $M = 2.5 \text{ kg}$	(A1)
3(c)(i)	loss of (total) energy (of system)	B1
	due to resistive forces	B1
3(c)(ii)	closed loop surrounding the origin with maximum x at $\pm 0.060 \text{ m}$ passing through $v = 0$	B1
	maximum velocity shown as $\pm 0.10 \text{ m s}^{-1}$ passing through $x = 0$	B1

PUBLISHED

Question	Answer	Marks
4(a)	(field line indicates) direction of force	B1
	force on a positive charge	B1
4(b)(i)	one straight line perpendicular to plates, starting on one plate and finishing on the other	B1
	five straight lines perpendicular to plates between the plates, uniformly spaced	B1
	downwards arrows on lines	B1
4(b)(ii)	$E = V / d$	C1
	$= 2400 / 0.046$	A1
	$= 5.2 \times 10^4 \text{ N C}^{-1}$	
4(c)(i)	smooth curve in region of field and straight line outside field	B1
	direction of deflection shown as downwards in region of field	B1
4(c)(ii)	helium nucleus has double the charge but four times the mass	B1
	velocity parallel to plates same and acceleration perpendicular to plates smaller (for helium)	B1
	final speed is lower (for helium)	B1

PUBLISHED

Question	Answer	Marks
5(a)(i)	$Q = CV$	C1
	$Q_0 = 24 \times 470 \times 10^{-6}$ $= 0.011 \text{ C}$	A1
5(a)(ii)	$I_0 = 24 / 5600$ $= 4.3 \times 10^{-3} \text{ A}$	A1
5(a)(iii)	$\tau = RC$	C1
	$= 5600 \times 470 \times 10^{-6}$ $= 2.6 \text{ s}$	A1
5(a)(iv)	line with negative gradient throughout passing through $(0, I_0)$	B1
	exponential decay curve asymptotic to t -axis	B1
5(b)(i)	current in wire P gives rise to a magnetic field	B1
	as current (in P) changes, wire Q cuts (magnetic) flux (of wire P)	B1
	cutting magnetic flux causes induced e.m.f. (across Q)	B1
5(b)(ii)	sketch shows line with a negative gradient throughout	B1

PUBLISHED

Question	Answer	Marks
6(a)(i)	PQRS and WXYZ	B1
6(a)(ii)	force on charge carriers is perpendicular to both (magnetic) field and current	B1
	as charge carriers are deflected to one side, an electric field is set up	B1
	(steady V_H when) electric and magnetic forces on charge carriers are equal (and opposite)	B1
6(b)(i)	n : number density of charge carriers	B1
	t : distance PW (or SZ or QX or RY)	B1
	q : charge on each charge carrier	B1
6(b)(ii)	V_H inversely proportional to t	B1
	(so t needs to be small for) V_H to be large enough to measure	B1

PUBLISHED

Question	Answer	Marks
7(a)(i)	peak voltage = $4.2 \times \sqrt{2}$ (= 5.9 V)	B1
	power = V^2 / R $= 5.9^2 / 760 = 0.046 \text{ W}$ or 46 mW	A1
7(a)(ii)	sketch shows peak(s) in power at 46 mW	B1
	correct shape (sinusoidal wave sitting on t -axis)	B1
	four cycles of repeating pattern shown, with $P = 0$ at 0, 10, 20, 30, 40 μs	B1
7(a)(iii)	line is symmetrical about 23 mW	B1
7(b)(i)	(alternating p.d. makes) the crystal vibrate	B1
	vibrations (of crystal) causes air to vibrate	B1
	frequency is in ultrasound range	B1
7(b)(ii)	(air makes) crystal vibrate, which causes an e.m.f. to be generated across the (second) crystal	B1

PUBLISHED

Question	Answer	Marks
8(a)	photon energy (to remove electron)	B1
	minimum energy to remove electron or energy to remove electron from surface or energy to remove electron with zero kinetic energy	B1
8(b)(i)	photon energy = hf	C1
	number per unit time = $8.36 \times 10^{-3} / (1.36 \times 10^{15} \times 6.63 \times 10^{-34})$ $= 9.27 \times 10^{15} \text{ s}^{-1}$	A1
8(b)(ii)	$hf = \phi + E_{\text{MAX}}$	C1
	$\phi = (1.36 \times 10^{15} \times 6.63 \times 10^{-34}) - (3.09 \times 10^{-19})$ $= 5.93 \times 10^{-19} \text{ J}$	A1
8(c)(i)	greater photon energy (and same work function)	M1
	so maximum kinetic energy is increased	A1
8(c)(ii)	(greater photon energy and same power so) lower number of photons (per unit time)	M1
	(each electron absorbs one photon) so lower rate of emission	A1

PUBLISHED

Question	Answer	Marks
9(a)	total power of radiation emitted (by the star)	B1
9(b)(i)	$F = L / (4\pi d^2)$	C1
	$= 9.86 \times 10^{27} / [4\pi \times (8.14 \times 10^{16})^2]$	A1
	$= 1.18 \times 10^{-7} \text{ W m}^{-2}$	
9(b)(ii)	$L = 4\pi\sigma r^2 T^4$	C1
	$9.86 \times 10^{27} = 4 \times \pi \times 5.67 \times 10^{-8} \times r^2 \times 9830^4$	
	radius = $1.22 \times 10^9 \text{ m}$	A1
9(c)	wavelength of peak intensity determined (from spectrum of star)	B1
	wavelength of peak intensity from object of known temperature determined	B1
	Wien's displacement law used or wavelength of peak intensity inversely proportional to temperature	B1

PUBLISHED

Question	Answer	Marks
10(a)(i)	cannot predict when a (particular) nucleus will decay or cannot predict which nucleus will decay next	B1
10(a)(ii)	not affected by external / environmental factors	B1
10(b)(i)	line fluctuates or trend is a straight line	B1
10(b)(ii)	straight line of best fit drawn on Fig. 10.1	B1
10(b)(iii)	$M = M_0 \exp(-\lambda t)$ so $\ln M = \ln M_0 - \lambda t$ so gradient = $-\lambda$ (and magnitude of gradient = λ)	B1
10(b)(iv)	gradient = $(-)(8.0 - 4.8) / (11.6 - 0)$ (allow any correct pair of values from Fig. 10.1)	C1
	$\lambda = 0.28 \text{ s}^{-1}$	A1
10(b)(v)	half-life = $0.693 / \lambda$ = $0.693 / 0.28$ = 2.5 s	A1
10(c)	(for reaction to occur,) energy is released	B1
	energy release comes from fall in mass so total mass of products must be less (than mass of carbon-15)	B1



Cambridge International AS & A Level

PHYSICS

9702/42

Paper 4 A Level Structured Questions

October/November 2022

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **16** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	force per unit mass	B1
1(b)(i)	lines drawn are radial from the surface	B1
	arrows show pointing towards planet	B1
1(b)(ii)	field lines show force (on satellite) is towards centre of planet or velocity of satellite is perpendicular to field lines	B1
	(gravitational) force perpendicular to velocity causes centripetal <u>acceleration</u>	B1
1(c)(i)	$T = 24 \text{ hours}$	C1
	$a = r\omega^2$ and $\omega = 2\pi / T$ or $a = v^2 / r$ and $v = 2\pi r / T$ or $a = 4\pi^2 r / T^2$	C1
	$a = (4\pi^2 \times 6.4 \times 10^6) / (24 \times 60 \times 60)^2$ $= 0.034 \text{ m s}^{-2}$	A1
1(c)(ii)	identification of the two forces acting on the object as gravitational force and (normal) contact force	M1
	gravitational force and normal contact force are in opposite directions, and their resultant causes the (centripetal) acceleration	A1

PUBLISHED

Question	Answer	Marks
2(a)	(thermal) energy per unit mass (to cause temperature change)	B1
	(thermal) energy per unit change in temperature	B1
2(b)(i)	work done correct (0)	B1
	increase in internal energy correct (+ E)	B1
2(b)(ii)	work done correct ($-W$) and increase in internal energy same as (b)(i)	B1
	thermal energy correct so that it adds to work done to give increase in internal energy	B1
2(c)	more thermal energy needed so specific heat capacity is greater	B1

PUBLISHED

Question	Answer	Marks
3(a)	p = pressure (of gas), V = volume (of gas) and k = Boltzmann constant	B1
	N = number of molecules	B1
	T = thermodynamic temperature	B1
3(b)	$(pV = NkT \text{ and } pV = \frac{1}{3}Nm\langle c^2 \rangle \text{ leading to } NkT = \frac{1}{3}Nm\langle c^2 \rangle)$	M1
	algebra leading to $(3/2)kT = \frac{1}{2}m\langle c^2 \rangle$ and use of $\frac{1}{2}m\langle c^2 \rangle = E_K$ leading to $(3/2)kT = E_K$	A1
3(c)(i)	$T = 296 \text{ K}$	C1
	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$	C1
	$\frac{1}{2} \times 5.31 \times 10^{-26} \times u^2 = (3/2) \times 1.38 \times 10^{-23} \times 296$	
	$u = 480 \text{ m s}^{-1}$	A1
3(c)(ii)	line passing through (P, u)	B1
	horizontal straight line	B1

PUBLISHED

Question	Answer	Marks
4(a)(i)	$x_0 = 8.0 \text{ cm}$	A1
4(a)(ii)	$\omega = 2\pi / T$	C1
	$= 2\pi / 4.0 = 1.6 \text{ rad s}^{-1}$	A1
4(a)(iii)	$E = \frac{1}{2}m\omega^2x_0^2$	C1
	$= \frac{1}{2} \times 36 \times 1.6^2 \times 0.080^2$	C1
	$= 0.29 \text{ J}$	A1
4(b)	dome-shaped curve, starting and ending at $E_K = 0$	B1
	maximum E_K shown as 0.29 J	B1
	position of peak shown at $h = 10.0 \text{ cm}$	B1
	line intercepts h -axis at $h = 2.0 \text{ cm}$ and at $h = 18.0 \text{ cm}$	B1

PUBLISHED

Question	Answer	Marks
5(a)	work done per unit charge	B1
	work done (on charge) in moving positive charge from infinity (to the point)	B1
5(b)(i)	radius = 0.060 m	A1
5(b)(ii)	$V = Q / 4\pi\epsilon_0 x$ $Q = (-) 850 \times 4\pi \times 8.85 \times 10^{-12} \times 0.060$ or $Q = (-) 850 \times 0.060 / 8.99 \times 10^9$ <i>(any correct pair of V and x values from curve)</i>	C1
	$Q = -5.7 \times 10^{-9} \text{ C}$	A1
5(c)(i)	$E_P = Q^2 / 4\pi\epsilon_0 x$ $= (5.67 \times 10^{-9})^2 / (4\pi \times 8.85 \times 10^{-12} \times 0.46)$	C1
	$= 6.3 \times 10^{-7} \text{ J}$	A1
5(c)(ii)	- force is repulsive so spheres move apart - force in direction of motion so speed increases - potential energy converted to kinetic energy so speed increases - force decreases with distance so acceleration decreases - momentum is conserved (at zero) (and masses are equal) so velocities are always equal and opposite <i>Any three points, 1 mark each</i>	B3

PUBLISHED

Question	Answer	Marks
6(a)	- p.d. across resistor = p.d. across capacitor - current (in resistor) proportional to p.d. across it - current causes capacitor to lose charge - charge (on capacitor) proportional to p.d. so p.d. decreases <i>Any two points, 1 mark each</i>	B2
	rate of change of p.d. decreases as p.d. decreases	B1
6(b)	$Q_0 = 0.90 \text{ mC}$ and at $t = \text{one time constant}$, $Q = Q_0 \exp(-1)$	B1
	at $t = \text{one time constant}$, $Q = 0.90 \exp(-1) = 0.33 \text{ mC}$	M1
	evidence of graph reading: when $Q = 0.33 \text{ mC}$, $t = 5.5 \text{ s}$	A1
	or	
	evidence of two correct sets of readings for Q and t from the graph	(B1)
	correct substitution of Q and t values into $Q_2 = Q_1 \exp[(t_1 - t_2) / \tau]$	(M1)
	calculation to give $\tau = 5.5 \text{ s}$	(A1)
	or	
	read-off of half-life as 3.75 s	(B1)
	use of $Q = Q_0 \exp(-t / \tau)$ to show that $\tau = \text{half-life} / \ln 2$	(M1)
	$\tau = 3.75 / \ln 2 = 5.4 \text{ s}$	(A1)

PUBLISHED

Question	Answer	Marks
6(b)	or	
	tangent drawn on $Q-t$ graph and value of Q at exact same time as tangent read from graph	(M1)
	gradient of tangent correctly calculated	(A1)
	$\tau = Q / \text{gradient}$ used to correctly calculate a value for τ as 5.5 s	(A1)
6(c)(i)	$C = Q / V$	C1
	$= [(0.90 \times 10^{-3}) / 7.5] = 1.2 \times 10^{-4} \text{ C}$	A1
	$= 120 \mu\text{F}$	
6(c)(ii)	$R = \tau / C$	C1
	$= 5.5 / (1.2 \times 10^{-4}) (= 45\,800 \Omega)$	A1
	$= 46 \text{ k}\Omega$	

PUBLISHED

Question	Answer	Marks
7(a)	force per unit current	M1
	force per unit length	M1
	current / wire is perpendicular to (magnetic) field (lines)	A1
7(b)(i)	current (in coil) is perpendicular to magnetic field (so force on wire)	B1
	force (on wire) is perpendicular to current and field (so is vertical) or current and field are both horizontal (so force is vertical)	B1
7(b)(ii)	$NBIL = mg$	C1
	$B = (2.16 \times 10^{-3} \times 9.81) / (40 \times 3.94 \times 0.0300)$	C1
	$= 4.48 \times 10^{-3} \text{ T}$	A1
7(b)(iii)	(magnetic) forces (on balance and newton meter) are (equal and) opposite	B1
	reading $= 0.563 - (2.16 \times 10^{-3} \times 9.81)$ $= 0.542 \text{ N}$	A1

PUBLISHED

Question	Answer	Marks
8(a)	direction of induced e.m.f.	M1
	such as to (produce effects that) oppose the change that caused it	A1
8(b)(i)	$X = 0.85 \text{ A}$	A1
	$Y = 2\pi / 0.040$	C1
	$= 160 \text{ rad s}^{-1}$	A1
8(b)(ii)	two cycles of a sinusoidal curve with a period of 0.040 s	B1
	correct phase (i.e. V_2 max / min at $t = 0, 0.02, 0.04, 0.06$ and 0.08 s , and V_2 zero at $t = 0.01, 0.03, 0.05, 0.07 \text{ s}$)	B1
	maximum / minimum V_2 shown (consistently) at $\pm 6.5 \text{ V}$	B1
8(b)(iii)	(magnitude of) V_2 is proportional to rate of change of (magnetic) flux	B1
	V_2 is proportional to <u>gradient</u> of I_1–t curve V_2 has maximum magnitude when I_1–t curve is steepest V_2 is zero when I_1–t curve is horizontal / a maximum or minimum V_2 changes sign when sign of gradient of I_1–t curve changes Any two points, 1 mark each	B2

Question	Answer	Marks
9(a)(i)	- energy of photon has a corresponding frequency - change in electron energy level emits a single photon - photon energy = difference in energy levels - discrete frequencies must have come from discrete energy gaps - discrete energy changes imply discrete energy levels <i>Any three points, 1 mark each</i>	B3
9(a)(ii)	transition (to -3.400 eV) from X corresponds to 658 nm line	C1
	$E_1 - E_2 = hc / \lambda$	C1
	$E_1 - (-3.400) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (658 \times 10^{-9} \times 1.60 \times 10^{-19})$ and so $E_1 = -1.51$ eV (<i>full substitution and answer needed</i>)	A1
9(b)(i)	redshift	B1
9(b)(ii)	moving away (from observer)	B1
9(b)(iii)	$\Delta\lambda / \lambda = v / c$ e.g. for 658 nm line: $\Delta\lambda = 686 - 658$ $(= 28 \text{ nm})$ (<i>other lines may be used</i>)	C1
	$28 / 658 = v / (3.00 \times 10^8)$ (<i>other lines may be used</i>)	C1
	$v = 1.3 \times 10^7 \text{ m s}^{-1}$	A1
9(c)	$v = H_0 d$	C1
	$H_0 = (1.3 \times 10^7) / (5.7 \times 10^{24})$ $= 2.3 \times 10^{-18} \text{ s}^{-1}$	A1

PUBLISHED

Question	Answer	Marks
10(a)(i)	introduction of tracer (into the body)	M1
	containing a β^+ emitter	A1
10(a)(ii)	positron interacts with electron	B1
	(pair) annihilation occurs	B1
	mass of particles converted into gamma photons	B1
10(b)	(annihilation of electron and positron) produces two photons	B1
	$E = (\Delta)mc^2$	B1
	$E = hf$ and $f = c / \lambda$ or $E = hc / \lambda$	B1
	$\lambda = \{[2 \times] 6.63 \times 10^{-34} \times 3.00 \times 10^8\} / \{[2 \times] 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2\}$ $= 2.4(3) \times 10^{-12} \text{ m or } 2.4(3) \text{ pm (full substitution and answer with unit needed)}$	B1



Cambridge International AS & A Level

PHYSICS

9702/43

Paper 4 A Level Structured Questions

October/November 2022

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **15** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	$F = (Gm_1m_2) / r^2$	M1
	where G is the gravitational constant	A1
1(b)	gravitational force provides the centripetal force	B1
	$mR\omega^2 = GMm / R^2$ and $\omega = 2\pi / T$ or $mv^2 / R = GMm / R^2$ and $v = 2\pi R / T$ or $4\pi^2 mR / T^2 = GMm / R^2$	M1
	correct completion of algebra to get $T^2 = (4\pi^2 / GM) R^3$, with identification of $(4\pi^2 / GM)$ as k	A1
1(c)(i)	$(24 \times 3600)^2 = (4\pi^2 \times R^3) / (6.67 \times 10^{-11} \times 6.0 \times 10^{24})$	C1
	$R = 4.2 \times 10^7 \text{ m}$	A1
1(c)(ii)	(orbit) must be above the Equator	B1
	(direction) must be from west to east	B1

PUBLISHED

Question	Answer	Marks
2(a)	• resistance of a metal • volume of a gas at constant pressure • e.m.f. of a thermocouple <i>Any two points, 1 mark each</i>	B2
2(b)(i)	$Q = mc\Delta T$	C1
	evidence of realisation that Q lost by water = Q gained by mercury	C1
	$18.7 \times 4.18 \times (37.4 - T) = 6.94 \times 0.140 \times (T - 23.0)$	C1
	$T = 37.2\text{ }^{\circ}\text{C}$	A1
2(b)(ii)	use a liquid with a lower (specific) heat capacity (than mercury) or use a smaller mass of mercury	B1
2(c)(i)	depends on properties of a real substance	B1
	$0\text{ }^{\circ}\text{C}$ is not absolute zero	B1
2(c)(ii)	ideal gas	B1

PUBLISHED

Question	Answer	Marks
3(a)	$a = -\omega^2 x$	M1
	a = acceleration, x = displacement from equilibrium position and ω = angular frequency	A1
3(b)(i)	$x_0 = 0.12 \text{ m}$	A1
3(b)(ii)	$v = \omega\sqrt{(x_0^2 - x^2)}$ two (x, v) pairs correctly read from Fig. 3.2 (one may be $(x_0, 0)$ or value of x_0 from (i))	C1
	e.g. $0.20 = \omega\sqrt{(0.12^2 - 0)}$ leading to $\omega = 1.7 \text{ rad s}^{-1}$	A1
3(b)(iii)	$E = \frac{1}{2}M\omega^2 x_0^2$	C1
	$0.050 = \frac{1}{2} \times M \times 1.67^2 \times 0.12^2$ $M = 2.5 \text{ kg}$	A1
	or	
	$(E_K)_{\max} = \frac{1}{2}Mv_0^2$	(C1)
	$0.050 = \frac{1}{2} M \times 0.20^2$ $M = 2.5 \text{ kg}$	(A1)
3(c)(i)	loss of (total) energy (of system)	B1
	due to resistive forces	B1
3(c)(ii)	closed loop surrounding the origin with maximum x at $\pm 0.060 \text{ m}$ passing through $v = 0$	B1
	maximum velocity shown as $\pm 0.10 \text{ m s}^{-1}$ passing through $x = 0$	B1

PUBLISHED

Question	Answer	Marks
4(a)	(field line indicates) direction of force	B1
	force on a positive charge	B1
4(b)(i)	one straight line perpendicular to plates, starting on one plate and finishing on the other	B1
	five straight lines perpendicular to plates between the plates, uniformly spaced	B1
	downwards arrows on lines	B1
4(b)(ii)	$E = V / d$	C1
	$= 2400 / 0.046$	A1
	$= 5.2 \times 10^4 \text{ N C}^{-1}$	
4(c)(i)	smooth curve in region of field and straight line outside field	B1
	direction of deflection shown as downwards in region of field	B1
4(c)(ii)	helium nucleus has double the charge but four times the mass	B1
	velocity parallel to plates same and acceleration perpendicular to plates smaller (for helium)	B1
	final speed is lower (for helium)	B1

PUBLISHED

Question	Answer	Marks
5(a)(i)	$Q = CV$	C1
	$Q_0 = 24 \times 470 \times 10^{-6}$ $= 0.011 \text{ C}$	A1
5(a)(ii)	$I_0 = 24 / 5600$ $= 4.3 \times 10^{-3} \text{ A}$	A1
5(a)(iii)	$\tau = RC$	C1
	$= 5600 \times 470 \times 10^{-6}$ $= 2.6 \text{ s}$	A1
5(a)(iv)	line with negative gradient throughout passing through $(0, I_0)$	B1
	exponential decay curve asymptotic to t -axis	B1
5(b)(i)	current in wire P gives rise to a magnetic field	B1
	as current (in P) changes, wire Q cuts (magnetic) flux (of wire P)	B1
	cutting magnetic flux causes induced e.m.f. (across Q)	B1
5(b)(ii)	sketch shows line with a negative gradient throughout	B1

PUBLISHED

Question	Answer	Marks
6(a)(i)	PQRS and WXYZ	B1
6(a)(ii)	force on charge carriers is perpendicular to both (magnetic) field and current	B1
	as charge carriers are deflected to one side, an electric field is set up	B1
	(steady V_H when) electric and magnetic forces on charge carriers are equal (and opposite)	B1
6(b)(i)	n : number density of charge carriers	B1
	t : distance PW (or SZ or QX or RY)	B1
	q : charge on each charge carrier	B1
6(b)(ii)	V_H inversely proportional to t	B1
	(so t needs to be small for) V_H to be large enough to measure	B1

PUBLISHED

Question	Answer	Marks
7(a)(i)	peak voltage = $4.2 \times \sqrt{2}$ (= 5.9 V)	B1
	power = V^2 / R $= 5.9^2 / 760 = 0.046 \text{ W}$ or 46 mW	A1
7(a)(ii)	sketch shows peak(s) in power at 46 mW	B1
	correct shape (sinusoidal wave sitting on t -axis)	B1
	four cycles of repeating pattern shown, with $P = 0$ at 0, 10, 20, 30, 40 μs	B1
7(a)(iii)	line is symmetrical about 23 mW	B1
7(b)(i)	(alternating p.d. makes) the crystal vibrate	B1
	vibrations (of crystal) causes air to vibrate	B1
	frequency is in ultrasound range	B1
7(b)(ii)	(air makes) crystal vibrate, which causes an e.m.f. to be generated across the (second) crystal	B1

PUBLISHED

Question	Answer	Marks
8(a)	photon energy (to remove electron)	B1
	minimum energy to remove electron or energy to remove electron from surface or energy to remove electron with zero kinetic energy	B1
8(b)(i)	photon energy = hf	C1
	number per unit time = $8.36 \times 10^{-3} / (1.36 \times 10^{15} \times 6.63 \times 10^{-34})$ $= 9.27 \times 10^{15} \text{ s}^{-1}$	A1
8(b)(ii)	$hf = \phi + E_{\text{MAX}}$	C1
	$\phi = (1.36 \times 10^{15} \times 6.63 \times 10^{-34}) - (3.09 \times 10^{-19})$ $= 5.93 \times 10^{-19} \text{ J}$	A1
8(c)(i)	greater photon energy (and same work function)	M1
	so maximum kinetic energy is increased	A1
8(c)(ii)	(greater photon energy and same power so) lower number of photons (per unit time)	M1
	(each electron absorbs one photon) so lower rate of emission	A1

PUBLISHED

Question	Answer	Marks
9(a)	total power of radiation emitted (by the star)	B1
9(b)(i)	$F = L / (4\pi d^2)$	C1
	$= 9.86 \times 10^{27} / [4\pi \times (8.14 \times 10^{16})^2]$	A1
	$= 1.18 \times 10^{-7} \text{ W m}^{-2}$	
9(b)(ii)	$L = 4\pi\sigma r^2 T^4$	C1
	$9.86 \times 10^{27} = 4 \times \pi \times 5.67 \times 10^{-8} \times r^2 \times 9830^4$	
	radius = $1.22 \times 10^9 \text{ m}$	A1
9(c)	wavelength of peak intensity determined (from spectrum of star)	B1
	wavelength of peak intensity from object of known temperature determined	B1
	Wien's displacement law used or wavelength of peak intensity inversely proportional to temperature	B1

PUBLISHED

Question	Answer	Marks
10(a)(i)	cannot predict when a (particular) nucleus will decay or cannot predict which nucleus will decay next	B1
10(a)(ii)	not affected by external / environmental factors	B1
10(b)(i)	line fluctuates or trend is a straight line	B1
10(b)(ii)	straight line of best fit drawn on Fig. 10.1	B1
10(b)(iii)	$M = M_0 \exp(-\lambda t)$ so $\ln M = \ln M_0 - \lambda t$ so gradient = $-\lambda$ (and magnitude of gradient = λ)	B1
10(b)(iv)	gradient = $(-)(8.0 - 4.8) / (11.6 - 0)$ (allow any correct pair of values from Fig. 10.1)	C1
	$\lambda = 0.28 \text{ s}^{-1}$	A1
10(b)(v)	half-life = $0.693 / \lambda$ = $0.693 / 0.28$ = 2.5 s	A1
10(c)	(for reaction to occur,) energy is released	B1
	energy release comes from fall in mass so total mass of products must be less (than mass of carbon-15)	B1



Cambridge International AS & A Level

PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

October/November 2022

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	A is the independent variable and s is the dependent variable or vary A and measure s	1
	keep B and t <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • pin / rod through hole • supported by a stand • sheet able to oscillate freely • at least one label from copper/sheet, hole, clamp, stand, rod, pin.	1
	drawn <u>clamped</u> rule(r) parallel to the direction of the oscillations (by eye) (to measure s)	1
	use rule(r) to measure lengths to determine A and A = length × breadth	1
	use of micrometer to measure t	1
	Method of Analysis	
	plot a graph of ln s against A or equivalent	1
	relationship valid <u>if</u> a straight line (with y-intercept = ln s ₀) is produced	1
	$K = -\frac{\text{gradient}}{Bt}$ $(K = -\frac{1}{Bt \times \text{gradient}} \text{ for A against ln s})$	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	use of cushion/sand box <u>in case sheet falls</u> or use gloves to protect hands from <u>cuts / sharp edges</u>	
D2	keep (initial) distance between (copper) sheet and (poles of) magnet <u>constant</u> or keep (initial) distance between (copper) sheet and coil(s) <u>constant</u>	
D3	keep s_0 <u>constant</u>	
D4	method to ensure s_0 is constant, e.g. initially line up (corner of) plate with fiducial marker / vertical pin to keep s_0 constant	
D5	method to determine s using video camera: • rule(r) in a position to measure s in the diagram • video camera shown in diagram or description of use of video camera • playback video recording by frame by frame / slow motion (to measure s)	
D6	repeat measurements of t <u>in different positions</u> and <u>average</u> t	
D7	measure B /magnetic flux density using a (calibrated) Hall probe	
D8	additional detail on use of Hall probe, e.g. adjust probe until maximum value or measure B using Hall probe first in one direction and then in the opposite direction and average	
D9	drawn method to create a magnetic field perpendicular to the area of the sheet, e.g. pair of magnets/horseshoe magnet/pair of coils connected to a (d.c.) supply	
D10	repeat experiment for each A and average s	

PUBLISHED

Question	Answer	Marks							
2(a)	$\text{gradient} = \frac{\pi Y Z d^2}{4\rho}$	1							
2(b)	<table><tr><td>$\frac{1}{R} / 10^{-3} \Omega^{-1}$</td></tr><tr><td>45 or 45.5</td></tr><tr><td>37 or 37.0</td></tr><tr><td>30 or 30.3</td></tr><tr><td>26 or 25.6</td></tr><tr><td>21 or 21.3</td></tr><tr><td>19 or 18.5</td></tr></table>	$\frac{1}{R} / 10^{-3} \Omega^{-1}$	45 or 45.5	37 or 37.0	30 or 30.3	26 or 25.6	21 or 21.3	19 or 18.5	1
$\frac{1}{R} / 10^{-3} \Omega^{-1}$									
45 or 45.5									
37 or 37.0									
30 or 30.3									
26 or 25.6									
21 or 21.3									
19 or 18.5									
	Absolute uncertainties in $\frac{1}{R}$ from ± 2 to ± 0.9 or ± 1 .	1							
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1							
	Error bars in $\frac{1}{R}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1							
2(c)(ii)	Straight line of best fit drawn. Points must be balanced. Do not accept line from top point to bottom point. Line must pass between (22.0, 30.0) and (23.0, 30.0) and (40.5, 65.0) and (42.0, 65.0).	1							
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1							

Question	Answer	Marks
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(d)	0.261 ± 0.003 (mm)	1
2(e)(i)	ρ determined using gradient and ρ given to two or three significant figures. $\rho = \frac{\pi Y Z d^2}{4 \times \text{gradient}} = \frac{\pi \times 22 \times 22 \times (\mathbf{d})^2}{4 \times (\mathbf{c})(\text{iii})}$	1
	ρ determined using gradient and given with correct SI unit ($\Omega \text{ m}$) and correct power of ten	1
2(e)(ii)	percentage uncertainty in ρ : percentage uncertainty = $\left(\frac{2 \times \Delta d}{d} + \frac{\Delta \text{gradient}}{\text{gradient}} + 0.05 + 0.05 \right) \times 100$ or correct substitution for max/min methods $\text{max } \rho = \frac{\pi \times (1.05 \times 22) \times (1.05 \times 22) \times (d + \Delta d)^2}{4 \times \text{min gradient}}$ $\text{min } \rho = \frac{\pi \times (0.95 \times 22) \times (0.95 \times 22) \times (d - \Delta d)^2}{4 \times \text{max gradient}}$	1

PUBLISHED

Question	Answer	Marks
2(f)	R determined to at least two significant figures from (c)(iii) or (d) and (e)(i) with correct substitution seen. $R = \frac{\text{gradient}}{0.950}$ or $R = \frac{\pi Y Z d^2}{4 \rho L} = \frac{\pi \times 22 \times 22 \times (\mathbf{d})^2}{4 \times (\mathbf{e})(\mathbf{i}) \times 0.950}$	1
	Absolute uncertainty in R determined. Method must be consistent with determination of R and correct substitution must be seen. for R determined using the gradient: $\Delta R = \frac{\Delta \text{gradient}}{\text{gradient}} \times R$ or for R determined using (d) and (e)(i): $\Delta R = \left(\frac{2 \times \Delta d}{d} + \frac{\Delta \rho}{\rho} + 0.05 + 0.05 \right) \times R$ or correct substitution for max/min methods: $\text{max } R = \frac{\pi \times (1.05 \times 22) \times (1.05 \times 22) \times (d + \Delta d)^2}{4 \times \text{min } \rho \times 0.950}$ $\text{min } R = \frac{\pi \times (0.95 \times 22) \times (0.95 \times 22) \times (d - \Delta d)^2}{4 \times \text{max } \rho \times 0.950}$	1



Cambridge International AS & A Level

PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

October/November 2022

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1	Defining the problem	
	z is the independent variable and t is the dependent variable or vary z and measure t	1
	keep B and A <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • pin/rod through hole • supported by a stand • sheet able to oscillate freely • at least one label from copper/sheet, hole, clamp stand, rod, pin	1
	use of stop-watch/timer to measure t (from release to stopping) or use of stop-watch/timer to measure time for the sheet (to stop) oscillating	1
	use of micrometer to measure z	1
	use of rule(r) to measure lengths to determine A and A = length × breadth	1
	Method of Analysis	
	plot a graph of lg t against lg z or equivalent (e.g. ln t against ln z)	1
	q = gradient	1
	$K = AB\rho \times 10^{y\text{-intercept}}$ ($K = AB\rho \times e^{y\text{-intercept}}$ for ln t against ln z)	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	use of cushion / sand box <u>in case sheet falls</u> or use gloves to protect hands from <u>cuts / sharp edges</u>	
D2	keep (initial) distance between (copper) sheet and (poles of) magnet <u>constant</u> or keep (initial) distance between (copper) sheet and coil(s) <u>constant</u>	
D3	keep initial displacement (of copper sheet) <u>constant</u>	
D4	method to ensure initial displacement (of copper sheet) is constant, e.g. initially line up (corner of) plate with fiducial marker/vertical pin	
D5	relationship valid <u>if</u> a straight line (with y-intercept = $\log\left(\frac{K}{AB\rho}\right)$) is produced	
D6	repeat measurements of z <u>in different positions</u> and <u>average</u> z	
D7	measure B / magnetic flux density using a (calibrated) Hall probe	
D8	additional detail on use of Hall probe, e.g. adjust (position of) probe until maximum value or measure B using Hall probe first in one direction and then in the opposite direction and average	
D9	drawn method to create a magnetic field perpendicular to the area of the sheet, e.g. pair of magnets/horseshoe magnet/pair of coils connected to a (d.c.) supply	
D10	repeat experiment for each z and average t	
D11	method to determine ρ , e.g. measure mass with balance and volume = Az and density = mass / volume	

Question	Answer	Marks														
2(a)	$\text{gradient} = \frac{c}{2h}$ $\text{y-intercept} = -\frac{c}{4h}$	1														
2(b)	<table><tr><th>T / ms</th><th>f / Hz</th></tr><tr><td>7.0 or 7.00 ± 1</td><td>140 or 143 $\pm (10\text{--}30)$</td></tr><tr><td>2.9 or 2.90 ± 0.2</td><td>340 or 345 $\pm (20\text{--}30)$</td></tr><tr><td>1.8 or 1.80 ± 0.1</td><td>560 or 556 ± 30</td></tr><tr><td>1.4 or 1.35 ± 0.1</td><td>710 or 714 $\pm (40\text{--}60)$ or 740 or 741</td></tr><tr><td>1.1 or 1.05 ± 0.1</td><td>910 or 909 $\pm (80\text{--}100)$ or 950 or 952</td></tr><tr><td>0.88 or 0.880 ± 0.02</td><td>1100 or 1140 $\pm (30\text{--}60)$</td></tr></table> Values of T and f correct as shown above. Absolute uncertainties in T and f correct as shown above.	T / ms	f / Hz	7.0 or 7.00 ± 1	140 or 143 $\pm (10\text{--}30)$	2.9 or 2.90 ± 0.2	340 or 345 $\pm (20\text{--}30)$	1.8 or 1.80 ± 0.1	560 or 556 ± 30	1.4 or 1.35 ± 0.1	710 or 714 $\pm (40\text{--}60)$ or 740 or 741	1.1 or 1.05 ± 0.1	910 or 909 $\pm (80\text{--}100)$ or 950 or 952	0.88 or 0.880 ± 0.02	1100 or 1140 $\pm (30\text{--}60)$	1
T / ms	f / Hz															
7.0 or 7.00 ± 1	140 or 143 $\pm (10\text{--}30)$															
2.9 or 2.90 ± 0.2	340 or 345 $\pm (20\text{--}30)$															
1.8 or 1.80 ± 0.1	560 or 556 ± 30															
1.4 or 1.35 ± 0.1	710 or 714 $\pm (40\text{--}60)$ or 740 or 741															
1.1 or 1.05 ± 0.1	910 or 909 $\pm (80\text{--}100)$ or 950 or 952															
0.88 or 0.880 ± 0.02	1100 or 1140 $\pm (30\text{--}60)$															
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1														
	Error bars in f plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1														

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Points must be balanced. Do not accept line from top point to bottom point. Line must pass between (2.20, 400) and (2.40, 400) and (5.20, 1000) and (5.60, 1000).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(d)	83.2 ± 0.3 (cm)	1
2(e)(i)	c determined using gradient and c given to two or three significant figures. $c = 2 \times h \times \text{gradient} = 2 \times \text{(d)} \times \text{(c)(iii)}$	1
	c determined using gradient and given with correct SI unit and correct power of ten: ms^{-1} or cms^{-1} .	1
2(e)(ii)	Percentage uncertainty in c from (c)(iii) and (d) with method shown. percentage uncertainty = $\left(\frac{\Delta h}{h} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or correct substitution for max/min methods: max $c = 2 \times \text{max } h \times \text{max gradient}$ min $c = 2 \times \text{min } h \times \text{min gradient}$	1

PUBLISHED

Question	Answer	Marks
2(f)	h determined to at least two significant figures from (e)(i) with correct substitution. $h = \frac{3 \times \mathbf{(e)(i)}}{4 \times 130}$	1
	Absolute uncertainty in h determined. Correct substitution must be seen. $\Delta h = \left(\frac{\Delta f}{f} + \frac{\Delta c}{c} \right) \times h = \left(\frac{5}{130} + \frac{\Delta c}{c} \right) \times h$ or correct substitution for max/min methods: $\max h = \frac{3 \times \max c}{4 \times \min f} = \frac{3 \times \max \mathbf{(e)(i)}}{4 \times 125}$ $\min h = \frac{3 \times \min c}{4 \times \max f} = \frac{3 \times \min \mathbf{(e)(i)}}{4 \times 135}$	1



Cambridge International AS & A Level

PHYSICS

9702/53

Paper 5 Planning, Analysis and Evaluation

October/November 2022

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	A is the independent variable and s is the dependent variable or vary A and measure s	1
	keep B and t <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • pin / rod through hole • supported by a stand • sheet able to oscillate freely • at least one label from copper/sheet, hole, clamp, stand, rod, pin.	1
	drawn <u>clamped</u> rule(r) parallel to the direction of the oscillations (by eye) (to measure s)	1
	use rule(r) to measure lengths to determine A and A = length × breadth	1
	use of micrometer to measure t	1
	Method of Analysis	
	plot a graph of ln s against A or equivalent	1
	relationship valid <u>if</u> a straight line (with y-intercept = ln s ₀) is produced	1
	$K = -\frac{\text{gradient}}{Bt}$ $(K = -\frac{1}{Bt \times \text{gradient}} \text{ for A against ln s})$	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	use of cushion/sand box <u>in case sheet falls</u> or use gloves to protect hands from <u>cuts / sharp edges</u>	
D2	keep (initial) distance between (copper) sheet and (poles of) magnet <u>constant</u> or keep (initial) distance between (copper) sheet and coil(s) <u>constant</u>	
D3	keep s_0 <u>constant</u>	
D4	method to ensure s_0 is constant, e.g. initially line up (corner of) plate with fiducial marker / vertical pin to keep s_0 constant	
D5	method to determine s using video camera: - rule(r) in a position to measure s in the diagram - video camera shown in diagram or description of use of video camera - playback video recording by frame by frame / slow motion (to measure s)	
D6	repeat measurements of t <u>in different positions</u> and <u>average</u> t	
D7	measure B /magnetic flux density using a (calibrated) Hall probe	
D8	additional detail on use of Hall probe, e.g. adjust probe until maximum value or measure B using Hall probe first in one direction and then in the opposite direction and average	
D9	drawn method to create a magnetic field perpendicular to the area of the sheet, e.g. pair of magnets/horseshoe magnet/pair of coils connected to a (d.c.) supply	
D10	repeat experiment for each A and average s	

PUBLISHED

Question	Answer	Marks							
2(a)	$\text{gradient} = \frac{\pi Y Z d^2}{4\rho}$	1							
2(b)	<table><tr><td>$\frac{1}{R} / 10^{-3} \Omega^{-1}$</td></tr><tr><td>45 or 45.5</td></tr><tr><td>37 or 37.0</td></tr><tr><td>30 or 30.3</td></tr><tr><td>26 or 25.6</td></tr><tr><td>21 or 21.3</td></tr><tr><td>19 or 18.5</td></tr></table>	$\frac{1}{R} / 10^{-3} \Omega^{-1}$	45 or 45.5	37 or 37.0	30 or 30.3	26 or 25.6	21 or 21.3	19 or 18.5	1
$\frac{1}{R} / 10^{-3} \Omega^{-1}$									
45 or 45.5									
37 or 37.0									
30 or 30.3									
26 or 25.6									
21 or 21.3									
19 or 18.5									
	Absolute uncertainties in $\frac{1}{R}$ from ± 2 to ± 0.9 or ± 1 .	1							
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1							
	Error bars in $\frac{1}{R}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1							
2(c)(ii)	Straight line of best fit drawn. Points must be balanced. Do not accept line from top point to bottom point. Line must pass between (22.0, 30.0) and (23.0, 30.0) and (40.5, 65.0) and (42.0, 65.0).	1							
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1							

Question	Answer	Marks
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient of worst acceptable line determined with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(d)	0.261 ± 0.003 (mm)	1
2(e)(i)	ρ determined using gradient and ρ given to two or three significant figures. $\rho = \frac{\pi Y Z d^2}{4 \times \text{gradient}} = \frac{\pi \times 22 \times 22 \times (\mathbf{d})^2}{4 \times (\mathbf{c})(\text{iii})}$	1
	ρ determined using gradient and given with correct SI unit ($\Omega \text{ m}$) and correct power of ten	1
2(e)(ii)	percentage uncertainty in ρ : percentage uncertainty = $\left(\frac{2 \times \Delta d}{d} + \frac{\Delta \text{gradient}}{\text{gradient}} + 0.05 + 0.05 \right) \times 100$ or correct substitution for max/min methods $\text{max } \rho = \frac{\pi \times (1.05 \times 22) \times (1.05 \times 22) \times (d + \Delta d)^2}{4 \times \text{min gradient}}$ $\text{min } \rho = \frac{\pi \times (0.95 \times 22) \times (0.95 \times 22) \times (d - \Delta d)^2}{4 \times \text{max gradient}}$	1

PUBLISHED

Question	Answer	Marks
2(f)	R determined to at least two significant figures from (c)(iii) or (d) and (e)(i) with correct substitution seen. $R = \frac{\text{gradient}}{0.950}$ or $R = \frac{\pi Y Z d^2}{4 \rho L} = \frac{\pi \times 22 \times 22 \times (\mathbf{d})^2}{4 \times (\mathbf{e})(\mathbf{i}) \times 0.950}$	1
	Absolute uncertainty in R determined. Method must be consistent with determination of R and correct substitution must be seen. for R determined using the gradient: $\Delta R = \frac{\Delta \text{gradient}}{\text{gradient}} \times R$ or for R determined using (d) and (e)(i): $\Delta R = \left(\frac{2 \times \Delta d}{d} + \frac{\Delta \rho}{\rho} + 0.05 + 0.05 \right) \times R$ or correct substitution for max/min methods: $\text{max } R = \frac{\pi \times (1.05 \times 22) \times (1.05 \times 22) \times (d + \Delta d)^2}{4 \times \text{min } \rho \times 0.950}$ $\text{min } R = \frac{\pi \times (0.95 \times 22) \times (0.95 \times 22) \times (d - \Delta d)^2}{4 \times \text{max } \rho \times 0.950}$	1



Cambridge International AS & A Level

PHYSICS

9702/41

Paper 4 A Level Structured Questions

October/November 2023

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)(i)	direction of the force acting on a (test) mass placed at the point	B1
1(a)(ii)	change in height negligible compared with radius (of Earth)	B1
	(so) field lines are (effectively) parallel	B1
1(b)(i)	$Y = GM / R^2$	M1
	G is the gravitational constant	A1
1(b)(ii)	gravitational force is (always) attractive or gravitational force (always) acts towards the centre of the sphere	B1
	force is in opposite direction to displacement or at a point to the right of the centre, force acts to the left or at a point to the left of the centre, force acts to the right	B1
1(b)(iii)	sketch: smooth curve with decreasing positive gradient, starting at $(R, -Y)$ and reaching $3R$ with g still negative or smooth curve with increasing positive gradient, ending at $(-R, Y)$ and reaching $-3R$ with g still positive	B1
	both of the above curves, in correct quadrants	B1
	curve passing through $(\pm 2R, \pm 0.25Y)$ and $(\pm 3R, \pm 0.11Y)$	B1

PUBLISHED

Question	Answer	Marks
2(a)	(thermal) energy per unit mass (to change temperature)	B1
	(thermal) energy per unit change in temperature	B1
2(b)(i)	work done = $p\Delta V$ $= (2.0 \times 10^5) \times (0.063 - 0.038) = 5000 \text{ J}$	A1
2(b)(ii)	gas is expanding (against external pressure)	B1
	gas does work / work is done by gas, so (work done on gas is) negative	B1
2(b)(iii)	$\Delta U = q + W$	C1
	$7600 = q + (-5000)$	A1
	$q = 12\,600 \text{ J}$	
2(b)(iv)	specific heat capacity = $q / m\Delta T$ $= 12600 / (0.35 \times 56)$	C1
	$= 640 \text{ J kg}^{-1} \text{ K}^{-1}$	A1
2(c)	same gain in internal energy so same temperature rise	B1
	no change in volume so no work done or no work done so less thermal energy needed (for same change in internal energy)	B1
	less thermal energy needed (for same temperature change) so lower specific heat capacity	B1

PUBLISHED

Question	Answer	Marks
3(a)(i)	N : number of molecules (of the gas)	B1
	m : mass of one molecule (of the gas)	B1
	$\langle c^2 \rangle$: mean square speed (of molecules)	B1
3(a)(ii)	$pV = NkT$	M1
	$NkT = \frac{1}{3}Nm\langle c^2 \rangle$ and $E_K = \frac{1}{2}m\langle c^2 \rangle$ leading to $E_K = (3/2) kT$	A1
3(b)	$\frac{1}{2} \times 3.34 \times 10^{-27} \times 9300^2 = (3/2) \times 1.38 \times 10^{-23} \times T$	C1
	$T = 6980 \text{ K}$	A1
3(c)(i)	$L = F \times 4\pi d^2$	C1
	$L = 2.52 \times 10^{-8} \times 4\pi \times (4.16 \times 10^{16})^2$	A1
	$= 5.48 \times 10^{26} \text{ W}$	
3(c)(ii)	$L = 4\pi\sigma r^2 T^4$	C1
	$5.48 \times 10^{26} = 4\pi \times 5.67 \times 10^{-8} \times r^2 \times 6980^4$	
	$r = 5.69 \times 10^8 \text{ m}$	A1
3(d)	(very high pressure so) molecules are (very) close together (<i>not just 'nearer'</i>)	B1
	forces between molecules are not negligible or volume of molecules not negligible compared with gas volume	B1

PUBLISHED

Question	Answer	Marks
4(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): acceleration is (always) in the opposite <u>direction</u> to displacement or acceleration is (always) directed towards a fixed point	B1
4(b)(i)	$\omega = 2\pi / T$	C1
	$\omega = 2\pi / 3.0$ $= 2.1 \text{ rad s}^{-1}$	A1
4(b)(ii)	$E = \frac{1}{2}m\omega^2x_0^2$	C1
	$= \frac{1}{2} \times 0.81 \times 2.1^2 \times 0.036^2$ $= 2.3 \times 10^{-3} \text{ J}$	A1
4(c)	sketch: line starting at (0, 0.036) and not reaching $x = \pm 0.036 \text{ m}$ at any other time	B1
	smooth curve, with no sudden changes in gradient, showing continuously decreasing magnitude of x from maximum displacement at $t = 0$ to final displacement of zero where the gradient is also zero	B1
	displacement reaches final value of zero between $t = 0.75 \text{ s}$ and $t = 3.0 \text{ s}$ at the latest	B1

PUBLISHED

Question	Answer	Marks
5(a)	work done per unit charge	B1
	work (done on charge) moving positive charge from infinity (to the point)	B1
5(b)	<i>Any three points from:</i> <i>Up to 2 points from:</i> • radius of sphere X is 2.0 m • radius of sphere Y is 4.0 m • radius of Y is double the radius of X <i>Up to 2 points from:</i> • charge on X is negative • charge on Y is positive • spheres carry opposite charges <i>Up to 1 point from:</i> • magnitudes of charges on the spheres are equal	B3
5(c)	particle is attracted to X or repelled from Y or resultant force on particle is towards X / away from Y / to the left	B1
	particle accelerates towards X / away from Y / to the left	B1
	(magnitude of) acceleration of particle increases	B1

PUBLISHED

Question	Answer	Marks
6(a)	• force per unit length • force per unit current • length / current perpendicular to field <i>1 mark for any two points, 2 marks for all three points</i>	B2
6(b)(i)	into the page	B1
6(b)(ii)	$F = Bqv$	C1
	$= 4.8 \times 10^{-3} \times 1.6 \times 10^{-19} \times 1.7 \times 10^7 = 1.3 \times 10^{-14} \text{ N}$	A1
6(b)(iii)	arrow at point X pointing down the page	B1
6(b)(iv)	$F = mv^2 / r$	C1
	$1.3 \times 10^{-14} = (9.11 \times 10^{-31}) \times (1.7 \times 10^7)^2 / r$	C1
	$(r = 0.020 \text{ m})$ $d = 2r$ $d = 0.040 \text{ m}$	A1
6(c)	path shows upwards deflection such that the curvature is always anticlockwise within the field	B1
	circular path with larger radius	B1
	line enters field at X and leaves field at distance $2d$ vertically from X	B1

PUBLISHED

Question	Answer	Marks
7(a)(i)	$P = I_0^2 R$	A1
7(a)(ii)	$P = 4I_0^2 R$	A1
7(b)	sketch: square wave of period T , with P always non-zero	B1
	horizontal lines, from 0 to $0.5T$ and from $1.0T$ to $1.5T$, all at the same level that the scale indicates to be $I_0^2 R$	B1
	horizontal lines, from $0.5T$ to $1.0T$ and from $1.5T$ to $2.0T$, at a level that is four times higher than the lower lines	B1
7(c)(i)	$\langle P \rangle = (5/2)I_0^2 R$	A1
7(c)(ii)	$\langle P \rangle = I_{\text{r.m.s.}}^2 R$	C1
	$I_{\text{r.m.s.}}^2 R = (5/2)I_0^2 R$	A1
	$I_{\text{r.m.s.}} = \sqrt{(5/2)} I_0$	

PUBLISHED

Question	Answer	Marks
8(a)(i)	$p = E / c$	M1
	$E = hc / \lambda$ and completion of algebra leading to $p = h / \lambda$	A1
8(a)(ii)	wavelength = $(6.63 \times 10^{-34}) / (9.5 \times 10^{-28}) = 700 \times 10^{-9} \text{ m}$ so red	B1
8(b)(i)	power = intensity $\times$ area	C1
	number per unit time = $(160 \times 2.5 \times 10^{-6}) / (9.5 \times 10^{-28} \times 3.00 \times 10^8) = 1.4 \times 10^{15} \text{ s}^{-1}$	A1
8(b)(ii)	pressure = force / area	C1
	force = rate of change of momentum $= 2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}$	C1
	pressure = $(2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}) / (2.5 \times 10^{-6})$ $= 1.1 \times 10^{-6} \text{ Pa}$	A1
8(c)	photons have greater momentum or fewer photons per unit time	B1
	greater photon momentum but smaller number of photons (per unit time) so pressure is the same	B1

PUBLISHED

Question	Answer	Marks
9(a)	(two small) nuclei join together	M1
	to form one larger nucleus	A1
9(b)	line with a peak at $A \approx 56$	B1
	line with steep initial positive gradient on the left of peak and shallower negative gradient at all points to the right of peak and line does not return to 0 binding energy	B1
9(c)(i)	X shown at value of A to the right of the peak	B1
9(c)(ii)	Y shown at value of A close to 1	B1
9(d)	energy from 1 nucleus = $(1.77 \times 10^{13}) / (6.02 \times 10^{23})$ $(= 2.94 \times 10^{-11} \text{ J})$	C1
	binding energy of Z = $[(1.25 + 1.81) \times 10^{-10}] - 2.94 \times 10^{-11}$ $(= 2.77 \times 10^{-10} \text{ J})$	C1
	nucleon number of Z = $93 + 139 + 2 - 1$ $(= 233)$	C1
	binding energy per nucleon = $(2.77 \times 10^{-10}) / (233 \times 1.60 \times 10^{-13})$ $= 7.43 \text{ MeV}$	A1

PUBLISHED

Question	Answer	Marks
10(a)	ultrasound production: vibrating quartz crystal	B1
	X-ray production: electrons hitting metal target	B1
	ultrasound detected wave: reflected	B1
	X-ray detected wave: transmitted	B1
10(b)(i)	$I = I_0 \exp(-\mu x)$	C1
	$\ln(0.72) = -6.2\mu$	A1
	$\mu = 0.053 \text{ cm}^{-1}$	
10(b)(ii)	$I / I_0 = \exp(-9.3 \times 0.053)$ (= 0.61)	C1
	percentage attenuated = $100 \times (1.00 - 0.61)$ = 39%	A1



Cambridge International AS & A Level

PHYSICS

9702/42

Paper 4 A Level Structured Questions

October/November 2023

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance
 For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	angle (subtended at centre of circle) when arc length = radius	B1
1(b)	$\omega = 2\pi / T$	C1
	$= 2\pi / (1.0 \times 60 \times 60)$ $= 1.7 \times 10^{-3} \text{ rad s}^{-1}$	A1
1(c)(i)	angle $= 1.7 \times 10^{-3} \times 1400$ $= 2.4 \text{ rad}$	A1
1(c)(ii)	$L = \text{arc length} / \text{angle}$ $= 0.44 / 2.4$ or $L = 0.44 \times (3600 / 1400) / 2\pi$	C1
	$L = 0.18 \text{ m}$	A1
1(c)(iii)	$a = r\omega^2$	C1
	$= 0.18 \times (1.745 \times 10^{-3})^2$ $= 5.5 \times 10^{-7} \text{ m s}^{-2}$	A1
1(d)	centripetal acceleration is negligible compared with acceleration of free fall or numerical comparison establishing answer to (c)(iii) $\ll 9.81$	B1
	resultant force is negligible compared with weight (of modelling clay) (so variation is negligible) or force exerted by minute hand (approximately) equal (and opposite) to weight of modelling clay	B1

PUBLISHED

Question	Answer	Marks
2(a)(i)	work done per unit mass	B1
	work (done) moving mass from infinity (to the point)	B1
2(a)(ii)	$\phi = -GM/r$ $= -(6.67 \times 10^{-11} \times 7.3 \times 10^{22}) / (1.7 \times 10^6)$	C1
	$= -2.9 \times 10^6 \text{ J kg}^{-1}$	A1
2(b)(i)	$E_P = m\phi$	B1
2(b)(ii)	$\frac{1}{2}mv^2 + m\phi = 0$	M1
	correct algebra leading to $v = \sqrt{-2\phi}$	A1
2(c)	speed $= \sqrt{2 \times 2.9 \times 10^6}$ $= 2400 \text{ m s}^{-1}$	A1
2(d)	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$	C1
	$3.34 \times 10^{-27} \times \langle c^2 \rangle = 3 \times 1.38 \times 10^{-23} \times 400$	C1
	$c_{\text{r.m.s.}} = 2200 \text{ m s}^{-1}$	A1
2(e)	r.m.s. speed is an average so many molecules have speeds greater than the escape speed or there is a distribution of molecular speeds (around the r.m.s. value) so many molecules have speeds greater than the escape speed	B1

PUBLISHED

Question	Answer	Marks
3(a)	sum of potential energy and kinetic energy (of particles)	B1
	(total) energy of random motion of particles	B1
3(b)(i)	no thermal energy <u>transferred</u>	B1
	work is done on the spring (increasing the potential energy of particles)	M1
	so internal energy increases	A1
3(b)(ii)	thermal energy transferred to water	B1
	work is done by water (expanding against atmosphere as it vaporises)	B1
	more thermal energy transferred than work done so internal energy increases	B1

PUBLISHED

Question	Answer	Marks
4(a)(i)	$\text{amplitude} = \frac{1}{2} \times 7.2 \times 10^{-15}$ $= 3.6 \times 10^{-15} \text{ m}$	A1
4(a)(ii)	$\omega = 2\pi / (0.20 \times 10^{-6})$ $= 3.1 \times 10^7 \text{ rad s}^{-1}$	A1
4(a)(iii)	$v_0 = \omega x_0$	C1
	$v_0 = 3.1 \times 10^7 \times 3.6 \times 10^{-15} = 1.1 \times 10^{-7} \text{ m s}^{-1}$	A1
4(b)(i)	$I_0 = nAv_0e$ $= 8.5 \times 10^{28} \times 4.3 \times 10^{-4} \times 1.1 \times 10^{-7} \times 1.60 \times 10^{-19}$	C1
	$= 0.64 \text{ A}$	A1
4(b)(ii)	sketch: two cycles of sinusoidal curve of amplitude I_0 and period $0.20 \mu\text{s}$	B1
	correct phase, with $I = +I_0$ at $t = 0$	B1
4(b)(iii)	equation of form $I = I_0 \cos \omega t$	M1
	value of I_0 used matches answer to (b)(i) and value of ω used matches answer to (a)(ii)	A1
	[if (a)(ii) and (b)(i) correct then $I = 0.64 \cos (3.1 \times 10^7 t)$]	
4(b)(iv)	$I_{\text{r.m.s.}} = I_0 / \sqrt{2}$ $= 0.64 / \sqrt{2}$ $= 0.45 \text{ A}$	A1

PUBLISHED

Question	Answer	Marks
5(a)	(electric) force is (directly) proportional to product of charges	B1
	(electric) force (between point charges) is inversely proportional to the square of their separation	B1
5(b)	$F = Q^2 / 4\pi\epsilon_0 X^2$	C1
	$6.3 \times 10^{-17} = Q^2 / [4\pi \times 8.85 \times 10^{-12} \times (3.8 \times 10^{-6})^2]$	
	charge = 3.2×10^{-19} C	A1
5(c)(i)	negative	B1
5(c)(ii)	four straight lines perpendicular to the plates, starting on one plate and finishing on the other	B1
	lines equally spaced	B1
	arrows indicating direction downwards	B1
5(c)(iii)	$E = V / d$	C1
	$mg = EQ$	C1
	mass = $(1200 \times 3.2 \times 10^{-19}) / (9.81 \times 0.052)$ $= 7.5 \times 10^{-16}$ kg	A1

PUBLISHED

Question	Answer	Marks
6(a)(i)	energy stored = area under graph	C1
	$= \frac{1}{2} \times 450 \times 10^{-6} \times 8.0 = 1.8 \times 10^{-3} \text{ J}$ or 1.8 mJ	A1
6(a)(ii)	$C = Q / V$ or $E = \frac{1}{2} CV^2$	C1
	$C = (450 \times 10^{-6}) / 8.0$ or $(2 \times 1.8 \times 10^{-3}) / 8.0^2$ $= 5.6 \times 10^{-5} \text{ F}$	A1
6(b)(i)	$V = V_0 \exp(-t / RC)$ and $\tau = RC$	C1
	$V = V_0 \exp(-t / \tau)$ $V_0 = 8.0 \text{ V}$, and at one time constant, $t = \tau$ $V / 8.0 = \exp(-\tau / \tau)$, so $\ln(V / 8.0) = -1.0$ or $-\ln(V / 8.0) = 1.0$	A1
6(b)(ii)	[t read from graph at $-\ln(V / 8.0) = 1.0$]: $\tau = 3.2 \text{ s}$	A1
6(b)(iii)	$\tau = RC$	C1
	$R = 3.2 / (5.6 \times 10^{-5})$ $= 5.7 \times 10^4 \Omega$	A1

PUBLISHED

Question	Answer	Marks
7(a)(i)	$V_H = BI / ntq$ $= (4.0 \times 10^{-6} \times 5.4) / (1.5 \times 10^{16} \times 1.8 \times 10^{-3} \times 1.60 \times 10^{-19}) = 5.0 \text{ V}$	A1
7(a)(ii)	sketch: straight diagonal line from (0, 0) to $t = 0.020 \text{ s}$ and straight diagonal line between two non-zero V_H values of same sign from $t = 0.040$ to 0.050 s	B1
	horizontal straight line at $V_H = 5.0 \text{ V}$ from $t = 0.020$ to 0.040 s	B1
	horizontal straight line at $V_H = 2.5 \text{ V}$ from $t = 0.050$ to 0.080 s	B1
7(b)(i)	e.m.f. = rate of change of (magnetic) flux (linkage)	C1
	$E = NA \Delta B / \Delta t$ or $E = NA \times \text{gradient (at } t = 0.010 \text{ s)}$	C1
	$E = 3000 \times 3.4 \times 10^{-4} \times (4.0 \times 10^{-6}) / (0.020) = 2.0 \times 10^{-4} \text{ V}$	A1
7(b)(ii)	sketch: line showing non-zero E from $t = 0$ to $t = 0.020 \text{ s}$ and from $t = 0.040 \text{ s}$ to $t = 0.050 \text{ s}$, and $E = 0$ at all other times	B1
	‘top hats’ showing constant non-zero E from $t = 0$ to $t = 0.020 \text{ s}$ and from $t = 0.040 \text{ s}$ to $t = 0.050 \text{ s}$	B1
	magnitude of E shown as $2.0 \times 10^{-4} \text{ V}$ in both non-zero sections	B1
	sign of E in the $t = 0$ to $t = 0.020 \text{ s}$ region opposite to the sign of E in the $t = 0.040 \text{ s}$ to $t = 0.050 \text{ s}$ region	B1

PUBLISHED

Question	Answer	Marks
8(a)	packet / quantum of <u>energy</u>	M1
	of electromagnetic radiation	A1
8(b)(i)	photoelectric effect	B1
8(b)(ii)	- electron needs a minimum energy to escape or - electron emitted if energy in packet is enough - energy must be absorbed in packets that are related to frequency - intensity relates to number of packets (not to energy in packet) - electron absorbs only a single whole packet <i>Any three points, 1 mark each</i>	B3
8(c)(i)	Planck constant	B1
8(c)(ii)	– work function (energy)	B1

PUBLISHED

Question	Answer	Marks
9(a)(i)	material introduced into the body and (position in body) can be detected or absorbed by the tissue (being studied)	B1
9(a)(ii)	$X = \beta^+ \text{ or } e^+ \text{ and } P = 1$	B1
	$Q = 0 \quad \text{and } R = 18$	B1
9(b)(i)	positrons (emitted in the decay) and electrons annihilate	B1
	mass of particles becomes energy of gamma photons	B1
9(b)(ii)	arrival times of photons are processed	B1
	image built up of tracer concentration in the tissue	B1
9(c)(i)	$A = \lambda N \text{ and } \lambda = \ln 2 / T$	C1
	$N = n \times N_A$	C1
	2 photons produced from each decay, so $R_0 = 2 \times \lambda \times n \times N_A$	A1
	$R_0 = (2 \ln 2) n N_A / T$ (allow 0.693 for $\ln 2$)	
9(c)(ii)	sketch: exponential decay curve from $t = 0$ to $t = 2T$, starting at $(0, R_0)$ and with a negative gradient of continuously decreasing magnitude	B1
	line with negative gradient passing through $(T, R_0/2)$ and $(2T, R_0/4)$	B1

PUBLISHED

Question	Answer	Marks
10(a)	temperature inversely proportional to wavelength	M1
	temperature is thermodynamic temperature of surface, and wavelength is the wavelength at which maximum emission rate occurs	A1
10(b)(i)	(astronomical) object of known luminosity	B1
10(b)(ii)	star / galaxy is moving away from the student	B1
10(b)(iii)	one tick placed in correct column in each row: wavelength: too high	B1
	surface temperature: too low	B1
	distance: unchanged	B1
	radius: too high	B1



Cambridge International AS & A Level

PHYSICS

9702/43

Paper 4 A Level Structured Questions

October/November 2023

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Due to a series-specific issue during the live exam series, all candidates were awarded full marks for questions 1 and 10.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)(i)	direction of the force acting on a (test) mass placed at the point	B1
1(a)(ii)	change in height negligible compared with radius (of Earth)	B1
	(so) field lines are (effectively) parallel	B1
1(b)(i)	$Y = GM / R^2$	M1
	G is the gravitational constant	A1
1(b)(ii)	gravitational force is (always) attractive or gravitational force (always) acts towards the centre of the sphere	B1
	force is in opposite direction to displacement or at a point to the right of the centre, force acts to the left or at a point to the left of the centre, force acts to the right	B1
1(b)(iii)	sketch: smooth curve with decreasing positive gradient, starting at $(R, -Y)$ and reaching $3R$ with g still negative or smooth curve with increasing positive gradient, ending at $(-R, Y)$ and reaching $-3R$ with g still positive	B1
	both of the above curves, in correct quadrants	B1
	curve passing through $(\pm 2R, \pm 0.25Y)$ and $(\pm 3R, \pm 0.11Y)$	B1

PUBLISHED

Question	Answer	Marks
2(a)	(thermal) energy per unit mass (to change temperature)	B1
	(thermal) energy per unit change in temperature	B1
2(b)(i)	work done = $p\Delta V$ $= (2.0 \times 10^5) \times (0.063 - 0.038) = 5000 \text{ J}$	A1
2(b)(ii)	gas is expanding (against external pressure)	B1
	gas does work / work is done by gas, so (work done on gas is) negative	B1
2(b)(iii)	$\Delta U = q + W$	C1
	$7600 = q + (-5000)$	A1
	$q = 12\,600 \text{ J}$	
2(b)(iv)	specific heat capacity = $q / m\Delta T$ $= 12600 / (0.35 \times 56)$	C1
	$= 640 \text{ J kg}^{-1} \text{ K}^{-1}$	A1
2(c)	same gain in internal energy so same temperature rise	B1
	no change in volume so no work done or no work done so less thermal energy needed (for same change in internal energy)	B1
	less thermal energy needed (for same temperature change) so lower specific heat capacity	B1

PUBLISHED

Question	Answer	Marks
3(a)(i)	N : number of molecules (of the gas)	B1
	m : mass of one molecule (of the gas)	B1
	$\langle c^2 \rangle$: mean square speed (of molecules)	B1
3(a)(ii)	$pV = NkT$	M1
	$NkT = \frac{1}{3}Nm\langle c^2 \rangle$ and $E_K = \frac{1}{2}m\langle c^2 \rangle$ leading to $E_K = (3/2)kT$	A1
3(b)	$\frac{1}{2} \times 3.34 \times 10^{-27} \times 9300^2 = (3/2) \times 1.38 \times 10^{-23} \times T$	C1
	$T = 6980 \text{ K}$	A1
3(c)(i)	$L = F \times 4\pi d^2$	C1
	$L = 2.52 \times 10^{-8} \times 4\pi \times (4.16 \times 10^{16})^2$	A1
	$= 5.48 \times 10^{26} \text{ W}$	
3(c)(ii)	$L = 4\pi\sigma r^2 T^4$	C1
	$5.48 \times 10^{26} = 4\pi \times 5.67 \times 10^{-8} \times r^2 \times 6980^4$	
	$r = 5.69 \times 10^8 \text{ m}$	A1
3(d)	(very high pressure so) molecules are (very) close together (<i>not just 'nearer'</i>)	B1
	forces between molecules are not negligible or volume of molecules not negligible compared with gas volume	B1

PUBLISHED

Question	Answer	Marks
4(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): acceleration is (always) in the opposite <u>direction</u> to displacement or acceleration is (always) directed towards a fixed point	B1
4(b)(i)	$\omega = 2\pi / T$	C1
	$\omega = 2\pi / 3.0$ $= 2.1 \text{ rad s}^{-1}$	A1
4(b)(ii)	$E = \frac{1}{2}m\omega^2x_0^2$	C1
	$= \frac{1}{2} \times 0.81 \times 2.1^2 \times 0.036^2$ $= 2.3 \times 10^{-3} \text{ J}$	A1
4(c)	sketch: line starting at (0, 0.036) and not reaching $x = \pm 0.036 \text{ m}$ at any other time	B1
	smooth curve, with no sudden changes in gradient, showing continuously decreasing magnitude of x from maximum displacement at $t = 0$ to final displacement of zero where the gradient is also zero	B1
	displacement reaches final value of zero between $t = 0.75 \text{ s}$ and $t = 3.0 \text{ s}$ at the latest	B1

PUBLISHED

Question	Answer	Marks
5(a)	work done per unit charge	B1
	work (done on charge) moving positive charge from infinity (to the point)	B1
5(b)	<i>Any three points from:</i> <i>Up to 2 points from:</i> • radius of sphere X is 2.0 m • radius of sphere Y is 4.0 m • radius of Y is double the radius of X <i>Up to 2 points from:</i> • charge on X is negative • charge on Y is positive • spheres carry opposite charges <i>Up to 1 point from:</i> • magnitudes of charges on the spheres are equal	B3
5(c)	particle is attracted to X or repelled from Y or resultant force on particle is towards X / away from Y / to the left	B1
	particle accelerates towards X / away from Y / to the left	B1
	(magnitude of) acceleration of particle increases	B1

PUBLISHED

Question	Answer	Marks
6(a)	• force per unit length • force per unit current • length / current perpendicular to field 1 mark for any two points, 2 marks for all three points	B2
6(b)(i)	into the page	B1
6(b)(ii)	$F = Bqv$	C1
	$= 4.8 \times 10^{-3} \times 1.6 \times 10^{-19} \times 1.7 \times 10^7 = 1.3 \times 10^{-14} \text{ N}$	A1
6(b)(iii)	arrow at point X pointing down the page	B1
6(b)(iv)	$F = mv^2 / r$	C1
	$1.3 \times 10^{-14} = (9.11 \times 10^{-31}) \times (1.7 \times 10^7)^2 / r$	C1
	$(r = 0.020 \text{ m})$ $d = 2r$ $d = 0.040 \text{ m}$	A1
6(c)	path shows upwards deflection such that the curvature is always anticlockwise within the field	B1
	circular path with larger radius	B1
	line enters field at X and leaves field at distance $2d$ vertically from X	B1

PUBLISHED

Question	Answer	Marks
7(a)(i)	$P = I_0^2 R$	A1
7(a)(ii)	$P = 4I_0^2 R$	A1
7(b)	sketch: square wave of period T , with P always non-zero	B1
	horizontal lines, from 0 to $0.5T$ and from $1.0T$ to $1.5T$, all at the same level that the scale indicates to be $I_0^2 R$	B1
	horizontal lines, from $0.5T$ to $1.0T$ and from $1.5T$ to $2.0T$, at a level that is four times higher than the lower lines	B1
7(c)(i)	$\langle P \rangle = (5/2)I_0^2 R$	A1
7(c)(ii)	$\langle P \rangle = I_{\text{r.m.s.}}^2 R$	C1
	$I_{\text{r.m.s.}}^2 R = (5/2)I_0^2 R$	A1
	$I_{\text{r.m.s.}} = \sqrt{(5/2)} I_0$	

PUBLISHED

Question	Answer	Marks
8(a)(i)	$p = E / c$	M1
	$E = hc / \lambda$ and completion of algebra leading to $p = h / \lambda$	A1
8(a)(ii)	wavelength = $(6.63 \times 10^{-34}) / (9.5 \times 10^{-28}) = 700 \times 10^{-9} \text{ m}$ so red	B1
8(b)(i)	power = intensity $\times$ area	C1
	number per unit time = $(160 \times 2.5 \times 10^{-6}) / (9.5 \times 10^{-28} \times 3.00 \times 10^8) = 1.4 \times 10^{15} \text{ s}^{-1}$	A1
8(b)(ii)	pressure = force / area	C1
	force = rate of change of momentum $= 2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}$	C1
	pressure = $(2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}) / (2.5 \times 10^{-6})$ $= 1.1 \times 10^{-6} \text{ Pa}$	A1
8(c)	photons have greater momentum or fewer photons per unit time	B1
	greater photon momentum but smaller number of photons (per unit time) so pressure is the same	B1

PUBLISHED

Question	Answer	Marks
9(a)	(two small) nuclei join together	M1
	to form one larger nucleus	A1
9(b)	line with a peak at $A \approx 56$	B1
	line with steep initial positive gradient on the left of peak and shallower negative gradient at all points to the right of peak and line does not return to 0 binding energy	B1
9(c)(i)	X shown at value of A to the right of the peak	B1
9(c)(ii)	Y shown at value of A close to 1	B1
9(d)	energy from 1 nucleus = $(1.77 \times 10^{13}) / (6.02 \times 10^{23})$ $(= 2.94 \times 10^{-11} \text{ J})$	C1
	binding energy of Z = $[(1.25 + 1.81) \times 10^{-10}] - 2.94 \times 10^{-11}$ $(= 2.77 \times 10^{-10} \text{ J})$	C1
	nucleon number of Z = $93 + 139 + 2 - 1$ $(= 233)$	C1
	binding energy per nucleon = $(2.77 \times 10^{-10}) / (233 \times 1.60 \times 10^{-13})$ $= 7.43 \text{ MeV}$	A1

PUBLISHED

Question	Answer	Marks
10(a)	ultrasound production: vibrating quartz crystal	B1
	X-ray production: electrons hitting metal target	B1
	ultrasound detected wave: reflected	B1
	X-ray detected wave: transmitted	B1
10(b)(i)	$I = I_0 \exp(-\mu x)$	C1
	$\ln(0.72) = -6.2\mu$	A1
	$\mu = 0.053 \text{ cm}^{-1}$	
10(b)(ii)	$I / I_0 = \exp(-9.3 \times 0.053)$ (= 0.61)	C1
	percentage attenuated = $100 \times (1.00 - 0.61)$ = 39%	A1



Cambridge International AS & A Level

PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

October/November 2023

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem.	
	f is the independent variable and E is the dependent variable or vary f and measure E	1
	keep V and R <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - coils C and D placed with their axes on a straight line - separate workable circuit for coil D (a.c.) voltmeter or oscilloscope connected across coil D (Do not accept a power supply connected to coil D.)	1
	<u>a.c.</u> power supply/signal generator connected to coil C	1
	workable circuit for coil C with power supply <u>and</u> (a.c.) voltmeter/oscilloscope in parallel with resistor <u>and</u> coil C	1
	method to determine f , e.g. read from signal generator or use of oscilloscope	1
	Method of Analysis	
	plot a graph of $\lg E$ against $\lg f$ or equivalent (e.g. $\ln E$ against $\ln f$)	1
	$q = \text{gradient}$	1
	$p = \frac{R}{V} \times 10^{\text{y-intercept}}$ (for $\ln E$ against $\ln f$: $p = \frac{R}{V} \times e^{\text{y-intercept}}$)	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	precaution (to prevent burns) <u>from hot coils/hot resistor</u> , e.g. use gloves to handle hot coil/resistor, switch off circuit <u>and</u> wait for hot coil/resistor to cool	
D2	keep the number of turns on each coil constant	
D3	keep distance between the coils constant	
D4	workable circuit diagram to determine R . e.g. circuit with ammeter connected in series and voltmeter in parallel with resistor or resistor connected to ohmmeter only	
D5	determination of resistance R : potential difference across $R \div$ current in R or use ohmmeter to measure R	
D6	method to keep distance between the coils constant, e.g. fix/clamp coils to bench	
D7	method to determine f from oscilloscope, e.g. period $T = \text{time-base} \times \text{horizontal distance}$ and $f = 1 / T$	
D8	method to determine V or E from oscilloscope, e.g. V or $E = y\text{-gain} \times \text{vertical distance}$	
D9	method to increase E e.g. use iron core, place coils closer, increase V , decrease R	
D10	relationship valid <u>if</u> a straight line is produced (passing through $\log \left(\frac{pV}{R} \right)$) Do not accept line passing through the origin.	

PUBLISHED

Question	Answer	Marks							
2(a)	$\text{gradient} = \frac{\sqrt{2g}}{uZ}$ $y\text{-intercept} = \frac{\sqrt{2g}}{u}$	1							
2(b)	<table><tr><td>$\sqrt{\frac{1}{h}} / \text{cm}^{\frac{1}{2}}$</td></tr><tr><td>0.218 or 0.2182</td></tr><tr><td>0.237 or 0.2370</td></tr><tr><td>0.248 or 0.2485</td></tr><tr><td>0.262 or 0.2617</td></tr><tr><td>0.282 or 0.2817</td></tr><tr><td>0.313 or 0.3131</td></tr></table> Values correct as shown above.	$\sqrt{\frac{1}{h}} / \text{cm}^{\frac{1}{2}}$	0.218 or 0.2182	0.237 or 0.2370	0.248 or 0.2485	0.262 or 0.2617	0.282 or 0.2817	0.313 or 0.3131	1
$\sqrt{\frac{1}{h}} / \text{cm}^{\frac{1}{2}}$									
0.218 or 0.2182									
0.237 or 0.2370									
0.248 or 0.2485									
0.262 or 0.2617									
0.282 or 0.2817									
0.313 or 0.3131									
	Uncertainties in $\sqrt{\frac{1}{h}}$ from ± 0.001 to ± 0.003 .	1							
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1							
	Error bars in $\sqrt{\frac{1}{h}}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1							

PUBLISHED

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (605, 0.230) and (615, 0.230) and between (845, 0.300) and (855, 0.300)	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

PUBLISHED

Question	Answer	Marks
2(d)(i)	u determined using y-intercept and u and Z given to 2, 3 or 4 significant figures. $u = \frac{\sqrt{2 \times 981}}{\text{y-intercept}} = \frac{44.29}{\text{(c)(iv)}}$	1
	Z determined using gradient with method shown and u and Z given with SI units with appropriate powers of ten. $Z = \frac{\sqrt{2 \times 981}}{u \times \text{gradient}} = \frac{44.29}{u \times \text{(c)(iii)}} \quad \text{or} \quad Z = \frac{\text{y-intercept}}{\text{gradient}} = \frac{\text{(c)(iv)}}{\text{(c)(iii)}}$	1
2(d)(ii)	Percentage uncertainty in Z with method shown. $\text{percentage uncertainty in } Z = \left(\frac{\Delta \text{y-intercept}}{\text{y-intercept}} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or Correct substitution for Δu and $\text{percentage uncertainty in } Z = \left(\frac{\Delta u}{u} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or Correct substitution for max/min methods.	1

PUBLISHED

Question	Answer	Marks
2(e)	M determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution. $M = \frac{\left(\frac{1}{\sqrt{25}} - y\text{-intercept} \right)}{\text{gradient}}$ or $M = \frac{uZ}{\sqrt{2gh}} - Z = \frac{uZ}{221.5} - Z$	1



Cambridge International AS & A Level

PHYSICS

9702/52

Paper 5 Planning, Analysis and Evaluation

October/November 2023

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	V is the independent variable and z is the dependent variable or vary V and measure z	1
	keep h <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • pulley supported by stand • stand placed on surface/bench/floor • minimum of two labels from stand, beaker, oil, surface/bench/floor, pulley, string	1
	use (metre) rule to measure h or (metre) rule correctly positioned with h marked on diagram	1
	use measuring cylinder to measure V	1
	timing method to measure time t of fall of beaker to determine z e.g. use timer/stopwatch or use light gate(s) <u>connected to a timer/data logger</u>	1

PUBLISHED

Question	Answer	Marks
1	Method of Analysis	
	plot a graph of $\frac{1}{z^2}$ against $\frac{1}{V}$ or equivalent (e.g. $\frac{1}{V}$ against $\frac{1}{z^2}$) Do not accept logarithms.	1
	$b = \frac{1}{2h \times y\text{-intercept}}$ (for $\frac{1}{V}$ against $\frac{1}{z^2}$: $b = \frac{M \times \text{gradient}}{ah}$ or $b = -\frac{\text{gradient}}{2h \times y\text{-intercept}}$)	1
	$a = \frac{M}{bh \times \text{gradient}}$ or $a = \frac{2M \times y\text{-intercept}}{\text{gradient}}$ (for $\frac{1}{V}$ against $\frac{1}{z^2}$: $a = -2M \times y\text{-intercept}$)	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	precaution linked to <u>oil spillage</u> , e.g. use of cushion/sand box/tray for falling beaker to land or use of bungs/lids on beakers or use foam on bench/floor or use foam to prevent rising beaker hitting pulley	
D2	precaution linked to <u>oil contact with skin</u> e.g. use gloves to avoid contact with oil	
D3	keep M constant	
D4	use a (top-pan) balance to measure M	
D5	method to keep h constant e.g. use a fiducial mark to release the beaker from the same position or release from the same position on the <u>clamped</u> rule each time	
D6	equation to determine z for method used, e.g. for timing h , $z = 2h / t$ or for one light gate, $z = L / t$ where L is the length of the interrupted beam or for two light gates, $z = \text{distance between light gates} / t$ Do not accept h / t .	
D7	additional detail on diagram to measure h , e.g. clamp (metre) rule with stand on surface or use of set squares positioned on the surface to side of rule or spirit level positioned to side of rule	
D8	use large value of h to increase time of fall of beaker	
D9	repeat measurements of z for the same V <u>and</u> average z	
D10	relationship valid <u>if</u> a straight line is produced (passing through $\left(\frac{1}{2bh}\right)$) Do not accept line passing through the origin.	

PUBLISHED

Question	Answer		Marks														
2(a)	gradient = $-\frac{t}{R}$ y-intercept = $\ln I_0R$		1														
2(b)	<table><tr><th>$1/C/10^4\text{ F}^{-1}$</th><th>$\ln(V/V)$</th></tr><tr><td>0.91 or 0.909</td><td>0.896 or 0.8961</td></tr><tr><td>0.76 or 0.758</td><td>1.012 or 1.0116</td></tr><tr><td>0.63 or 0.633</td><td>1.115 or 1.1151</td></tr><tr><td>0.61 or 0.606</td><td>1.131 or 1.1314</td></tr><tr><td>0.48 or 0.482</td><td>1.253 or 1.2528</td></tr><tr><td>0.36 or 0.357</td><td>1.348 or 1.3481</td></tr></table>		$1/C/10^4\text{ F}^{-1}$	$\ln(V/V)$	0.91 or 0.909	0.896 or 0.8961	0.76 or 0.758	1.012 or 1.0116	0.63 or 0.633	1.115 or 1.1151	0.61 or 0.606	1.131 or 1.1314	0.48 or 0.482	1.253 or 1.2528	0.36 or 0.357	1.348 or 1.3481	1
	$1/C/10^4\text{ F}^{-1}$	$\ln(V/V)$															
	0.91 or 0.909	0.896 or 0.8961															
0.76 or 0.758	1.012 or 1.0116																
0.63 or 0.633	1.115 or 1.1151																
0.61 or 0.606	1.131 or 1.1314																
0.48 or 0.482	1.253 or 1.2528																
0.36 or 0.357	1.348 or 1.3481																
	Values correct as shown above.																
	Uncertainties in $\ln(V/V)$ from ± 0.021 or ± 0.020 to ± 0.010 or ± 0.013		1														
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.		1														
	Error bars in $\ln(V/V)$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.		1														

PUBLISHED

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (0.820, 0.95) and (0.845, 0.95) and between (0.400, 1.30) and (0.425, 1.30)	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line. Gradient must be negative.	1
	Gradient determined of worst acceptable line. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and x into $y = mx + c$.	1

PUBLISHED

Question	Answer	Marks
2(d)(i)	R determined using gradient. $R = -\frac{30.0}{\text{gradient}} = \frac{30.0}{\text{(c)(iii)}}$	1
	I_0 determined using y-intercept with method shown. $I_0 = \frac{e^{y\text{-intercept}}}{R} = \frac{e^{\text{(c)(iv)}}}{\text{(d)(i)}}$	1
	R and I_0 determined correctly using gradient and y-intercept and R and I_0 given to 2 or 3 significant figures and R and I_0 given with SI units with appropriate powers of ten. Units: R: Ω or s F^{-1} I_0: A or V F s^{-1} or $\text{V } \Omega^{-1}$	1
2(d)(ii)	Percentage uncertainty in R with method shown. $\text{percentage uncertainty in } R = \left(\frac{\Delta t}{t} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or Correct substitution for max/min methods.	1

PUBLISHED

Question	Answer	Marks
2(e)	C determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitutions. $C = \frac{\text{gradient}}{\ln V - y\text{-intercept}}$ or $C = -\frac{\text{gradient}}{y\text{-intercept} - \ln V}$ or $C = -\frac{t}{R(\ln V - \ln I_0 R)}$ or $C = \frac{t}{R(\ln I_0 R - \ln V)}$	1



Cambridge International AS & A Level

PHYSICS

9702/53

Paper 5 Planning, Analysis and Evaluation

October/November 2023

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance
 For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem.	
	f is the independent variable and E is the dependent variable or vary f and measure E	1
	keep V and R <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - coils C and D placed with their axes on a straight line - separate workable circuit for coil D (a.c.) voltmeter or oscilloscope connected across coil D (Do not accept a power supply connected to coil D.)	1
	<u>a.c.</u> power supply/signal generator connected to coil C	1
	workable circuit for coil C with power supply <u>and</u> (a.c.) voltmeter/oscilloscope in parallel with resistor <u>and</u> coil C	1
	method to determine f , e.g. read from signal generator or use of oscilloscope	1
	Method of Analysis	
	plot a graph of $\lg E$ against $\lg f$ or equivalent (e.g. $\ln E$ against $\ln f$)	1
	$q = \text{gradient}$	1
	$p = \frac{R}{V} \times 10^{\text{y-intercept}}$ (for $\ln E$ against $\ln f$: $p = \frac{R}{V} \times e^{\text{y-intercept}}$)	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	precaution (to prevent burns) <u>from hot coils/hot resistor</u> , e.g. use gloves to handle hot coil/resistor, switch off circuit <u>and</u> wait for hot coil/resistor to cool	
D2	keep the number of turns on each coil constant	
D3	keep distance between the coils constant	
D4	workable circuit diagram to determine R . e.g. circuit with ammeter connected in series and voltmeter in parallel with resistor or resistor connected to ohmmeter only	
D5	determination of resistance R : potential difference across $R \div$ current in R or use ohmmeter to measure R	
D6	method to keep distance between the coils constant, e.g. fix/clamp coils to bench	
D7	method to determine f from oscilloscope, e.g. period $T = \text{time-base} \times \text{horizontal distance}$ and $f = 1 / T$	
D8	method to determine V or E from oscilloscope, e.g. V or $E = y\text{-gain} \times \text{vertical distance}$	
D9	method to increase E e.g. use iron core, place coils closer, increase V , decrease R	
D10	relationship valid <u>if</u> a straight line is produced (passing through $\log \left(\frac{pV}{R} \right)$) Do not accept line passing through the origin.	

PUBLISHED

Question	Answer	Marks							
2(a)	$\text{gradient} = \frac{\sqrt{2g}}{uZ}$ $y\text{-intercept} = \frac{\sqrt{2g}}{u}$	1							
2(b)	<table><tr><td>$\sqrt{\frac{1}{h}} / \text{cm}^{-\frac{1}{2}}$</td></tr><tr><td>0.218 or 0.2182</td></tr><tr><td>0.237 or 0.2370</td></tr><tr><td>0.248 or 0.2485</td></tr><tr><td>0.262 or 0.2617</td></tr><tr><td>0.282 or 0.2817</td></tr><tr><td>0.313 or 0.3131</td></tr></table> Values correct as shown above.	$\sqrt{\frac{1}{h}} / \text{cm}^{-\frac{1}{2}}$	0.218 or 0.2182	0.237 or 0.2370	0.248 or 0.2485	0.262 or 0.2617	0.282 or 0.2817	0.313 or 0.3131	1
$\sqrt{\frac{1}{h}} / \text{cm}^{-\frac{1}{2}}$									
0.218 or 0.2182									
0.237 or 0.2370									
0.248 or 0.2485									
0.262 or 0.2617									
0.282 or 0.2817									
0.313 or 0.3131									
	Uncertainties in $\sqrt{\frac{1}{h}}$ from ± 0.001 to ± 0.003 .	1							
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1							
	Error bars in $\sqrt{\frac{1}{h}}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1							

PUBLISHED

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (605, 0.230) and (615, 0.230) and between (845, 0.300) and (855, 0.300)	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

PUBLISHED

Question	Answer	Marks
2(d)(i)	u determined using y-intercept and u and Z given to 2, 3 or 4 significant figures. $u = \frac{\sqrt{2 \times 981}}{\text{y-intercept}} = \frac{44.29}{\text{(c)(iv)}}$	1
	Z determined using gradient with method shown and u and Z given with SI units with appropriate powers of ten. $Z = \frac{\sqrt{2 \times 981}}{u \times \text{gradient}} = \frac{44.29}{u \times \text{(c)(iii)}} \quad \text{or} \quad Z = \frac{\text{y-intercept}}{\text{gradient}} = \frac{\text{(c)(iv)}}{\text{(c)(iii)}}$	1
2(d)(ii)	Percentage uncertainty in Z with method shown. $\text{percentage uncertainty in } Z = \left(\frac{\Delta \text{y-intercept}}{\text{y-intercept}} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or Correct substitution for Δu and $\text{percentage uncertainty in } Z = \left(\frac{\Delta u}{u} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or Correct substitution for max/min methods.	1

PUBLISHED

Question	Answer	Marks
2(e)	M determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution. $M = \frac{\left(\frac{1}{\sqrt{25}} - y\text{-intercept} \right)}{\text{gradient}}$ or $M = \frac{uZ}{\sqrt{2gh}} - Z = \frac{uZ}{221.5} - Z$	1

Cambridge International AS & A Level

PHYSICS**9702/41**

Paper 4 A Level Structured Questions

October/November 2024

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **15** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)(i)	(gravitational) force acts perpendicular to direction of motion	B1
	gravitational force provides centripetal acceleration	B1
1(b)(ii)	$(F =) GMm / x^2 = m\omega^2$ and $\omega = 2\pi / T$ or $GMm / x^2 = 4\pi^2 mx / T^2$	M1
	completion of algebra leading to $x^3 = GMT^2 / 4\pi^2$	A1
	clear indication that B = radius of planet and that A = mass (of planet)	B1
1(b)(iii)	gradient = $\sqrt[3]{(4\pi^2 / GA)}$	C1
	e.g. $(1280 - 360) / (12 \times 10^6) = \sqrt[3]{(4\pi^2 / [6.67 \times 10^{-11} \times A])}$	C1
	$A = 1.3 \times 10^{24} \text{ kg}$	A1
	intercept = gradient $\times B$	C1
	e.g. $360 = ((1280 - 360) \times B) / (12 \times 10^6)$ $B = 4.7 \times 10^6 \text{ m}$	A1

PUBLISHED

Question	Answer	Marks
2(a)	(thermal) energy per unit mass (to change temperature)	B1
	(thermal) energy per unit change in temperature	B1
2(b)(i)	<i>Any three bulleted points from:</i> - the blocks end up in thermal equilibrium - heat capacity of Y is larger than heat capacity of X - no heat loss to the surroundings <i>Up to 2 points from these six:</i> - initial temperature of X = 85 °C - initial temperature of Y = 25 °C - the temperature change of X = 45 °C - the temperature change of Y = 15 °C - the temperature change in X is three times that in Y - final temperature of both = 40 °C	B3
2(b)(ii)	$\Delta\theta = 45\text{ °C}$ for X and 15 °C for Y	C1
	$mc \times 45 = 1.3 \times m \times 901 \times 15$	C1
	$c = 390\text{ J kg K}^{-1}$	A1

PUBLISHED

Question	Answer	Marks
3(a)(i)	number of particles per unit amount of substance	B1
3(a)(ii)	$N_A = R / k$	B1
3(b)(i)	X pressure and Y pressure <u>both</u> = NkT / V	B1
	X amount = N / N_A and Y amount = $2N / N_A$	B1
	X mean-square speed = $3kT / m$ and Y mean-square speed = $3kT / 2m$	B1
	X internal energy = $3NkT / 2$ and Y internal energy = $3NkT$	B1
3(b)(ii)	line passing through the origin and not returning to either axis	B1
	curve with positive decreasing gradient	B1

PUBLISHED

Question	Answer	Marks
4(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): acceleration is (always) in the opposite <u>direction</u> to displacement or acceleration is (always) <u>directed</u> towards a fixed point	B1
4(b)(i)	amplitude = $(9.5 - 3.5) / 2$ = 3.0 cm	A1
4(b)(ii)	$\omega = v_0 / x_0$	C1
	= $9.5 / 3.0 = 3.2 \text{ rad s}^{-1}$	A1
4(b)(iii)	$T = 2\pi / \omega$	C1
	= $2\pi / 3.2$ = 2.0 s	A1
4(b)(iv)	attempted sinusoidal curve starting with a minimum at $t = 0$	B1
	sinusoidal curve of period 2.0 s from $t = 0$ to $t = 6.0$ s	B1
	all peaks shown at $h = 9.5$ cm	B1
	all troughs shown at $h = 3.5$ cm	B1

PUBLISHED

Question	Answer	Marks
5(a)	(electric) field equals (electric) potential gradient	M1
	reference to minus sign	A1
5(b)	- for potential to be zero, one potential must be positive and the other potential must be negative - for potential to be zero, the charges must have opposite sign - for field to be zero, the fields (due to X and Y) must be in opposite directions - for field to be zero, the charges must have the same sign - the signs of the charges cannot (simultaneously) be both the same and opposite (so not possible) <i>Any three points, 1 mark each</i>	B3
5(c)(i)	$V_X = (-) Q / 4\pi\epsilon_0 x$ and $V_Y = (-) 2Q / 4\pi\epsilon_0 y$	C1
	$(V_X + V_Y = 0 \text{ so } Q / 4\pi\epsilon_0 x = 2Q / 4\pi\epsilon_0 y \text{ leading to } y = 2x)$	A1
5(c)(ii)	$E_X = Q / 4\pi\epsilon_0 x^2$	A1
5(c)(iii)	$E_Y = 2Q / 4\pi\epsilon_0 (2x)^2$ $(= Q / 8\pi\epsilon_0 x^2)$	C1
	(opposite charges so fields in same direction so magnitudes add): $E = (Q / 4\pi\epsilon_0 x^2) + (Q / 8\pi\epsilon_0 x^2)$ $= 3Q / 8\pi\epsilon_0 x^2$	A1

PUBLISHED

Question	Answer	Marks
6(a)(i)	conversion (from a.c.) to d.c.	B1
6(a)(ii)	half-wave: voltage in one direction is removed	B1
	full-wave: voltage in one direction is reversed	B1
6(b)(i)	one gap connected by a single diode and other gap connected directly	B1
	diode drawn (in a circuit) with correct circuit symbol	B1
6(b)(ii)	smoothing	B1
6(c)(i)	$E = \frac{1}{2}CV^2$	C1
	$C = 2 \times 0.041 / 12^2 = 5.7 \times 10^{-4} \text{ F} = 570 \mu\text{F}$	A1
6(c)(ii)	$8.0 = 12.0 \exp(-0.010 / RC)$	C1
	$\ln(8.0 / 12.0) = -0.010 / (R \times 5.7 \times 10^{-4})$	C1
	$R = 43 \Omega$	A1

PUBLISHED

Question	Answer	Marks
7(a)	• force per unit length • force per unit current • length / current perpendicular to field <i>1 mark for any two points, 2 marks for all three points</i>	B2
7(b)	concentric circles around the wire (at least two circles needed)	B1
	spacing between circles increases with distance from wire (at least four circles needed)	B1
	arrows showing direction of field is clockwise	B1
7(c)(i)	(each) wire sits in the (magnetic) field created by the other	B1
	current (in one wire) is perpendicular to (magnetic) field (due to other wire) so (magnetic) force acts (on wire)	B1
7(c)(ii)	arrow drawn, starting from X and pointing towards Y, labelled F	B1
7(c)(iii)	(forces have) equal magnitudes	B1
	(forces are in) opposite directions	B1
7(c)(iv)	no change (in the direction of the force) since both the current in X and the field due to Y have reversed	B1

PUBLISHED

Question	Answer	Marks
8(a)	photoelectric effect	B1
8(b)(i)	$E = hf$	C1
	work function = $6.63 \times 10^{-34} \times 8.8 \times 10^{14}$ $= 5.8 \times 10^{-19} \text{ J}$	A1
8(b)(ii)	$hf = \phi + \frac{1}{2}mv_{\text{MAX}}^2$	C1
	$6.63 \times 10^{-34} \times 11 \times 10^{14} = (5.8 \times 10^{-19}) + (\frac{1}{2} \times 9.11 \times 10^{-31} \times v_{\text{MAX}}^2)$	C1
	$v_{\text{MAX}} = 5.7 \times 10^5 \text{ m s}^{-1}$	A1
8(c)	E_{MAX} shown as zero from $f = 8.0$ to 8.8 and non-zero from $f = 8.8$ to 11	B1
	all non-zero E_{MAX} shown as a single straight line with a positive gradient	B1
	line passing through (11, 1.45)	B1

PUBLISHED

Question	Answer	Marks
9(a)(i)	positron	B1
9(a)(ii)	$\lambda = \ln 2 / (110 \times 60) = 1.05 \times 10^{-4} \text{ s}^{-1}$	A1
9(a)(iii)	$N = M / (18 \text{ u})$ or $(M \text{ in grams} \times N_A / 18)$ $N = (2.1 \times 10^{-12}) / (18 \times 1.66 \times 10^{-27})$ or $(2.1 \times 10^{-9} \times 6.02 \times 10^{23}) / 18$ $(= 7.0 \times 10^{13})$	C1
	$A = \lambda N$	C1
	$= 1.05 \times 10^{-4} \times 7.0 \times 10^{13}$ $= 7.4 \times 10^9 \text{ Bq}$	A1
9(b)(i)	(pair) annihilation occurs - the <u>mass</u> of the two particles is converted into energy - two <u>gamma</u> photons are formed and travel in opposite directions or two <u>gamma</u> photons are formed and leave the body - difference in arrival times of photons (at detector) is processed <i>Any three points, 1 mark each</i>	B3
9(b)(ii)	with a shorter half-life: sample would (almost) fully decay before the test is complete	B1
	a longer half-life: exposes patient to harmful/ionising radiation unnecessarily or with a longer half-life: a larger dose (of tracer) needed to produce detectable activity	B1

PUBLISHED

Question	Answer	Marks
10(a)	<u>redshift</u> is the increase in observed wavelength / decrease in observed frequency (caused by Doppler effect) - radiation from distant galaxies is observed to be redshifted - redshift provides evidence that galaxies are moving apart <u>galaxies</u> moving apart means Universe must be expanding <i>Any three points, 1 mark each</i>	B3
10(b)(i)	$F = L / 4\pi d^2$	C1
	$d = \sqrt{(1.90 \times 10^{36} / [4\pi \times 8.42 \times 10^{-16}])}$ $= 1.34 \times 10^{25} \text{ m}$	A1
10(b)(ii)	$\Delta\lambda / \lambda = v / c$ $(726 - 658) / 658 = v / (3.00 \times 10^8)$	C1
	$v = 3.1 \times 10^7 \text{ m s}^{-1}$	A1
10(c)(i)	line with positive gradient passing through the origin	B1
	straight line with positive gradient	B1
10(c)(ii)	Hubble constant	B1

Cambridge International AS & A Level

PHYSICS**9702/42**

Paper 4 A Level Structured Questions

October/November 2024

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **14** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

PUBLISHED**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Question	Answer	Marks
1(a)(i)	$v = r\omega$	C1
	$= 0.85 \times 140$	A1
	$= 120 \text{ m s}^{-1}$	
1(a)(ii)	$a = r\omega^2$ or $a = v^2 / r$	C1
	$a = 0.85 \times 140^2$ or $120^2 / 0.85$	A1
	$= 1.7 \times 10^4 \text{ m s}^{-2}$	
1(b)(i)	direction of (induced) e.m.f.	M1
	is such as to (produce effects that) oppose the <u>change</u> that caused it	A1
1(b)(ii)	$T = 2\pi / \omega$	A1
	$= 2\pi / 140 = 0.045 \text{ s} = 45 \text{ ms}$	
1(b)(iii)	$\Phi = BA$	C1
	$= 0.18 \times \pi \times 0.85^2$	C1
	$= 0.41 \text{ Wb}$	A1
1(b)(iv)	$E = \Phi / t$	C1
	$= 0.41 / 0.045$	A1
	$= 9.1 \text{ V}$	
1(b)(v)	force (on spoke) must be anticlockwise, so current is from A to X (by Fleming's left hand rule), so X is at the higher potential	B1

PUBLISHED

Question	Answer	Marks
2(a)	force per unit mass	B1
2(b)(i)	$g = GM/x^2$	C1
	$= (6.67 \times 10^{-11} \times 1.99 \times 10^{30}) / (1.47 \times 10^{11})^2$	A1
	$= 6.14 \times 10^{-3} \text{ N kg}^{-1}$	
2(b)(ii)	$E_p = -GMm/x$	C1
	$= - (6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 2.63) / (1.47 \times 10^{11})$	
	$= - 2.37 \times 10^9 \text{ J}$	A1
2(c)(i)	$F = L / 4\pi x^2$	C1
	$(g = GM/x^2 \text{ and so } x^2 = GM/g)$	M1
	and	
	$x^2 = L / 4\pi F$	
	elimination of x and subsequent algebra shown leading to $g = 4\pi GMF / L$	A1
2(c)(ii)	correct read-off of pair of values of g and F and full substitution of values of g , G , M and F into equation e.g. $L = (4\pi \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 1.83 \times 10^3) / (8.0 \times 10^{-3})$	C1
	$L = 3.8 \times 10^{26} \text{ W}$	A1
2(c)(iii)	$L = 4\pi\sigma r^2 T^4$	C1
	$3.8 \times 10^{26} = (4\pi \times 5.67 \times 10^{-8} \times 5780^4) \times r^2$	
	$r = 6.9 \times 10^8 \text{ m}$	A1

PUBLISHED

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause change of state)	B1
	(thermal) energy to change state at constant temperature	B1
3(b)(i)	$W = p\Delta V$	C1
	$= 1.0 \times 10^5 \times 0.017 = 1700 \text{ J} = 1.7 \text{ kJ}$	A1
3(b)(ii)	$\Delta U = Q + W$	C1
	$Q = 17.6 + 1.7$ $= 19.3 \text{ kJ}$	A1
3(b)(iii)	mass $= 710 \times 7.2 \times 10^{-5}$ ($= 0.051 \text{ kg}$)	C1
	$L = 19.3 / 0.051$ $= 380 \text{ kJ kg}^{-1}$	A1
3(c)	fusion involves (much) smaller volume change (than vaporisation)	B1
	smaller change in intermolecular spacing so smaller change in internal energy	B1
	negligible work done (by substance during fusion) so L_F is less (than L_V)	B1

Question	Answer	Marks
4(a)	- molecules are in (constant) random motion (all) collisions between molecules are (perfectly) elastic - no forces between molecules (except during collisions) - volume of molecules is negligible (compared with volume of gas) - collisions involving molecules are instantaneous <i>Any three points, 1 mark each</i>	B3
4(b)	- molecules collide with (walls of) container - momentum of molecule changes during collision (with walls) - change in momentum is caused by force on molecule by wall - molecule experiences force from wall so molecule exerts force on wall - many molecules exerting force across the area of the wall leads to pressure (on the wall) <i>Any three points, 1 mark each</i>	B3
4(c)	<i>Any three bulleted points from:</i> - both gases are ideal <i>Up to 2 points from:</i> - mass of one molecule of gas X is 3.3×10^{-27} kg - mass of one molecule of gas Y is 6.6×10^{-27} kg - mass of one molecule of gas Y is double mass of one molecule of gas X <i>Up to 2 points from:</i> - sample of X contains 0.27 mol / 1.6×10^{23} molecules - sample of Y contains 0.81 mol / 4.9×10^{23} molecules - sample of Y contains treble the amount of gas / number of molecules as sample of X <i>Up to 2 points from:</i> - mass of gas X is 5.4×10^{-4} kg - mass of gas Y is 3.2×10^{-3} kg - mass of gas Y is six times mass of gas X	B3

PUBLISHED

Question	Answer	Marks
5(a)	arrow from sphere, perpendicular to string, pointing left and down	B1
5(b)(i)	amplitude = 0.016 m	A1
5(b)(ii)	angular frequency = $2\pi / T$	C1
	$= 2\pi / 0.40$ $= 16 \text{ rad s}^{-1}$	A1
5(b)(iii)	total energy = $\frac{1}{2}m\omega^2x_0^2$	C1
	$= \frac{1}{2} \times 0.15 \times 15.7^2 \times 0.016^2$ $= 4.7 \times 10^{-3} \text{ J}$	A1
5(c)	dome-shaped curve starting and ending on the x-axis, with peak at $x = 0$	B1
	maximum E_K shown as $4.7 \times 10^{-3} \text{ J}$	B1
	minimum x shown as -0.016 m and maximum x shown as $+0.016 \text{ m}$ at the ends of the line	B1

PUBLISHED

Question	Answer	Marks
6(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
6(b)	at least four straight, radial lines to/from surface of sphere	B1
	at least four straight radial lines drawn, approximately equally spaced	B1
	arrows pointing away from the surface of the sphere	B1
6(c)(i)	radius = 3.2 cm	A1
6(c)(ii)	$E = Q / (4\pi\epsilon_0 X^2)$	C1
	$Q = \text{e.g. } 2.2 \times 10^5 \times 4\pi \times 8.85 \times 10^{-12} \times 0.032^2$	C1
	$= 2.5 \times 10^{-8} \text{ C}$	A1
6(c)(iii)	- the (positive) charge is all the way around the surface - a charge placed inside the sphere is pulled equally in all directions - if the field was not zero, the charges would move (until field is zero) - electric field lines go from positive charge to negative charge, and there are no negative charges inside the sphere <i>Any point, 1 mark</i>	B1

PUBLISHED

Question	Answer	Marks							
7(a)	charge / potential (difference)	M1							
	charge is charge on one plate, <u>and</u> potential is p.d. between the plates	A1							
7(b)(i)	straight line starting at the origin	B1							
	line with positive gradient ending at (V, Q)	B1							
7(b)(ii)	work done is the area under the graph	B1							
	$W = \frac{1}{2}QV$	A1							
7(c)(i)	final p.d. shown as $V/4$ for both capacitors	B1							
	final charges add together to give Q	B1							
	charge on Y = $3 \times$ charge on X (and both charges shown as a multiple of Q)	B1							
	Fully correct answer: <table border="1"> <tr> <td></td><td>X</td><td>Y</td></tr> <tr> <td>final p.d.</td><td>$V/4$</td><td>$V/4$</td></tr> <tr> <td>final charge</td><td>$Q/4$</td><td>$3Q/4$</td></tr> </table>			X	Y	final p.d.	$V/4$	$V/4$	final charge
	X	Y							
final p.d.	$V/4$	$V/4$							
final charge	$Q/4$	$3Q/4$							
7(c)(ii)	less than	B1							

PUBLISHED

Question	Answer	Marks
8(a)	number of cycles per unit time	B1
8(b)(i)	period = $2\pi / 40\pi = 0.050 \text{ s} = 50 \text{ ms}$	A1
8(b)(ii)	sinusoidal curve, starting at (0, 0) and initially increasing from there	B1
	periodic line showing 2 cycles with period 50 ms from $t = 0$ to $t = 100 \text{ ms}$	B1
	all peaks shown at $I = +3.5 \text{ A}$ and all troughs shown at $I = -3.5 \text{ A}$	B1
8(b)(iii)	$I_{\text{r.m.s}} = 3.5 / \sqrt{2}$ $= 2.5 \text{ A}$	A1
8(c)	$P = I^2 R$	C1
	peak power = $3.5^2 \times 680 (= 8330 \text{ W})$ or mean power = $2.47^2 \times 680 (= 4170 \text{ W})$	M1
	peak and mean powers both calculated correctly, with supporting working, and compared leading to conclusion that mean power is half the peak power	A1

PUBLISHED

Question	Answer	Marks
9(a)	diffraction is characteristic of wave behaviour so shows that electrons can behave like waves	B1
9(b)	$qV = \frac{1}{2}mv^2$	C1
	$p = mv$	C1
	$p = m \times \sqrt{(2qV/m)}$ $= \sqrt{(2qVm)}$	A1
9(c)	(electrons have) greater momentum <u>so</u> smaller (de Broglie) wavelength	B1
	fringes become closer together	B1
9(d)(i)	straight line with positive gradient	B1
	line with positive gradient passing through the origin	B1
9(d)(ii)	Planck constant	B1

PUBLISHED

Question	Answer	Marks
10(a)(i)	cannot predict when a particular <u>nucleus</u> will decay or cannot predict which <u>nucleus</u> will decay next	B1
10(a)(ii)	(decay is) not affected by external (environmental) factors	B1
10(a)(iii)	fluctuations in (measured) count rate	B1
10(b)(i)	- large nuclei undergo fission whereas small nuclei undergo fusion - fission involves one nucleus splitting into two (or more) (smaller) nuclei - fusion involves two nuclei joining together to form one (larger) nucleus - fission is (usually) initiated by neutron bombardment - fusion is (usually) initiated by (very) high temperatures <i>Any three points, 1 mark each</i>	B3
10(b)(ii)	binding energy <u>per nucleon</u> is greatest for intermediate nucleon numbers (may be shown on sketch graph with axes labelled 'binding energy <u>per nucleon</u> ' and 'nucleon number')	B1
	both fusion and fission involve an increase in binding energy (per nucleon)	B1

Cambridge International AS & A Level

PHYSICS**9702/43**

Paper 4 A Level Structured Questions

October/November 2024

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **15** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)(i)	(gravitational) force acts perpendicular to direction of motion	B1
	gravitational force provides centripetal acceleration	B1
1(b)(ii)	$(F =) GMm / x^2 = m\omega^2$ and $\omega = 2\pi / T$ or $GMm / x^2 = 4\pi^2 mx / T^2$	M1
	completion of algebra leading to $x^3 = GMT^2 / 4\pi^2$	A1
	clear indication that B = radius of planet and that A = mass (of planet)	B1
1(b)(iii)	gradient = $\sqrt[3]{(4\pi^2 / GA)}$	C1
	e.g. $(1280 - 360) / (12 \times 10^6) = \sqrt[3]{(4\pi^2 / [6.67 \times 10^{-11} \times A])}$	C1
	$A = 1.3 \times 10^{24} \text{ kg}$	A1
	intercept = gradient $\times B$	C1
	e.g. $360 = ((1280 - 360) \times B) / (12 \times 10^6)$ $B = 4.7 \times 10^6 \text{ m}$	A1

PUBLISHED

Question	Answer	Marks
2(a)	(thermal) energy per unit mass (to change temperature)	B1
	(thermal) energy per unit change in temperature	B1
2(b)(i)	<i>Any three bulleted points from:</i> - the blocks end up in thermal equilibrium - heat capacity of Y is larger than heat capacity of X - no heat loss to the surroundings <i>Up to 2 points from these six:</i> - initial temperature of X = 85 °C - initial temperature of Y = 25 °C - the temperature change of X = 45 °C - the temperature change of Y = 15 °C - the temperature change in X is three times that in Y - final temperature of both = 40 °C	B3
2(b)(ii)	$\Delta\theta = 45\text{ °C}$ for X and 15 °C for Y	C1
	$mc \times 45 = 1.3 \times m \times 901 \times 15$	C1
	$c = 390\text{ J kg K}^{-1}$	A1

PUBLISHED

Question	Answer	Marks
3(a)(i)	number of particles per unit amount of substance	B1
3(a)(ii)	$N_A = R / k$	B1
3(b)(i)	X pressure and Y pressure <u>both</u> = NkT / V	B1
	X amount = N / N_A and Y amount = $2N / N_A$	B1
	X mean-square speed = $3kT / m$ and Y mean-square speed = $3kT / 2m$	B1
	X internal energy = $3NkT / 2$ and Y internal energy = $3NkT$	B1
3(b)(ii)	line passing through the origin and not returning to either axis	B1
	curve with positive decreasing gradient	B1

PUBLISHED

Question	Answer	Marks
4(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): acceleration is (always) in the opposite <u>direction</u> to displacement or acceleration is (always) <u>directed</u> towards a fixed point	B1
4(b)(i)	amplitude = $(9.5 - 3.5) / 2$ = 3.0 cm	A1
4(b)(ii)	$\omega = v_0 / x_0$	C1
	= $9.5 / 3.0 = 3.2 \text{ rad s}^{-1}$	A1
4(b)(iii)	$T = 2\pi / \omega$	C1
	= $2\pi / 3.2$	A1
	= 2.0 s	
4(b)(iv)	attempted sinusoidal curve starting with a minimum at $t = 0$	B1
	sinusoidal curve of period 2.0 s from $t = 0$ to $t = 6.0$ s	B1
	all peaks shown at $h = 9.5$ cm	B1
	all troughs shown at $h = 3.5$ cm	B1

PUBLISHED

Question	Answer	Marks
5(a)	(electric) field equals (electric) potential gradient	M1
	reference to minus sign	A1
5(b)	- for potential to be zero, one potential must be positive and the other potential must be negative - for potential to be zero, the charges must have opposite sign - for field to be zero, the fields (due to X and Y) must be in opposite directions - for field to be zero, the charges must have the same sign - the signs of the charges cannot (simultaneously) be both the same and opposite (so not possible) <i>Any three points, 1 mark each</i>	B3
5(c)(i)	$V_X = (-) Q / 4\pi\epsilon_0 x$ and $V_Y = (-) 2Q / 4\pi\epsilon_0 y$	C1
	$(V_X + V_Y = 0 \text{ so } Q / 4\pi\epsilon_0 x = 2Q / 4\pi\epsilon_0 y \text{ leading to } y = 2x)$	A1
5(c)(ii)	$E_X = Q / 4\pi\epsilon_0 x^2$	A1
5(c)(iii)	$E_Y = 2Q / 4\pi\epsilon_0 (2x)^2$ $(= Q / 8\pi\epsilon_0 x^2)$	C1
	(opposite charges so fields in same direction so magnitudes add): $E = (Q / 4\pi\epsilon_0 x^2) + (Q / 8\pi\epsilon_0 x^2)$ $= 3Q / 8\pi\epsilon_0 x^2$	A1

PUBLISHED

Question	Answer	Marks
6(a)(i)	conversion (from a.c.) to d.c.	B1
6(a)(ii)	half-wave: voltage in one direction is removed	B1
	full-wave: voltage in one direction is reversed	B1
6(b)(i)	one gap connected by a single diode and other gap connected directly	B1
	diode drawn (in a circuit) with correct circuit symbol	B1
6(b)(ii)	smoothing	B1
6(c)(i)	$E = \frac{1}{2}CV^2$	C1
	$C = 2 \times 0.041 / 12^2 = 5.7 \times 10^{-4} \text{ F} = 570 \mu\text{F}$	A1
6(c)(ii)	$8.0 = 12.0 \exp(-0.010 / RC)$	C1
	$\ln(8.0 / 12.0) = -0.010 / (R \times 5.7 \times 10^{-4})$	C1
	$R = 43 \Omega$	A1

PUBLISHED

Question	Answer	Marks
7(a)	• force per unit length • force per unit current • length / current perpendicular to field <i>1 mark for any two points, 2 marks for all three points</i>	B2
7(b)	concentric circles around the wire (at least two circles needed)	B1
	spacing between circles increases with distance from wire (at least four circles needed)	B1
	arrows showing direction of field is clockwise	B1
7(c)(i)	(each) wire sits in the (magnetic) field created by the other	B1
	current (in one wire) is perpendicular to (magnetic) field (due to other wire) so (magnetic) force acts (on wire)	B1
7(c)(ii)	arrow drawn, starting from X and pointing towards Y, labelled F	B1
7(c)(iii)	(forces have) equal magnitudes	B1
	(forces are in) opposite directions	B1
7(c)(iv)	no change (in the direction of the force) since both the current in X and the field due to Y have reversed	B1

PUBLISHED

Question	Answer	Marks
8(a)	photoelectric effect	B1
8(b)(i)	$E = hf$	C1
	work function = $6.63 \times 10^{-34} \times 8.8 \times 10^{14}$ $= 5.8 \times 10^{-19} \text{ J}$	A1
8(b)(ii)	$hf = \phi + \frac{1}{2}mv_{\text{MAX}}^2$	C1
	$6.63 \times 10^{-34} \times 11 \times 10^{14} = (5.8 \times 10^{-19}) + (\frac{1}{2} \times 9.11 \times 10^{-31} \times v_{\text{MAX}}^2)$	C1
	$v_{\text{MAX}} = 5.7 \times 10^5 \text{ m s}^{-1}$	A1
8(c)	E_{MAX} shown as zero from $f = 8.0$ to 8.8 and non-zero from $f = 8.8$ to 11	B1
	all non-zero E_{MAX} shown as a single straight line with a positive gradient	B1
	line passing through (11, 1.45)	B1

PUBLISHED

Question	Answer	Marks
9(a)(i)	positron	B1
9(a)(ii)	$\lambda = \ln 2 / (110 \times 60) = 1.05 \times 10^{-4} \text{ s}^{-1}$	A1
9(a)(iii)	$N = M / (18 \text{ u})$ or $(M \text{ in grams} \times N_A / 18)$ $N = (2.1 \times 10^{-12}) / (18 \times 1.66 \times 10^{-27})$ or $(2.1 \times 10^{-9} \times 6.02 \times 10^{23}) / 18$ $(= 7.0 \times 10^{13})$	C1
	$A = \lambda N$	C1
	$= 1.05 \times 10^{-4} \times 7.0 \times 10^{13}$ $= 7.4 \times 10^9 \text{ Bq}$	A1
9(b)(i)	(pair) annihilation occurs - the <u>mass</u> of the two particles is converted into energy - two <u>gamma</u> photons are formed and travel in opposite directions or - two <u>gamma</u> photons are formed and leave the body - difference in arrival times of photons (at detector) is processed <i>Any three points, 1 mark each</i>	B3
9(b)(ii)	with a shorter half-life: sample would (almost) fully decay before the test is complete	B1
	a longer half-life: exposes patient to harmful/ionising radiation unnecessarily or with a longer half-life: a larger dose (of tracer) needed to produce detectable activity	B1

PUBLISHED

Question	Answer	Marks
10(a)	<u>redshift</u> is the increase in observed wavelength / decrease in observed frequency (caused by Doppler effect) - radiation from distant galaxies is observed to be redshifted - redshift provides evidence that galaxies are moving apart <u>galaxies</u> moving apart means Universe must be expanding <i>Any three points, 1 mark each</i>	B3
10(b)(i)	$F = L / 4\pi d^2$	C1
	$d = \sqrt{(1.90 \times 10^{36} / [4\pi \times 8.42 \times 10^{-16}])}$ $= 1.34 \times 10^{25} \text{ m}$	A1
10(b)(ii)	$\Delta\lambda / \lambda = v / c$ $(726 - 658) / 658 = v / (3.00 \times 10^8)$	C1
	$v = 3.1 \times 10^7 \text{ m s}^{-1}$	A1
10(c)(i)	line with positive gradient passing through the origin	B1
	straight line with positive gradient	B1
10(c)(ii)	Hubble constant	B1

Cambridge International AS & A Level

PHYSICS**9702/51**

Paper 5 Planning, Analysis and Evaluation

October/November 2024

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **11** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	s is the independent variable and v is the dependent variable or vary s and measure v	1
	keep D <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - light gate positioned at P - light gate connected to timer / data logger - labels for light gate and P and data logger / timer and at least one other label from block, magnet(s), trolley, s and D	1
	measure D with a rule(r) and measure L with a rule(r) or calipers	1
	description to determine v at P, e.g. (measure length of) card to interrupt beam	1
	method to measure s , e.g. use calipers	1

PUBLISHED

Question	Answer	Marks
1	Method of Analysis	
	plot a graph of v^2 against $\frac{1}{s^4}$ or equivalent (e.g. $\frac{1}{s^4}$ against v^2) Do not accept logarithms.	1
	$K = \frac{m \times \text{gradient}}{2DA^2B^2L^2}$ (or $K = \frac{m}{2DA^2B^2L^2 \times \text{gradient}}$ for $\frac{1}{s^4}$ against v^2)	1
	$Q = -\frac{m \times y\text{-intercept}}{2D}$ (or $Q = KA^2B^2L^2 \times y\text{-intercept}$ or $Q = \frac{m \times y\text{-intercept}}{2D \times \text{gradient}}$ for $\frac{1}{s^4}$ against v^2)	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
	D1 method to <u>stop</u> the trolley (after passing point P), e.g. labelled block / buffer / cushion drawn after P or place a block / buffer / cushion after P to <u>stop</u> the trolley	
	D2 keep L , A , m and B <u>constant</u>	
	D3 use micrometer / calipers to measure diameter (d) of the magnet and $A = \frac{\pi d^2}{4}$	
	D4 method to secure block to bench, e.g. clamp block to bench or (heavy) mass on top of block or method to secure magnets, e.g. use glue to stick magnets to trolley / block	
	D5 method to increase the accuracy of measuring s or D , e.g. use a marker to left of the trolley	
	D6 measure B using a (calibrated) Hall probe and adjust / rotate probe until <u>maximum</u> value or measure B using Hall probe first in one direction, then in the opposite direction and average	
	D7 use a (top-pan) balance to measure m	
	D8 use of strong magnets to increase v	
	D9 repeat measurements of v for each value of s and average v	
	D10 relationship valid <u>if</u> a straight line is produced (passing through $\left(-\frac{2DQ}{m}\right)$) Do not accept line passing through the origin.	

PUBLISHED

Question	Answer	Marks							
2(a)	$\text{gradient} = \frac{3}{3E - E_s}$ $y\text{-intercept} = \frac{2Z}{3E - E_s}$	1							
2(b)	<table><tr><td>$\frac{1}{I} / \text{A}^{-1}$</td></tr><tr><td>5150 or 5155</td></tr><tr><td>5560 or 5556</td></tr><tr><td>5810 or 5814</td></tr><tr><td>6250</td></tr><tr><td>6670 or 6667</td></tr><tr><td>6940 or 6944</td></tr></table> Values correct as shown above.	$\frac{1}{I} / \text{A}^{-1}$	5150 or 5155	5560 or 5556	5810 or 5814	6250	6670 or 6667	6940 or 6944	1
$\frac{1}{I} / \text{A}^{-1}$									
5150 or 5155									
5560 or 5556									
5810 or 5814									
6250									
6670 or 6667									
6940 or 6944									
	Uncertainties in $\frac{1}{I}$ from ± 50 or ± 60 to ± 90 or ± 100 .	1							
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1							
	Error bars in $\frac{1}{I}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1							

PUBLISHED

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (1.63, 5400) and (1.67, 5400) and between (2.58, 6800) and (2.62, 6800).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and x into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

PUBLISHED

Question	Answer	Marks
2(d)(i)	E determined using gradient and E and Z given to 2 or 3 or 4 significant figures. $E = \frac{1}{3} \left(\frac{3}{\text{gradient}} + E_s \right) = \frac{3 + \text{gradient} \times E_s}{3 \times \text{gradient}} = \frac{1}{\text{gradient}} + \frac{E_s}{3}$ $E = \frac{1}{\text{gradient}} + 0.733$	1
	Z determined using y-intercept and E and Z given with SI units with correct powers of ten. $Z = \frac{(3E - E_s) \times \text{y-intercept}}{2} \text{ or } Z = \frac{3 \times \text{y-intercept}}{2 \times \text{gradient}}$ Unit of E: V Unit of Z: Ω	1
2(d)(ii)	Absolute uncertainty in E with method shown. $\text{uncertainty} = \left(\frac{\Delta \text{gradient}}{\text{gradient}} \times \frac{1}{\text{gradient}} \right) + \frac{0.05}{3}$ or correct substitution for max/min methods.	1

PUBLISHED

Question	Answer	Marks
2(e)	Value of R determined to a minimum of two significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution and correct use of power of ten. $R = \frac{\frac{1}{250 \times 10^{-6}} - y\text{-intercept}}{\text{gradient}}$ or $R = \frac{1}{\text{gradient} \times 250 \times 10^{-6}} - \frac{2Z}{3}$ or $R = \frac{3E - 2.2}{3 \times 250 \times 10^{-6}} - \frac{2Z}{3}$	1

Cambridge International AS & A Level

PHYSICS**9702/52**

Paper 5 Planning, Analysis and Evaluation

October/November 2024

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **10** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	L is the independent variable and $\Delta\theta$ or <u>temperature change/increase</u> is the dependent variable or vary L and measure $\Delta\theta$ or <u>temperature change/increase</u>	1
	keep t <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - oil in a beaker/container (on a bench) - coil fully submerged in oil (bulb of) thermometer in the oil - at least three labels from thermometer, coil or resistance wire, oil, beaker/container, clamp/stand, bench Do not accept other heating sources.	1
	method to determine V – diagram of workable circuit including: - power supply connected to wire - voltmeter positioned to measure V across the coil	1
	measure the initial and final temperature <u>and</u> find the difference $\Delta\theta$	1
	method to determine t , e.g. use stopwatch/timer and method to determine L e.g. use a rule(r) to measure L / length of wire or e.g. using number of turns and measure the diameter of the coil with rule(r) / calipers	1

PUBLISHED

Question	Answer	Marks				
1	Method of Analysis					
	plot a graph of $\Delta\theta$ against $\frac{1}{L}$ or equivalent, e.g. $\frac{1}{L}$ against $\Delta\theta$ Do not accept logarithms.	1				
	<table><tr><td>for $\Delta\theta$ against $\frac{1}{L}$</td><td>for $\frac{1}{L}$ against $\Delta\theta$</td></tr><tr><td>$K = \frac{AtV^2}{m \times \text{gradient}}$</td><td>$K = \frac{AtV^2 \times \text{gradient}}{m}$</td></tr></table>	for $\Delta\theta$ against $\frac{1}{L}$	for $\frac{1}{L}$ against $\Delta\theta$	$K = \frac{AtV^2}{m \times \text{gradient}}$	$K = \frac{AtV^2 \times \text{gradient}}{m}$	1
	for $\Delta\theta$ against $\frac{1}{L}$	for $\frac{1}{L}$ against $\Delta\theta$				
	$K = \frac{AtV^2}{m \times \text{gradient}}$	$K = \frac{AtV^2 \times \text{gradient}}{m}$				
<table><tr><td>for $\Delta\theta$ against $\frac{1}{L}$</td><td>for $\frac{1}{L}$ against $\Delta\theta$</td></tr><tr><td>$Z = -mK \times y\text{-intercept}$ or $Z = -\frac{AtV^2 \times y\text{-intercept}}{\text{gradient}}$</td><td>$Z = AtV^2 \times y\text{-intercept}$</td></tr></table>	for $\Delta\theta$ against $\frac{1}{L}$	for $\frac{1}{L}$ against $\Delta\theta$	$Z = -mK \times y\text{-intercept}$ or $Z = -\frac{AtV^2 \times y\text{-intercept}}{\text{gradient}}$	$Z = AtV^2 \times y\text{-intercept}$	1	
for $\Delta\theta$ against $\frac{1}{L}$	for $\frac{1}{L}$ against $\Delta\theta$					
$Z = -mK \times y\text{-intercept}$ or $Z = -\frac{AtV^2 \times y\text{-intercept}}{\text{gradient}}$	$Z = AtV^2 \times y\text{-intercept}$					

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	precaution linked to <u>hot</u> oil / beaker / wire, e.g. use of gloves to prevent burns from oil or precaution linked to spillage of oil, e.g. perform experiment in a tray	
D2	keep A and m and V <u>constant</u>	
D3	use a micrometer to measure the <u>diameter</u> (d) of the <u>wire</u> and $A = \frac{\pi d^2}{4}$	
D4	repeat measurements of d <u>along the wire</u> and average	
D5	method to reduce heat loss e.g. add insulation around the container / add a lid to the container	
D6	method to keep V constant, e.g. adjust / change a variable resistor / power supply to keep V or voltmeter reading constant	
D7	use a balance to determine the mass of the oil and mass of oil = mass of (beaker + oil) – mass of beaker or place beaker on balance and zero balance, then add oil and read balance	
D8	stir the oil for uniform temperature or keep the initial temperature (of oil) constant	
D9	repeat the experiment for the same value of L and average $\Delta\theta$ / average temperature <u>change</u>	
D10	relationship valid <u>if</u> a straight line is produced (passing through $\left(-\frac{Z}{mK}\right)$) Do not accept line passing through the origin.	
D11	method to determine L accurately, e.g. measure length of unwound coil	

PUBLISHED

Question	Answer	Marks							
2(a)	gradient = Bn^2 y -intercept = $-B$	1							
2(b)	<table><tr><td>d^2 / cm^2</td></tr><tr><td>615 or 615.0</td></tr><tr><td>458 or 458.0</td></tr><tr><td>292 or 292.4</td></tr><tr><td>216 or 216.1</td></tr><tr><td>166 or 166.4</td></tr><tr><td>139 or 139.2</td></tr></table> Values correct as shown above.	d^2 / cm^2	615 or 615.0	458 or 458.0	292 or 292.4	216 or 216.1	166 or 166.4	139 or 139.2	1
d^2 / cm^2									
615 or 615.0									
458 or 458.0									
292 or 292.4									
216 or 216.1									
166 or 166.4									
139 or 139.2									
	Uncertainties in d^2 decreasing from ± 10 to ± 4 or ± 5 .	1							
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1							
	Error bars in d^2 plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1							
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (1.90, 250) and (2.00, 250) and between (3.55, 500) and (3.65, 500).	1							
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1							

Question	Answer	Marks
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and x into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1
2(d)(i)	B determined using y-intercept ($B = -y\text{-intercept}$) and B and n given to 2 or 3 or 4 significant figures.	1
	n determined using gradient and B and n given with SI units with correct powers of ten. $n = \sqrt{\frac{\text{gradient}}{B}} \quad \text{or} \quad n = \sqrt{\frac{\text{gradient}}{-y\text{-intercept}}}$ Unit for B : cm^2 No unit for n .	1

PUBLISHED

Question	Answer	Marks
2(d)(ii)	Percentage uncertainty in n determined with method shown. $\text{percentage uncertainty} = \frac{1}{2} \left(\frac{\Delta y\text{-intercept}}{y\text{-intercept}} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times 100$ or correct substitution for max/min methods.	1
2(e)	θ determined to a minimum of two significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution and correct power of ten. $\theta = \sin^{-1} \sqrt{\frac{\text{gradient}}{-y\text{-intercept} + 900}}$ or $\theta = \sin^{-1} \sqrt{\frac{n^2 B}{B + 900}}$	1

Cambridge International AS & A Level

PHYSICS**9702/53**

Paper 5 Planning, Analysis and Evaluation

October/November 2024

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **11** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	s is the independent variable and v is the dependent variable or vary s and measure v	1
	keep D <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - light gate positioned at P - light gate connected to timer / data logger - labels for light gate and P and data logger / timer and at least one other label from block, magnet(s), trolley, s and D	1
	measure D with a rule(r) and measure L with a rule(r) or calipers	1
	description to determine v at P, e.g. (measure length of) card to interrupt beam	1
	method to measure s , e.g. use calipers	1

PUBLISHED

Question	Answer	Marks
1	Method of Analysis	
	plot a graph of v^2 against $\frac{1}{s^4}$ or equivalent (e.g. $\frac{1}{s^4}$ against v^2) Do not accept logarithms.	1
	$K = \frac{m \times \text{gradient}}{2DA^2B^2L^2}$ (or $K = \frac{m}{2DA^2B^2L^2 \times \text{gradient}}$ for $\frac{1}{s^4}$ against v^2)	1
	$Q = -\frac{m \times y\text{-intercept}}{2D}$ (or $Q = KA^2B^2L^2 \times y\text{-intercept}$ or $Q = \frac{m \times y\text{-intercept}}{2D \times \text{gradient}}$ for $\frac{1}{s^4}$ against v^2)	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
	D1 method to <u>stop</u> the trolley (after passing point P), e.g. labelled block / buffer / cushion drawn after P or place a block / buffer / cushion after P to <u>stop</u> the trolley	
	D2 keep L , A , m and B <u>constant</u>	
	D3 use micrometer / calipers to measure diameter (d) of the magnet and $A = \frac{\pi d^2}{4}$	
	D4 method to secure block to bench, e.g. clamp block to bench or (heavy) mass on top of block or method to secure magnets, e.g. use glue to stick magnets to trolley / block	
	D5 method to increase the accuracy of measuring s or D , e.g. use a marker to left of the trolley	
	D6 measure B using a (calibrated) Hall probe and adjust / rotate probe until <u>maximum</u> value or measure B using Hall probe first in one direction, then in the opposite direction and average	
	D7 use a (top-pan) balance to measure m	
	D8 use of strong magnets to increase v	
	D9 repeat measurements of v for each value of s and average v	
	D10 relationship valid <u>if</u> a straight line is produced (passing through $\left(-\frac{2DQ}{m}\right)$) Do not accept line passing through the origin.	

PUBLISHED

Question	Answer	Marks							
2(a)	$\text{gradient} = \frac{3}{3E - E_s}$ $y\text{-intercept} = \frac{2Z}{3E - E_s}$	1							
2(b)	<table><tr><td>$\frac{1}{I} / \text{A}^{-1}$</td></tr><tr><td>5150 or 5155</td></tr><tr><td>5560 or 5556</td></tr><tr><td>5810 or 5814</td></tr><tr><td>6250</td></tr><tr><td>6670 or 6667</td></tr><tr><td>6940 or 6944</td></tr></table> Values correct as shown above.	$\frac{1}{I} / \text{A}^{-1}$	5150 or 5155	5560 or 5556	5810 or 5814	6250	6670 or 6667	6940 or 6944	1
$\frac{1}{I} / \text{A}^{-1}$									
5150 or 5155									
5560 or 5556									
5810 or 5814									
6250									
6670 or 6667									
6940 or 6944									
	Uncertainties in $\frac{1}{I}$ from ± 50 or ± 60 to ± 90 or ± 100 .	1							
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1							
	Error bars in $\frac{1}{I}$ plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1							

PUBLISHED

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Do not accept line from top point to bottom point. Points must be balanced. Line must pass between (1.63, 5400) and (1.67, 5400) and between (2.58, 6800) and (2.62, 6800).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and x into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

PUBLISHED

Question	Answer	Marks
2(d)(i)	E determined using gradient and E and Z given to 2 or 3 or 4 significant figures. $E = \frac{1}{3} \left(\frac{3}{\text{gradient}} + E_s \right) = \frac{3 + \text{gradient} \times E_s}{3 \times \text{gradient}} = \frac{1}{\text{gradient}} + \frac{E_s}{3}$ $E = \frac{1}{\text{gradient}} + 0.733$	1
	Z determined using y-intercept and E and Z given with SI units with correct powers of ten. $Z = \frac{(3E - E_s) \times \text{y-intercept}}{2} \text{ or } Z = \frac{3 \times \text{y-intercept}}{2 \times \text{gradient}}$ Unit of E: V Unit of Z: Ω	1
2(d)(ii)	Absolute uncertainty in E with method shown. $\text{uncertainty} = \left(\frac{\Delta \text{gradient}}{\text{gradient}} \times \frac{1}{\text{gradient}} \right) + \frac{0.05}{3}$ or correct substitution for max/min methods.	1

PUBLISHED

Question	Answer	Marks
2(e)	Value of R determined to a minimum of two significant figures from (c)(iii) and (c)(iv) or (d)(i) with correct substitution and correct use of power of ten. $R = \frac{\frac{1}{250 \times 10^{-6}} - y\text{-intercept}}{\text{gradient}}$ or $R = \frac{1}{\text{gradient} \times 250 \times 10^{-6}} - \frac{2Z}{3}$ or $R = \frac{3E - 2.2}{3 \times 250 \times 10^{-6}} - \frac{2Z}{3}$	1

Cambridge International AS & A Level

PHYSICS**9702/41**

Paper 4 A Level Structured Questions

October/November 2025**MARK SCHEME**Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **17** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	arithmetic error
	benefit of the doubt given
	contradiction in response, mark not awarded
	correct point or mark awarded
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect physics
	incorrect point or mark not awarded
	information missing or insufficient for credit

Annotation	Meaning
MO	mandatory mark not awarded
SEEN	point has been noted, but no credit has been given or blank page seen
POT	power of ten error
TE	transcription error

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	velocity and acceleration both have constant magnitude	B1
	velocity is (always) perpendicular to acceleration	B1
1(b)(i)	$v = R\omega$	A1
1(b)(ii)	$a = R\omega^2$ or $a = v^2 / R$	C1
	$a = v\omega$	A1
1(c)(i)	$x = R \sin \theta$	A1
1(c)(ii)	$\theta = \omega t$	A1
1(c)(iii)	clear substitution of $\theta = \omega t$ into $x = R \sin \theta$ leading to $x = R \sin \omega t$	A1
1(c)(iv)	equation is of the form $x = x_0 \sin \omega t$ (so simple harmonic motion)	B1
1(d)(i)	amplitude = $0.46 / 2$ = 0.23 m	A1
1(d)(ii)	$\omega = 2\pi / T$	C1
	period = $2\pi / 1.9$ = 3.3 s	A1

PUBLISHED

Question	Answer	Marks
1(d)(iii)	$a_0 = \omega^2 x_0$	C1
	$= 1.9^2 \times 0.23$	A1
	$= 0.83 \text{ m s}^{-2}$	
1(e)	shadow on screen, labelled A, above left-hand edge of the circular path	B1

Question	Answer	Marks
2(a)	work done on / by system	B1
	thermal energy supplied to / removed from system	B1
2(b)(i)	no thermal energy transferred to / from system (due to lack of time)	B1
	work is done on the gas to compress it / to decrease its volume	B1
	internal energy increases so temperature increases	B1
2(b)(ii)	(during vaporisation) molecular separation increases	B1
	(heating causes) potential energy of molecules to increase	B1
	kinetic energy of molecules unchanged so temperature unchanged	B1

PUBLISHED

Question	Answer	Marks
3(a)	force per unit mass	B1
3(b)(i)	$F = GMm / x^2$	C1
	$g = F / m$	A1
	$g = [GMm / x^2] / m = GM / x^2$ and G = gravitational constant	
3(b)(ii)	arrow drawn at P pointing directly towards the point mass	B1
3(b)(iii)	fields are in opposite directions	B1
	field strength at Q is four times the field strength at P	B1
3(c)	line starting at $(R, -g_0)$ and ending at $(L - R, +g_0)$	B1
	line passing through $(L / 2, 0)$	B1
	curve becoming shallower from R to $(L / 2)$ and then steeper from $(L / 2)$ to $(L - R)$	B1

PUBLISHED

Question	Answer	Marks
4(a)(i)	temperature = $-273.15\text{ }^{\circ}\text{C}$	A1
4(a)(ii)	temperature = 0 K	A1
4(b)(i)	gas is ideal	B1
4(b)(ii)	$pV = NkT$	C1
	$N = 270 / (8.0 \times 10^{-21})$ $= 3.4 \times 10^{22}$	A1
4(b)(iii)	$n = (3.4 \times 10^{22}) / (6.02 \times 10^{23})$ $= 0.056\text{ mol}$	A1
4(c)	$\frac{1}{2} m\langle c^2 \rangle = (3 / 2) kT$	C1
	$\frac{1}{2} \times m \times 1900^2 = 1.5 \times 8.0 \times 10^{-21}$ $(m = 6.65 \times 10^{-27}\text{ kg})$	C1
	$m = (6.65 \times 10^{-27}) / (1.66 \times 10^{-27})$	C1
	$= 4.0\text{ u}$	A1

PUBLISHED

Question	Answer	Marks
5(a)	work done per unit charge	B1
	work (done) moving positive charge from infinity (to the point)	B1
5(b)(i)	potential (due to proton) = $(1.60 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12} \times 10 \times 10^{-12})$ or potential (due to electron) = $(-1.60 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12} \times 110 \times 10^{-12})$	C1
	$V = [(1.60 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12})] \times [(10^{-1} - 110^{-1}) \times 10^{12}] = 130 \text{ V}$	A1
5(b)(ii)	$V = [(1.60 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12})] \times [(30^{-1} - 90^{-1}) \times 10^{12}]$	C1
	= (+) 32 V	A1
5(b)(iii)	cross drawn midway between the electron and the proton	B1
5(b)(iv)	line from (10, +130) to (110, -130)	B1
	curve getting shallower until $x = 60 \text{ pm}$, crossing $V = 0$ at (60, 0) and then getting steeper after $x = 60 \text{ pm}$	B1
	curve passing through (30, ± 32) and (90, ± 32)	B1

PUBLISHED

Question	Answer	Marks
6(a)	series charges: $Q_s = Q_1 = Q_2$	B1
	series p.d.s: $V_s = V_1 + V_2$	B1
	parallel charges: $Q_s = Q_1 + Q_2$	B1
	parallel p.d.s: $V_s = V_1 = V_2$	B1
6(b)(i)	$E = \frac{1}{2} CV^2$	C1
	p.d. = $[(2 \times 19 \times 10^{-3}) / (470 \times 10^{-6})]^{\frac{1}{2}}$ $= 9.0 \text{ V}$	A1
6(b)(ii)	$E = Q^2 / 2C$ or $C = Q / V$	C1
	$Q = (19 \times 10^{-3} \times 2 \times 470 \times 10^{-6})^{\frac{1}{2}}$ or $Q = 470 \times 10^{-6} \times 9.0$ $Q = 4.2 \times 10^{-3} \text{ C}$	A1
6(b)(iii)	total charge unchanged	C1
	total capacitance = $(470 + 180) \times 10^{-6} \text{ (F)}$	C1
	$E = Q^2 / 2C = (4.23 \times 10^{-3})^2 / (2 \times 650 \times 10^{-6})$ (= 0.014 J) $E = 14 \text{ mJ}$	A1

PUBLISHED

Question	Answer	Marks
7(a)	(induced) e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)(i)	flux = e.m.f. $\times$ time	C1
	flux = 0.54×15 = 8.1 Wb	A1
7(b)(ii)	$\Phi = BA$	C1
	area = $8.1 / (38 \times 10^{-6})$ = $2.1 \times 10^5 \text{ m}^2$	A1
7(b)(iii)	area = speed $\times$ time $\times$ width	C1
	$v = (2.1 \times 10^5) / (15 \times 68)$ = 210 m s^{-1}	A1
7(b)(iv)	opposing force (due to current in wings) must be backwards	B1
	from Fleming's left-hand rule, current (in wings) must be from Q to P	B1
	current is from – to + inside an e.m.f. source so P is at higher potential	B1

PUBLISHED

Question	Answer	Marks
8(a)	packet / quantum of <u>energy</u>	M1
	of electromagnetic radiation	A1
8(b)(i)	$E = c^2 \Delta m$	C1
	$\Delta m = (4.274 \times 10^6 \times 1.60 \times 10^{-19}) / (1.66 \times 10^{-27} \times (3.00 \times 10^8)^2)$ (= 0.00458 u)	C1
	$m = 233.915174 + 4.000407 + 0.00458$ = 237.92016 u	A1
8(b)(ii)	$E = hc / \lambda$ or $E = hf$ and $c = f\lambda$	C1
	$(4.274 - 4.200) \times 1.60 \times 10^{-13} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$	C1
	$\lambda = 1.7 \times 10^{-11} \text{ m}$	A1
8(b)(iii)	(true) energy of gamma photon is smaller so (true) wavelength is larger	B1
8(c)	(anti)neutrinos are emitted during beta decay	B1
	particles emitted during beta decay carry varying amounts of energy, so energy of gamma photon is also variable (between decays)	B1

PUBLISHED

Question	Answer	Marks
9(a)	temperature inversely proportional to wavelength	M1
	temperature is thermodynamic temperature of surface of star and wavelength is the wavelength at which maximum emission rate from star occurs	A1
9(b)	<i>Any three points from:</i> • (surface) temperature of star X = 7000 K or star X has a higher temperature than the Sun • star X has a higher luminosity than the Sun • luminosity of star X = 2.7×10^{27} W • radius of star X = 1.3×10^9 m	B3
9(c)	light (from star X) is redshifted	B1
	wavelength of peak emission rate would be greater (using observed data)	B1

PUBLISHED

Question	Answer	Marks
10(a)	product of density and speed	M1
	speed of sound in medium (and density of the medium)	A1
10(b)	ultrasound waves cause crystal to vibrate	B1
	vibrations (of crystal) cause induced e.m.f. (across crystal)	B1
10(c)(i)	intensity reflection coefficient= $(40.4 - 1.48)^2 / (40.4 + 1.48)^2$	C1
	= 0.86	A1
10(c)(ii)	Z values are very similar	B1
	(almost) all the ultrasound will be transmitted or (almost) none of the ultrasound will be reflected	B1

Cambridge International AS & A Level

PHYSICS**9702/42**

Paper 4 A Level Structured Questions

October/November 2025**MARK SCHEME**Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **17** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	arithmetic error
	benefit of the doubt given
	contradiction in response, mark not awarded
	correct point or mark awarded
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect physics
	incorrect point or mark not awarded
	information missing or insufficient for credit

PUBLISHED

Annotation	Meaning
MO	mandatory mark not awarded
SEEN	point has been noted, but no credit has been given or blank page seen
POT	power of ten error
TE	transcription error

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)(i)	radius = $6.37 \times 10^6 \times \cos 52.2^\circ = 3.90 \times 10^6 \text{ m}$	A1
1(a)(ii)	period = 24 hours	C1
	$v = 2\pi r / T$ or $v = r\omega$ and $\omega = 2\pi / T$	C1
	$v = (2\pi \times 3.90 \times 10^6) / (24 \times 60 \times 60)$ $= 280 \text{ m s}^{-1}$	A1
1(b)(i)	$F = mv^2 / r$	C1
	$= (58.6 \times 280^2) / (3.90 \times 10^6)$ $= 1.2 \text{ N}$	A1
1(b)(ii)	arrow pointing horizontally to the left	B1
1(b)(iii)	arrow from student pointing along the dotted line, labelled 'weight'	B1
	upwards arrow from student pointing in a direction to the left of normal and above the tangent to the Earth, labelled 'contact force'	B1

PUBLISHED

Question	Answer	Marks
2(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
2(b)	<i>Any two points from:</i> • molecules are in <u>continuous</u> random motion • molecules have negligible volume compared with volume of gas • collisions (involving molecules) are (perfectly) elastic • collisions (of molecules) are instantaneous	B2
2(c)(i)	$pV = nRT$	C1
	$p = (0.0160 \times 8.31 \times 282) / (1.87 \times 10^{-4})$ $= 2.01 \times 10^5 \text{ Pa}$	A1
2(c)(ii)	number of molecules = $0.0160 \times 6.02 \times 10^{23}$	C1
	separation = $\sqrt[3]{[(1.87 \times 10^{-4}) / (0.0160 \times 6.02 \times 10^{23})]}$ $= 2.7 \times 10^{-9} \text{ m}$ (<i>allow any answer that is $3 \times 10^{-9} \text{ m}$ to one significant figure</i>)	A1
2(d)(i)	$F = 6.67 \times 10^{-11} \times (3.34 \times 10^{-27})^2 / (2.7 \times 10^{-9})^2$	C1
	$= 1.0 \times 10^{-46} \text{ N}$	A1
2(d)(ii)	numerical comparison between 10^{-46} N (F) and 10^{-26} N (the weight of molecule) leading to a conclusion that the assumption is supported	B1

PUBLISHED

Question	Answer	Marks
3(a)	same temperature	B1
	no <u>net</u> transfer of thermal energy (between them)	B1
3(b)(i)	density	B1
3(b)(ii)	<i>Any two points from:</i> - large response time / large time to reach equilibrium or cannot measure rapidly changing temperatures - reaching equilibrium requires (significant) transfer of energy or changes temperature of environment being measured or cannot measure temperature of small objects - bulky / difficult to set up or difficult to take readings / scale not calibrated to read temperature or cannot measure temperature of solid objects	B2
3(b)(iii)	substance with large mass or temperature that is constant (over time) or to calibrate other thermometers (in a laboratory)	B1

PUBLISHED

Question	Answer	Marks
3(b)(iv)	$0\text{ }^{\circ}\text{C} = 273\text{ K}$	C1
	$T = 273 \times (7.83 - 2.31) / (8.69 - 2.31)$ $(= 236\text{ K})$	C1
	$\theta = 236 - 273$ $= -37\text{ }^{\circ}\text{C}$	A1

Question	Answer	Marks
4(a)	<u>change</u> in internal energy = work done + energy transfer by heating	C1
	<u>increase</u> in internal energy = work done <u>on</u> system + energy transferred <u>to</u> the system by heating	A1
4(b)(i)	$U = (3/2) pV$	A1
4(b)(ii)	$pV = NkT$ <u>and</u> k identified as Boltzmann constant	B1
	$U = (3/2) NkT$	A1
4(c)(i)	$W = (+)8XY$	A1
4(c)(ii)	$W = -20XY$	A1
4(d)	work done during stages BC and DA = 0	B1
	change in internal energy (over complete cycle) = 0	C1
	thermal energy supplied = $20XY - 8XY$ $= (+)12XY$	A1

PUBLISHED

Question	Answer	Marks
5(a)(i)	straight line through the origin shows that a is proportional to x	B1
	negative gradient shows that a is always in the opposite direction to x	B1
5(a)(ii)	$a_0 = \omega^2 x_0$	C1
	$\omega = 2\pi / T$	C1
	$T = 2\pi \sqrt{(x_0 / a_0)}$ $= 2\pi \sqrt{(1.2 / 13)}$ $= 1.9 \text{ s}$	A1
5(b)(i)	loss of energy of oscillations	B1
	due to <u>resistive</u> force(s)	B1
5(b)(ii)	line starting from $x = \pm 1.2 \text{ cm}$ at $t = 0$	B1
	line starting from non-zero value of x from $t = 0$ to $t = 2T$ that is entirely either above or below the t -axis	B1
	curve from $t = 0$ starting from non-zero x value, with both magnitude of x value and magnitude of gradient continuously decreasing	B1

PUBLISHED

Question	Answer	Marks
6(a)	force per unit positive charge	B1
6(b)(i)	radial lines	B1
	arrows pointing away from the sphere	B1
6(b)(ii)	$C = Q / V$	C1
	$V = 83 / 69$ $= (+)1.2 \text{ V}$	A1
6(b)(iii)	$V = Q / 4\pi\epsilon_0 r$ $r = (83 \times 10^{-12}) / (4\pi \times 8.85 \times 10^{-12} \times 1.2)$	C1
	$= 0.62 \text{ m}$	A1
6(b)(iv)	$E = Q / 4\pi\epsilon_0 r^2$	C1
	$= (83 \times 10^{-12}) / (4\pi \times 8.85 \times 10^{-12} \times 0.62^2)$ $= 1.9 \text{ N C}^{-1}$	A1
6(c)	$26 = 83 \exp [-t / (120 \times 10^6 \times 69 \times 10^{-12})]$	C1
	$t = 9.6 \times 10^{-3} \text{ s}$	A1

PUBLISHED

Question	Answer	Marks
7(a)(i)	$T = 2\pi / 40\pi = 0.050 \text{ s}$	A1
7(a)(ii)	$V_{\text{r.m.s.}} = 18 / \sqrt{2}$ $= 13 \text{ V}$	A1
7(b)	sinusoidal curve of period 50 ms from $t = 0$ to $t = 100 \text{ ms}$	B1
	correct phase (V_{MAX} at $t = 0, 50, 100 \text{ ms}$ and $-V_{\text{MAX}}$ at 25, 75 ms etc.)	B1
	maximum and minimum voltages shown as $\pm 18 \text{ V}$	B1
7(c)	<i>Any three points from:</i> - rectification is full-wave - mean power = 14 W - resistance of $R = 12 \Omega$ - peak current in $R = 1.6 \text{ A}$ or r.m.s. current in $R = 1.1 \text{ A}$ - period of <u>output</u> voltage / power = 25 ms or frequency of <u>output</u> voltage / power = 40 Hz or angular frequency of <u>output</u> voltage / power = 250 rad s^{-1}	B3

PUBLISHED

Question	Answer	Marks
8(a)	<i>Any three points from:</i> • electrons moving between levels emit a single photon • energy of photon = difference between energy levels • energy of photon depends on frequency • discrete frequencies (in spectrum) so differences between electron energies must be discrete • discrete differences between electron energies means energy levels must be discrete	B3
8(b)(i)	energy = $-(13.6 \times 1.60 \times 10^{-19})$ $= -2.18 \times 10^{-18} \text{ J}$	A1
8(b)(ii)	$\Delta E = hf$	C1
	$= (6.63 \times 10^{-34} \times 2.47 \times 10^{15}) / (1.60 \times 10^{-19}) = 10.2 \text{ eV}$	A1
8(b)(iii)	$n = 2$ energy level = -3.4 eV	A1
	$n = 3$ energy difference = 12.1 eV	A1
	$n = 4$ energy difference = 12.8 eV	A1
	$n = 3$ energy level = -1.5 eV and $n = 4$ energy level = -0.8 eV	A1

PUBLISHED

Question	Answer	Marks
9(a)	difference between mass of nucleus and mass of (constituent) nucleons	M1
	when nucleons are separated to infinity	A1
9(b)	$\Delta m = (2 \times 2.013553) - (4.001505) \text{ (u)}$ $(= 0.025601 \text{ u})$	C1
	$E = c^2 \Delta m$	C1
	energy from one He-4 nucleus = $0.025601 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ $(= 3.82 \times 10^{-12} \text{ J})$	C1
	energy to form 1.00 mol = $3.82 \times 10^{-12} \times 6.02 \times 10^{23}$ $= 2.30 \times 10^{12} \text{ J}$	A1
9(c)(i)	$L = 1.09 \times 10^{11} \times (3.00 \times 10^8)^2$	C1
	$= 9.81 \times 10^{27} \text{ W}$	A1
9(c)(ii)	$L = 4\pi\sigma r^2 T^4$ $9.81 \times 10^{27} = 4\pi \times 5.67 \times 10^{-8} \times (1.19 \times 10^9)^2 \times T^4$	C1
	$T = 9930 \text{ K}$	A1
9(d)	standard candles have known luminosity	B1
	radiant flux intensity (from star) measured (on the Earth)	B1
	distance found from $F = L / (4\pi d^2)$	B1

PUBLISHED

Question	Answer	Marks
10(a)	<u>difference</u> in degrees of blackening	B1
10(b)(i)	$I = I_0 \exp(-\mu x)$ $= I_0 \exp(-5.8 \times 0.35) = 0.13 I_0$	A1
10(b)(ii)	use of $\exp\{-(0.35 \times 3.7)\}$ factor	C1
	$0.053 I_0 = I_0 \exp\{-[(0.35 \times 3.7) + 2.1\mu]\}$	C1
	$\mu = 0.78 \text{ cm}^{-1}$	A1
10(b)(iii)	factor of only 2.5 between the (detected) <u>intensities</u> (so not good contrast)	B1
10(c)	(structure) scanned in (thin) sections	B1
	(many) scans (of each section) taken from different angles	B1
	scanning repeated for all sections and (data) compiled (to form 3D image)	B1

Cambridge International AS & A Level

PHYSICS**9702/43**

Paper 4 A Level Structured Questions

October/November 2025**MARK SCHEME**Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **17** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	arithmetic error
	benefit of the doubt given
	contradiction in response, mark not awarded
	correct point or mark awarded
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect physics
	incorrect point or mark not awarded
	information missing or insufficient for credit

Annotation	Meaning
MO	mandatory mark not awarded
SEEN	point has been noted, but no credit has been given or blank page seen
POT	power of ten error
TE	transcription error

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

PUBLISHED

Question	Answer	Marks
1(a)	velocity and acceleration both have constant magnitude	B1
	velocity is (always) perpendicular to acceleration	B1
1(b)(i)	$v = R\omega$	A1
1(b)(ii)	$a = R\omega^2$ or $a = v^2 / R$	C1
	$a = v\omega$	A1
1(c)(i)	$x = R \sin \theta$	A1
1(c)(ii)	$\theta = \omega t$	A1
1(c)(iii)	clear substitution of $\theta = \omega t$ into $x = R \sin \theta$ leading to $x = R \sin \omega t$	A1
1(c)(iv)	equation is of the form $x = x_0 \sin \omega t$ (so simple harmonic motion)	B1
1(d)(i)	amplitude = $0.46 / 2$ = 0.23 m	A1
1(d)(ii)	$\omega = 2\pi / T$	C1
	period = $2\pi / 1.9$ = 3.3 s	A1

PUBLISHED

Question	Answer	Marks
1(d)(iii)	$a_0 = \omega^2 x_0$	C1
	$= 1.9^2 \times 0.23$	A1
	$= 0.83 \text{ m s}^{-2}$	
1(e)	shadow on screen, labelled A, above left-hand edge of the circular path	B1

Question	Answer	Marks
2(a)	work done on / by system	B1
	thermal energy supplied to / removed from system	B1
2(b)(i)	no thermal energy transferred to / from system (due to lack of time)	B1
	work is done on the gas to compress it / to decrease its volume	B1
	internal energy increases so temperature increases	B1
2(b)(ii)	(during vaporisation) molecular separation increases	B1
	(heating causes) potential energy of molecules to increase	B1
	kinetic energy of molecules unchanged so temperature unchanged	B1

PUBLISHED

Question	Answer	Marks
3(a)	force per unit mass	B1
3(b)(i)	$F = GMm / x^2$	C1
	$g = F / m$	A1
	$g = [GMm / x^2] / m = GM / x^2$ and G = gravitational constant	
3(b)(ii)	arrow drawn at P pointing directly towards the point mass	B1
3(b)(iii)	fields are in opposite directions	B1
	field strength at Q is four times the field strength at P	B1
3(c)	line starting at $(R, -g_0)$ and ending at $(L - R, +g_0)$	B1
	line passing through $(L / 2, 0)$	B1
	curve becoming shallower from R to $(L / 2)$ and then steeper from $(L / 2)$ to $(L - R)$	B1

PUBLISHED

Question	Answer	Marks
4(a)(i)	temperature = $-273.15\text{ }^{\circ}\text{C}$	A1
4(a)(ii)	temperature = 0 K	A1
4(b)(i)	gas is ideal	B1
4(b)(ii)	$pV = NkT$	C1
	$N = 270 / (8.0 \times 10^{-21})$ $= 3.4 \times 10^{22}$	A1
4(b)(iii)	$n = (3.4 \times 10^{22}) / (6.02 \times 10^{23})$ $= 0.056\text{ mol}$	A1
4(c)	$\frac{1}{2} m\langle c^2 \rangle = (3 / 2) kT$	C1
	$\frac{1}{2} \times m \times 1900^2 = 1.5 \times 8.0 \times 10^{-21}$ $(m = 6.65 \times 10^{-27}\text{ kg})$	C1
	$m = (6.65 \times 10^{-27}) / (1.66 \times 10^{-27})$	C1
	$= 4.0\text{ u}$	A1

PUBLISHED

Question	Answer	Marks
5(a)	work done per unit charge	B1
	work (done) moving positive charge from infinity (to the point)	B1
5(b)(i)	potential (due to proton) = $(1.60 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12} \times 10 \times 10^{-12})$ or potential (due to electron) = $(-1.60 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12} \times 110 \times 10^{-12})$	C1
	$V = [(1.60 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12})] \times [(10^{-1} - 110^{-1}) \times 10^{12}] = 130 \text{ V}$	A1
5(b)(ii)	$V = [(1.60 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12})] \times [(30^{-1} - 90^{-1}) \times 10^{12}]$	C1
	= (+) 32 V	A1
5(b)(iii)	cross drawn midway between the electron and the proton	B1
5(b)(iv)	line from (10, +130) to (110, -130)	B1
	curve getting shallower until $x = 60 \text{ pm}$, crossing $V = 0$ at (60, 0) and then getting steeper after $x = 60 \text{ pm}$	B1
	curve passing through (30, ± 32) and (90, ± 32)	B1

PUBLISHED

Question	Answer	Marks
6(a)	series charges: $Q_s = Q_1 = Q_2$	B1
	series p.d.s: $V_s = V_1 + V_2$	B1
	parallel charges: $Q_s = Q_1 + Q_2$	B1
	parallel p.d.s: $V_s = V_1 = V_2$	B1
6(b)(i)	$E = \frac{1}{2} CV^2$	C1
	p.d. = $[(2 \times 19 \times 10^{-3}) / (470 \times 10^{-6})]^{\frac{1}{2}}$ $= 9.0 \text{ V}$	A1
6(b)(ii)	$E = Q^2 / 2C$ or $C = Q / V$	C1
	$Q = (19 \times 10^{-3} \times 2 \times 470 \times 10^{-6})^{\frac{1}{2}}$ or $Q = 470 \times 10^{-6} \times 9.0$ $Q = 4.2 \times 10^{-3} \text{ C}$	A1
6(b)(iii)	total charge unchanged	C1
	total capacitance = $(470 + 180) \times 10^{-6} \text{ (F)}$	C1
	$E = Q^2 / 2C = (4.23 \times 10^{-3})^2 / (2 \times 650 \times 10^{-6})$ (= 0.014 J) $E = 14 \text{ mJ}$	A1

PUBLISHED

Question	Answer	Marks
7(a)	(induced) e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)(i)	flux = e.m.f. $\times$ time	C1
	flux = 0.54×15 = 8.1 Wb	A1
7(b)(ii)	$\Phi = BA$	C1
	area = $8.1 / (38 \times 10^{-6})$ = $2.1 \times 10^5 \text{ m}^2$	A1
7(b)(iii)	area = speed $\times$ time $\times$ width	C1
	$v = (2.1 \times 10^5) / (15 \times 68)$ = 210 m s^{-1}	A1
7(b)(iv)	opposing force (due to current in wings) must be backwards	B1
	from Fleming's left-hand rule, current (in wings) must be from Q to P	B1
	current is from – to + inside an e.m.f. source so P is at higher potential	B1

PUBLISHED

Question	Answer	Marks
8(a)	packet / quantum of <u>energy</u>	M1
	of electromagnetic radiation	A1
8(b)(i)	$E = c^2 \Delta m$	C1
	$\Delta m = (4.274 \times 10^6 \times 1.60 \times 10^{-19}) / (1.66 \times 10^{-27} \times (3.00 \times 10^8)^2)$ (= 0.00458 u)	C1
	$m = 233.915174 + 4.000407 + 0.00458$ = 237.92016 u	A1
8(b)(ii)	$E = hc / \lambda$ or $E = hf$ and $c = f\lambda$	C1
	$(4.274 - 4.200) \times 1.60 \times 10^{-13} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$	C1
	$\lambda = 1.7 \times 10^{-11} \text{ m}$	A1
8(b)(iii)	(true) energy of gamma photon is smaller so (true) wavelength is larger	B1
8(c)	(anti)neutrinos are emitted during beta decay	B1
	particles emitted during beta decay carry varying amounts of energy, so energy of gamma photon is also variable (between decays)	B1

PUBLISHED

Question	Answer	Marks
9(a)	temperature inversely proportional to wavelength	M1
	temperature is thermodynamic temperature of surface of star and wavelength is the wavelength at which maximum emission rate from star occurs	A1
9(b)	<i>Any three points from:</i> • (surface) temperature of star X = 7000 K or star X has a higher temperature than the Sun • star X has a higher luminosity than the Sun • luminosity of star X = 2.7×10^{27} W • radius of star X = 1.3×10^9 m	B3
9(c)	light (from star X) is redshifted	B1
	wavelength of peak emission rate would be greater (using observed data)	B1

PUBLISHED

Question	Answer	Marks
10(a)	product of density and speed	M1
	speed of sound in medium (and density of the medium)	A1
10(b)	ultrasound waves cause crystal to vibrate	B1
	vibrations (of crystal) cause induced e.m.f. (across crystal)	B1
10(c)(i)	intensity reflection coefficient= $(40.4 - 1.48)^2 / (40.4 + 1.48)^2$	C1
	= 0.86	A1
10(c)(ii)	Z values are very similar	B1
	(almost) all the ultrasound will be transmitted or (almost) none of the ultrasound will be reflected	B1

Cambridge International AS & A Level

PHYSICS**9702/51**

Paper 5 Planning, Analysis and Evaluation

October/November 2025

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **11** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	benefit of the doubt given
	correct awarding one mark from additional detail 1. similar numbered ticks are used for additional detail 2, 3, 4 etc.
	correct point or mark awarded
	defining the problem mark
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect point or mark not awarded
	incorrect unit
	information missing or insufficient for credit

Annotation	Meaning
MO	methods of data collection mark
SEEN	point has been noted, but no credit has been given or blank page seen
R	repeat of point previously awarded mark

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	vary r and measure v or r is the independent variable and v is the dependent variable	1
	keep x <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - one end of spring resting against block clamped to bench using G-clamp - light gate positioned at P - light gate connected to timer - apparatus shown on bench - labels for light gate and P and at least one other label from bench, block, stand, spring, ball, timer	1
	method to determine r , e.g. use calipers or micrometer to measure diameter d and $r = d / 2$	1
	description of method to determine v , use diameter of ball (to interrupt beam) $\div$ measured time at light gate positioned at P	1
	instrument to determine x , e.g. rule(r) or calipers	1
	Method of Analysis	
	plot a graph of $(2 \lg v)$ or $(\lg v^2)$ against $\lg r$ or plot a graph of $(\lg v)$ against $(\lg r)$ or equivalent, e.g. $(\ln v)$ against $(\ln r)$	1
	$n = -\text{gradient}$ for $(2 \lg v)$ or $(\lg v^2)$ against $\lg r$ or $n = -2 \times \text{gradient}$ for $(\lg v)$ against $\lg r$	1

PUBLISHED

Question	Answer	Marks
1	$Y = \frac{\rho \times 10^{y\text{-intercept}}}{kx^2} \quad \text{for } (2 \lg v) \text{ or } (\lg v^2) \text{ against } \lg r$ or $Y = \frac{\rho \times 10^{2 \times y\text{-intercept}}}{kx^2} \quad \text{for } (\lg v) \text{ against } \lg r$	1
	Additional detail including safety considerations	6
	D1 precaution to <u>prevent ball leaving bench</u> , e.g. screens around apparatus / cushions on bench (to stop the ball)	
	D2 keep k and ρ constant	
	D3 description of method to determine k , e.g. add mass to spring and $k = mg / \text{extension}$ or use newton meter to measure force applied to spring and $k = \text{force} / \text{extension}$ or take several readings of force and extension, plot a force–extension graph and $k = \text{gradient}$	
	D4 description of experimental method to determine ρ , e.g. measure mass of ball using a balance and $\rho = \frac{m}{\frac{4}{3}\pi r^3}$	
	D5 repeat measurements of <u>diameter</u> or <u>d</u> in different directions <u>and</u> determine the average value of d	
	D6 method to keep x constant, e.g. use a pin / ruler / card to indicate the starting point each time <u>to keep x constant</u>	
	D7 $x = \text{original length of spring} - \text{compressed length of spring}$	
	D8 adjust (vertical) position of light gate so that the diameter of (each) ball cuts the beam	
	D9 repeat experiment for the same value of r and determine the average v	

Question	Answer	Marks
1	D10 relationship valid <u>if</u> a straight line is produced (with y-intercept = $\lg\left(\frac{Ykx^2}{\rho}\right)$ or $\frac{1}{2}\lg\left(\frac{Ykx^2}{\rho}\right)$). Do not accept line through the origin.	

Question	Answer	Marks							
2(a)	gradient = $H\lambda_0$ y-intercept = λ_0	1							
2(b)	<table><tr><td>$\frac{d}{c} / 10^{15} \text{ s}$</td></tr><tr><td>$(1.6 \text{ or } 1.60) \pm 0.40$</td></tr><tr><td>$(3.47 \text{ or } 3.467) \pm 0.40$</td></tr><tr><td>$(4.83 \text{ or } 4.833) \pm 0.40$</td></tr><tr><td>$(6.00 \text{ or } 6.000) \pm 0.40$</td></tr><tr><td>$(9.50 \text{ or } 9.500) \pm 0.40$</td></tr><tr><td>$(12.5 \text{ or } 12.50) \pm 0.40$</td></tr></table> Values of $\frac{d}{c}$ correct as shown above.	$\frac{d}{c} / 10^{15} \text{ s}$	$(1.6 \text{ or } 1.60) \pm 0.40$	$(3.47 \text{ or } 3.467) \pm 0.40$	$(4.83 \text{ or } 4.833) \pm 0.40$	$(6.00 \text{ or } 6.000) \pm 0.40$	$(9.50 \text{ or } 9.500) \pm 0.40$	$(12.5 \text{ or } 12.50) \pm 0.40$	1
	$\frac{d}{c} / 10^{15} \text{ s}$								
$(1.6 \text{ or } 1.60) \pm 0.40$									
$(3.47 \text{ or } 3.467) \pm 0.40$									
$(4.83 \text{ or } 4.833) \pm 0.40$									
$(6.00 \text{ or } 6.000) \pm 0.40$									
$(9.50 \text{ or } 9.500) \pm 0.40$									
$(12.5 \text{ or } 12.50) \pm 0.40$									
	Uncertainties in $\frac{d}{c}$ correct as shown above.	1							

PUBLISHED

Question	Answer	Marks
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1
	Error bars in $\frac{d}{c}$ plotted correctly. All error bars must be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1
2(c)(ii)	Straight line of best fit drawn. Thickness of the line must be less than half a small square. Do not accept line from top point to bottom point. Line must pass between (2.5, 660.0) and (2.9, 660.0) and between (11.2, 676.0) and (11.6, 676.0).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). Thickness of the line must be less than half a small square. All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and x and y into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

PUBLISHED

Question	Answer	Marks
2(d)	λ_0 determined using y-intercept and λ_0 given to 3 or 4 significant figures and H given to 2, 3 or 4 significant figures. $\lambda_0 = \text{y-intercept}$	1
	H determined using gradient and λ_0 and H given with SI units with appropriate powers of ten. $H = \frac{\text{gradient}}{\text{y-intercept}}$ or $H = \frac{\text{gradient}}{\lambda_0}$ Unit of λ_0 : m, nm, μm Unit of H : s^{-1}	1
2(e)	Value of T determined to a minimum of two significant figures from (d) and correct power of ten. $T = \frac{1}{H}$	1
	Absolute uncertainty determined with correct substitution. $\Delta T = \left(\frac{\Delta \text{y-intercept}}{\text{y-intercept}} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times T$ or $\Delta T = \left(\frac{\text{max } \lambda_0}{\text{min gradient}} \right) - T \quad \text{or} \quad \Delta T = \left(\frac{\text{max y-intercept}}{\text{min gradient}} \right) - T$ or $\Delta T = \left(\frac{\text{min } \lambda_0}{\text{max gradient}} \right) - T \quad \text{or} \quad \Delta T = \left(\frac{\text{min y-intercept}}{\text{max gradient}} \right) - T$	1

Cambridge International AS & A Level

PHYSICS**9702/52**

Paper 5 Planning, Analysis and Evaluation

October/November 2025

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	benefit of the doubt given
	correct awarding one mark from additional detail 1. similar numbered ticks are used for additional detail 2, 3, 4 etc.
	correct point or mark awarded
	defining the problem mark
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect point or mark not awarded
	incorrect unit
	information missing or insufficient for credit

PUBLISHED

Annotation	Meaning
MO	methods of data collection mark
SEEN	point has been noted, but no credit has been given or blank page seen
R	repeat of point previously awarded mark

Question	Answer	Marks
1	Defining the problem	
	vary v and measure I or v is the independent variable and I is the dependent variable	1
	keep A and R <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - fan positioned in line with the turbine so that blades of both fan and turbine overlap - fan on bench - fan labelled and one other label from bench, turbine, blade(s) (of turbine), terminals, L	1
	workable circuit diagram showing resistor connected to an ammeter in series with the terminals of the turbine using correct circuit symbols	1
	method to vary v , e.g. change speed of fan / change distance between fan and blades / vary current in or p.d. across fan	1
	method to determine temperature T , e.g. use a thermometer	1
	Method of Analysis	
	plot a graph of I^2 against v^3 or equivalent (e.g. $\lg I$ against $\lg v$ or $2 \ln I$ against $\ln v$)	1
	relationship valid <u>if</u> a straight line is produced passing through the origin (For $\lg I$ against $\lg v$: relationship valid <u>if</u> a straight line <u>with gradient = 1.5</u> is produced)	1
	$Q = \frac{2TR \times \text{gradient}}{AP}$ (For $\lg I$ against $\lg v$: $Q = \frac{2TR \times 10^{2 \times y\text{-intercept}}}{AP}$)	1

PUBLISHED

Question	Answer	Marks
1	Additional detail including safety considerations	6
	D1 precaution with reason linked to (moving) fan blades / turbine blades, e.g. keep away from the fan to avoid (moving) blades or use a screen around the fan / turbine to avoid (moving) blades or precaution with reason linked to prevent air / dust particles in eye, e.g. use goggles to avoid air stream (into eye)	
	D2 clamp turbine / fan <u>to bench</u>	
	D3 keep P and T <u>constant</u>	
	D4 $T = t + 273$	
	D5 method to determine A : use a rule(r) / calipers to measure L and $A = \pi L^2$	
	D6 repeat measurements of L in different positions / different blades and average	
	D7 method to measure v , e.g. use an anemometer or <u>air speed</u> meter or method to measure P , e.g. use a manometer or barometer or <u>pressure</u> gauge	
	D8 wait for steady / constant air flow / movement of blades / current	
	D9 method to determine R , e.g.: separate circuit showing ohmmeter connected to R only or terminals of turbine connected correctly to resistor and ammeter and voltmeter across resistor and $R = V / I$ or separate workable circuit with power supply resistor, ammeter and voltmeter across R and $R = V / I$	
	D10 method to check temperature / pressure is constant, e.g. measure temperature / pressure several times / before and after	

PUBLISHED

Question	Answer	Marks														
2(a)	gradient = n y-intercept = $\lg \frac{2\pi}{k}$	1														
2(b)	<table><tr><th>$\lg (r / 10^8 \text{ m})$</th><th>$\lg (T / 10^3 \text{ s})$</th></tr><tr><td>0.152 or 0.1523</td><td>$(1.72 \text{ or } 1.716) \pm 0.04$</td></tr><tr><td>0.270 or 0.2695</td><td>$(1.91 \text{ or } 1.908) \pm 0.03$</td></tr><tr><td>0.377 or 0.3766</td><td>$(2.08 \text{ or } 2.079 \text{ or } 2.0792) \pm 0.04$</td></tr><tr><td>0.470 or 0.4698</td><td>$(2.23 \text{ or } 2.230 \text{ or } 2.2304) \pm 0.03$</td></tr><tr><td>0.576 or 0.5763</td><td>$(2.38 \text{ or } 2.380 \text{ or } 2.3802) \pm 0.04$</td></tr><tr><td>0.723 or 0.7226</td><td>$(2.59 \text{ or } 2.591 \text{ or } 2.5911) \pm 0.03$</td></tr></table> Values of $\lg (r / 10^8 \text{ m})$ and $\lg (T / 10^3 \text{ s})$ correct as shown above. Uncertainties in $\lg (T / 10^3 \text{ s})$ correct as shown above.	$\lg (r / 10^8 \text{ m})$	$\lg (T / 10^3 \text{ s})$	0.152 or 0.1523	$(1.72 \text{ or } 1.716) \pm 0.04$	0.270 or 0.2695	$(1.91 \text{ or } 1.908) \pm 0.03$	0.377 or 0.3766	$(2.08 \text{ or } 2.079 \text{ or } 2.0792) \pm 0.04$	0.470 or 0.4698	$(2.23 \text{ or } 2.230 \text{ or } 2.2304) \pm 0.03$	0.576 or 0.5763	$(2.38 \text{ or } 2.380 \text{ or } 2.3802) \pm 0.04$	0.723 or 0.7226	$(2.59 \text{ or } 2.591 \text{ or } 2.5911) \pm 0.03$	1
$\lg (r / 10^8 \text{ m})$	$\lg (T / 10^3 \text{ s})$															
0.152 or 0.1523	$(1.72 \text{ or } 1.716) \pm 0.04$															
0.270 or 0.2695	$(1.91 \text{ or } 1.908) \pm 0.03$															
0.377 or 0.3766	$(2.08 \text{ or } 2.079 \text{ or } 2.0792) \pm 0.04$															
0.470 or 0.4698	$(2.23 \text{ or } 2.230 \text{ or } 2.2304) \pm 0.03$															
0.576 or 0.5763	$(2.38 \text{ or } 2.380 \text{ or } 2.3802) \pm 0.04$															
0.723 or 0.7226	$(2.59 \text{ or } 2.591 \text{ or } 2.5911) \pm 0.03$															
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1														
	Error bars in $\lg (T / 10^3 \text{ s})$ plotted correctly. All error bars must be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1														

PUBLISHED

Question	Answer	Marks
2(c)(ii)	Straight line of best fit drawn. Thickness of the line must be less than half a small square. Do not accept line from top point to bottom point. Line must pass between (0.19, 1.80) and (0.21, 1.80) and between (0.645, 2.50) and (0.66, 2.50).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). Thickness of the line must be less than half a small square. All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent powers of ten into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

Question	Answer	Marks
2(d)	Value of n determined using gradient and n and k given to 2 or 3 significant figures. $n = \text{gradient} = \text{(c)(iii)}$	1
	Value of k determined using y -intercept. Correct method must be seen. $k = \frac{2\pi}{10^{y\text{-intercept}}} = \frac{2\pi}{10^{\text{(c)(iv)}}$	1
	Absolute uncertainties in n and k determined. Absolute uncertainty in $n = \text{absolute uncertainty in gradient}$ and $\Delta k = \frac{2\pi}{10^{y\text{-intercept}}} - \frac{2\pi}{10^{\text{WAL } y\text{-intercept}}}$ Correct method must be seen.	1
2(e)	r determined to a minimum of 2 significant figures from (c)(iii) and (c)(iv) or (d) with correct substitution and correct power of ten. $r = 10^{\frac{\lg 1380 - y\text{-intercept}}{\text{gradient}}} \times 10^8$ or $r = \sqrt[n]{\frac{k \times 1380}{2\pi}} \times 10^8$	1

Cambridge International AS & A Level

PHYSICS**9702/53**

Paper 5 Planning, Analysis and Evaluation

October/November 2025

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	benefit of the doubt given
	correct awarding one mark from additional detail 1. similar numbered ticks are used for additional detail 2, 3, 4 etc.
	correct point or mark awarded
	defining the problem mark
	error carried forward applied
	error in number of significant figures
	incorrect or insufficient point ignored while marking the rest of the response
	incorrect point or mark not awarded
	incorrect unit
	information missing or insufficient for credit

Annotation	Meaning
MO	methods of data collection mark
SEEN	point has been noted, but no credit has been given or blank page seen
R	repeat of point previously awarded mark

PUBLISHED

Question	Answer	Marks
1	Defining the problem	
	vary r and measure v or r is the independent variable and v is the dependent variable	1
	keep x <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: - one end of spring resting against block clamped to bench using G-clamp - light gate positioned at P - light gate connected to timer - apparatus shown on bench - labels for light gate and P and at least one other label from bench, block, stand, spring, ball, timer	1
	method to determine r , e.g. use calipers or micrometer to measure diameter d and $r = d / 2$	1
	description of method to determine v , use diameter of ball (to interrupt beam) $\div$ measured time at light gate positioned at P	1
	instrument to determine x , e.g. rule(r) or calipers	1
	Method of Analysis	
	plot a graph of $(2 \lg v)$ or $(\lg v^2)$ against $\lg r$ or plot a graph of $(\lg v)$ against $(\lg r)$ or equivalent, e.g. $(\ln v)$ against $(\ln r)$	1
	$n = -\text{gradient}$ for $(2 \lg v)$ or $(\lg v^2)$ against $\lg r$ or $n = -2 \times \text{gradient}$ for $(\lg v)$ against $\lg r$	1

PUBLISHED

Question	Answer	Marks
1	$Y = \frac{\rho \times 10^{y\text{-intercept}}}{kx^2} \quad \text{for } (2 \lg v) \text{ or } (\lg v^2) \text{ against } \lg r$ or $Y = \frac{\rho \times 10^{2 \times y\text{-intercept}}}{kx^2} \quad \text{for } (\lg v) \text{ against } \lg r$	1
	Additional detail including safety considerations	6
	D1 precaution to <u>prevent ball leaving bench</u> , e.g. screens around apparatus / cushions on bench (to stop the ball)	
	D2 keep k and ρ constant	
	D3 description of method to determine k , e.g. add mass to spring and $k = mg / \text{extension}$ or use newton meter to measure force applied to spring and $k = \text{force} / \text{extension}$ or take several readings of force and extension, plot a force–extension graph and $k = \text{gradient}$	
	D4 description of experimental method to determine ρ , e.g. measure mass of ball using a balance and $\rho = \frac{m}{\frac{4}{3}\pi r^3}$	
	D5 repeat measurements of <u>diameter</u> or <u>d</u> in different directions <u>and</u> determine the average value of d	
	D6 method to keep x constant, e.g. use a pin / ruler / card to indicate the starting point each time <u>to keep x constant</u>	
	D7 $x = \text{original length of spring} - \text{compressed length of spring}$	
	D8 adjust (vertical) position of light gate so that the diameter of (each) ball cuts the beam	
	D9 repeat experiment for the same value of r and determine the average v	

Question	Answer	Marks
1	D10 relationship valid <u>if</u> a straight line is produced (with y-intercept = $\lg\left(\frac{Ykx^2}{\rho}\right)$ or $\frac{1}{2}\lg\left(\frac{Ykx^2}{\rho}\right)$). Do not accept line through the origin.	

Question	Answer	Marks							
2(a)	gradient = $H\lambda_0$ y-intercept = λ_0	1							
2(b)	<table><tr><td>$\frac{d}{c} / 10^{15} \text{ s}$</td></tr><tr><td>(1.6 or 1.60) $\pm$ 0.40</td></tr><tr><td>(3.47 or 3.467) $\pm$ 0.40</td></tr><tr><td>(4.83 or 4.833) $\pm$ 0.40</td></tr><tr><td>(6.00 or 6.000) $\pm$ 0.40</td></tr><tr><td>(9.50 or 9.500) $\pm$ 0.40</td></tr><tr><td>(12.5 or 12.50) $\pm$ 0.40</td></tr></table> Values of $\frac{d}{c}$ correct as shown above.	$\frac{d}{c} / 10^{15} \text{ s}$	(1.6 or 1.60) $\pm$ 0.40	(3.47 or 3.467) $\pm$ 0.40	(4.83 or 4.833) $\pm$ 0.40	(6.00 or 6.000) $\pm$ 0.40	(9.50 or 9.500) $\pm$ 0.40	(12.5 or 12.50) $\pm$ 0.40	1
$\frac{d}{c} / 10^{15} \text{ s}$									
(1.6 or 1.60) $\pm$ 0.40									
(3.47 or 3.467) $\pm$ 0.40									
(4.83 or 4.833) $\pm$ 0.40									
(6.00 or 6.000) $\pm$ 0.40									
(9.50 or 9.500) $\pm$ 0.40									
(12.5 or 12.50) $\pm$ 0.40									
	Uncertainties in $\frac{d}{c}$ correct as shown above.	1							

PUBLISHED

Question	Answer	Marks
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1
	Error bars in $\frac{d}{c}$ plotted correctly. All error bars must be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	1
2(c)(ii)	Straight line of best fit drawn. Thickness of the line must be less than half a small square. Do not accept line from top point to bottom point. Line must pass between (2.5, 660.0) and (2.9, 660.0) and between (11.2, 676.0) and (11.6, 676.0).	1
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). Thickness of the line must be less than half a small square. All error bars must be plotted.	1
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be greater than half the length of the drawn line.	1
	Gradient determined of worst acceptable line with clear substitution of data points into $\Delta y / \Delta x$. uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined by substitution of correct point with consistent power of ten in m and x and y into $y = mx + c$.	1
	y-intercept of worst acceptable line determined by substitution into $y = mx + c$. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) Do not accept ECF from false origin method.	1

PUBLISHED

Question	Answer	Marks
2(d)	λ_0 determined using y-intercept and λ_0 given to 3 or 4 significant figures and H given to 2, 3 or 4 significant figures. $\lambda_0 = \text{y-intercept}$	1
	H determined using gradient and λ_0 and H given with SI units with appropriate powers of ten. $H = \frac{\text{gradient}}{\text{y-intercept}}$ or $H = \frac{\text{gradient}}{\lambda_0}$ Unit of λ_0 : m, nm, μm Unit of H : s^{-1}	1
2(e)	Value of T determined to a minimum of two significant figures from (d) and correct power of ten. $T = \frac{1}{H}$	1
	Absolute uncertainty determined with correct substitution. $\Delta T = \left(\frac{\Delta \text{y-intercept}}{\text{y-intercept}} + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times T$ or $\Delta T = \left(\frac{\text{max } \lambda_0}{\text{min gradient}} \right) - T$ or $\Delta T = \left(\frac{\text{max y-intercept}}{\text{min gradient}} \right) - T$ or $\Delta T = \left(\frac{\text{min } \lambda_0}{\text{max gradient}} \right) - T$ or $\Delta T = \left(\frac{\text{min y-intercept}}{\text{max gradient}} \right) - T$	1